ECL701

RIZVI EDUCATION SOCIETY'S

Rizvi College of Engineering

Department of Electronics and Computer Science Engineering

VLSI Design Lab Journal



NAME:

CLASS:

ROLL NO. / UIN:

RIZVI EDUCATION SOCIETY'S

Rizvi College of Engineering

Department of Electronics and Computer Science Engineering

Lab File

Class & Semester: *BE (Sem-VII)*

Subject: *VLSI Design Lab*

Name:……

Roll No. :……..

UIN:…….

Document Revised on: ***10 / 07 / 2023***

RIZVI EDUCATION SOCIETY'S

Rizvi College of Engineering

Department of Electronics and Computer Science Engineering

Certificate

This is to certify that …….. Of BE Final year Electronics and Computer Science Engineering Department has performed all the experiments of VLSI Design Lab with satisfactory results.

……..

Subject In-charge

INSTITUTE VISION

To become a leading entity in transforming the diverse class of learners into innovators, analyzers and entrepreneurs competent to develop eco-friendly sustainable solutions and work in multi-disciplinary environment to meet the global challenges and contribute towards nation building.

DEPARTMENT VISION

To create industry-ready engineers in the field of Electronics and Computer Science, blending expertise in software development with emerging technologies like AI, data science, IoT, and robotics, while building a strong base in automation and electronics through state-of-the-art infrastructure, hands-on learning, and expert guidance, empowering them to deliver innovative, sustainable, and impactful solutions for global societal and industrial challenges.

PROGRAM OUTCOMES

PO1. Engineering knowledge
PO2. Problem analysis
PO3. Design/development of solutions
PO4. Conduct investigations of complex problems
PO5. Modern tool usage
PO6. The engineer and society
PO7. Environment and sustainability
PO8. Ethics
PO9. Individual and team work
PO10. Communication
PO11. Project management and finance
PO12. Life-long learning

PROGRAM SPECIFIC OUTCOMES

PSO1. Intelligent and Connected System Design: Engineer innovative, secure, and scalable systems by synergizing electronics and computing technologies—including AI, IoT, robotics, and embedded systems—to solve complex industrial and societal challenges.

PSO2. Ethical and Sustainable Technological Solutions: Develop responsible, data-driven innovations that integrate ethics, environmental sustainability, and cutting-edge technologies to create meaningful impact in real-world applications.

INSTITUTE MISSION

IM1. Impart Core Fundamental Principles: To impart core fundamental principles of engineering and science that will enable the learner to develop solutions to complex engineering problems, through conventional and innovative teaching learning methods and mentoring.

IM2. Bridge the Technical Skill Gap: To bridge the technical skill gap through curriculum enrichment activities for industry readiness.

IM3. Inculcate Professional Etiquettes and Ethics: To groom the learner through dedicated training, placement and extension activities and to inculcate professional etiquettes and ethics aimed at holistic development of the learners enabling them to acquire distinguished positions in the leading industries or be eligible for higher studies in globally recognized universities.

IM4. Research and Development: To provide modern infrastructure and the necessary resources, for planning and implementing innovative ideas, leading to meaningful research and development and entrepreneurship.

DEPARTMENT MISSION

DM1. Industry-Ready Engineers: Equip students with cutting-edge skills in Electronics and Computer Science through hands-on learning, industry projects, and emerging technologies like AI, Data Science, IoT, and Robotics.

DM2. Bridging Academia & Industry: Integrate real-world applications, internships, and research-driven education to enhance industry readiness and career success.

DM3. Innovation & Entrepreneurship: Foster problem-solving, leadership, and entrepreneurial thinking to develop sustainable and impactful technological solutions.

DM4. Ethical & Sustainable Engineering: Instill technical excellence, ethical values, and a commitment to sustainability for global societal and industrial advancement.

PROGRAM EDUCATIONAL OBJECTIEVES

PEO1. Successful Career: To build a successful career in leading industries related to the field of Computer Engineering wherein the engineer will be able to provide the necessary solutions to the challenges witnessed in existing and new business models.

PEO2. Leadership Qualities: To exhibit the qualities of team spirit, leadership, and problem-solving skills to achieve top positions in the organization or to enhance entrepreneurial skills.

PEO3. Adaptability to New Technology: To be able to adapt to new technologies and platforms and share their knowledge with their peers in the allied fields.

PEO4. Research & Higher Studies: To develop proficiency in computer engineering and related fields to be able to work in multi-disciplinary areas with a strong focus on innovation and research.

Laboratory Outcomes

LO1: Demonstrate transfer, dynamic characteristics of various digital circuits.

LO2: Understand the circuit design using various simulation tools

LO3: Demonstrate layouts for various circuits and doing simulations.

LO4: Understand the variation in the behaviour after extraction.

Rubrics

Following rubrics will be used to assess the work submitted by the students.

1. **For Experiment 1 to 10**

Criteria 1 : Timely Submission			
5	**4**	**3**	**1**
Excellent	**Good**	**Average**	**Poor**
The entire Experiment assessed before the deadline specified.	The Experiment was assessed on deadline.	1-week late submission of Experiment.	Too late submission of Experiment.
Criteria 2 : Result Interpretation and Output			
5	**4**	**3**	**1**
Excellent	**Good**	**Average**	**Poor**
Clear and good interpretation of all data. Finding of results are clear and logic of experiment is correct.	Interpretation of all data is correct. Finding of results are OK	Interpretation of all data is correct. Finding of results requires some suggestions.	Major part of expected output is not found. Or wrong output

Criteria 3 : Program Logic & Layout understanding			
5	**4**	**3**	**1**
Excellent	**Good**	**Average**	**Poor**
Excellent understanding of Program Logic and small viva shows complete understanding of the Concept.	Good understanding of Program Logic & small viva shows the good understanding Concept of experiments	Average understanding of Program Logic & small viva shows some questions about Concept are not answered	Response shows a complete lack of understanding for the Experiment.

2. For Assignments 1 to 3

Criteria 1 : General description			
Criteria description in detail			
5	**4**	**3**	**1**
Excellent	Good	Average	Poor
Demonstrates an appreciation of the limitations an temporary nature of conceptual Knowledge in the discipline or field.	Demonstrates an appreciation of them limits of their own understanding.	Have adequate breadth, but limited depth of understanding of basic concepts	Cannot discuss concepts in their own words.
Criteria 2 : Knowledge and understanding			
Criteria description in detail			
5	**4**	**3**	**1**
Excellent	Good	Average	Poor
Able to recall the basic theoretical knowledge of the experiment and able to follow the practical procedure to hardware software co-design	Able to recall the basic theoretical knowledge of the experiment and able to follow the practical procedure	Able to recall the basic theoretical knowledge of the experiment and unable to follow the practical procedure	Unable to recall the basic theoretical knowledge of the experiment and unable to follow the practical procedure
Criteria 3 : Levels of attainment			
Criteria description in detail			
5	**4**	3	**1**
Excellent	Good	Average	Poor
Has gone beyond the basic expected standards. Exhibits high levels of independence and can use principles to	Has completely reached the expected standards of thinking and practice. Can function independently in novel contexts, adapting	Reached basic academic standards Accurate knowledge of some facts and capable of limited safe practice.	Not yet to desired standard of knowledge or safe practice

generate new understandings and ways of completing procedures and can provide theoretically defensible arguments for their new interpretation and adaptations	concepts, skills and procedures to meet situational factors		

Lab Guidelines

1. Students should report to the concerned labs as per time table schedule.

2. Students who turn up late to the labs will in no case be permitted to do the program scheduled for the day.

3. You may use the computers in the lab only when a teacher is present.

4. Report any hardware fault immediately to your teacher.

5. Never attempt to dismantle the different parts of the computer.

6. Each student must log in to his/her account. No sharing of accounts is permitted.

7. The computers are for your academic use. Playing computer games for entertainment is strictly not allowed.

8. Shut down the computer properly after use.

9. Do not charge your personal mobile devices in the lab.

INDEX

Sr. No.	Title	Page No.	Performed On	Sign & Date	Rubrics Points
1	To study and verification of Output and Transfer characteristics of NMOS transistor.	**9-22**			
2	To study and verification of Output and Transfer characteristics of PMOS transistor.	**23-35**			
3	Design and verify the layout of NMOS and PMOS devices in micro-wind	**36-55**			
4	To design and verify the layout of CMOS inverter.	**56-62**			
5	To design layout of NAND and NOR gate using micro-wind	**63-73**			
6	To design layout of S-R flip-flop using Micro-wind.	**74-77**			
7	To design the layout of the D latch using micro-wind	**78-83**			
8	To design the layout of 6T SRAM using micro-wind.	**84-88**			
9	To design the layout of Ripple Carry Adder using DSCH and Micro-wind	**89-97**			
10	To design the layout of Carry look ahead adder using DSCH and Micro-wind	**98-107**			
1.	Assignment 2				
2.	Assignment 3				
				TOTAL	

EXPERIMENTS

Expt. No. 1

Title: *To study and verification of Output and Transfer characteristics of NMOS transistor*

Rubric	*Score (1 to 5)*
Timely Submission	
Result Interpretation and Output	
Program Logic and Concept Understanding	
Total	

Performed On: _____

Sign:_____

Experiment No. 1

Aim: To study and verification of Output and Transfer characteristics of NMOS transistor

software: Computer with Or-cad P-spice

Link for virtual lab:
http://www.vlsi-iitg.vlabs.ac.in

Theory:

Introduction

The metal–oxide–semiconductor field-effect transistor (MOSFET) is a transistor used for amplifying or switching electronic signals. In MOSFETs, a voltage on the oxide-insulated gate electrode can induce a conducting channel between the two other contacts called source and drain. The channel can be of n-type or p-type, and is accordingly called an n-MOSFET or a p-MOSFET. Figure 1 shows the schematic diagram of the structure of an n--MOS device before and after channel formation.

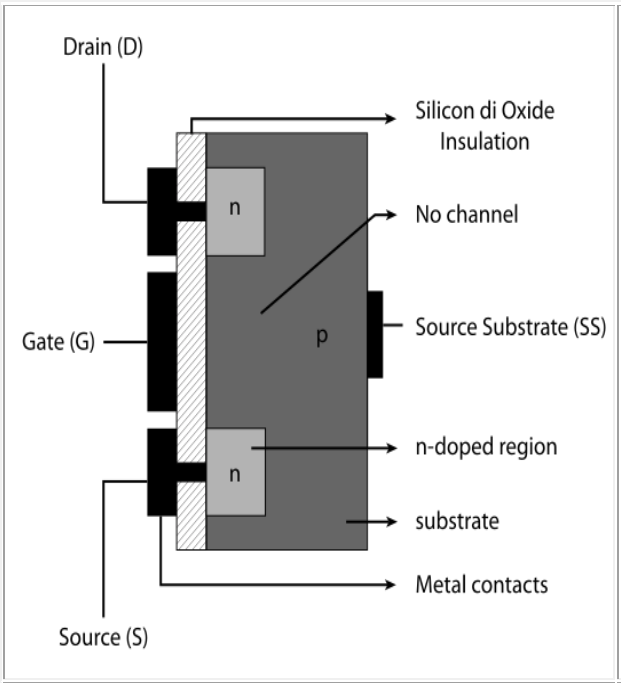


Fig. (1a): n-MOSFET before channel formation

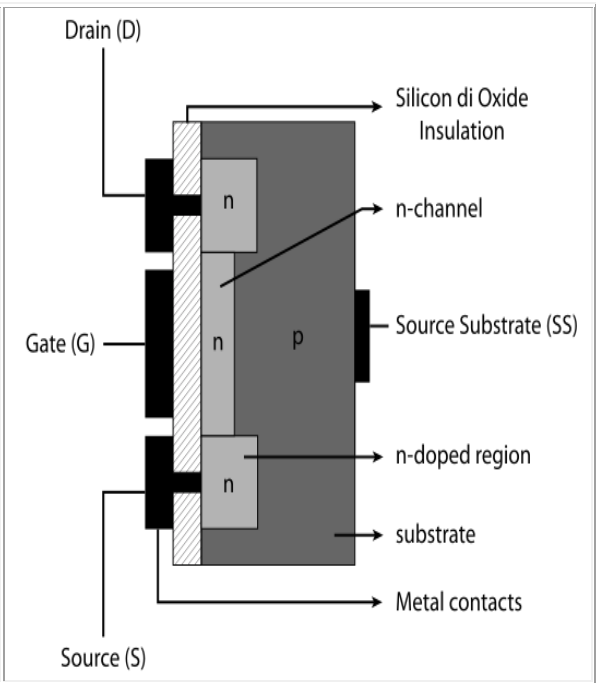


Fig. (1b): n-MOSFET structure after channel formation

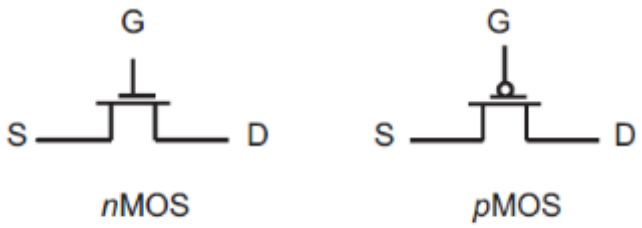
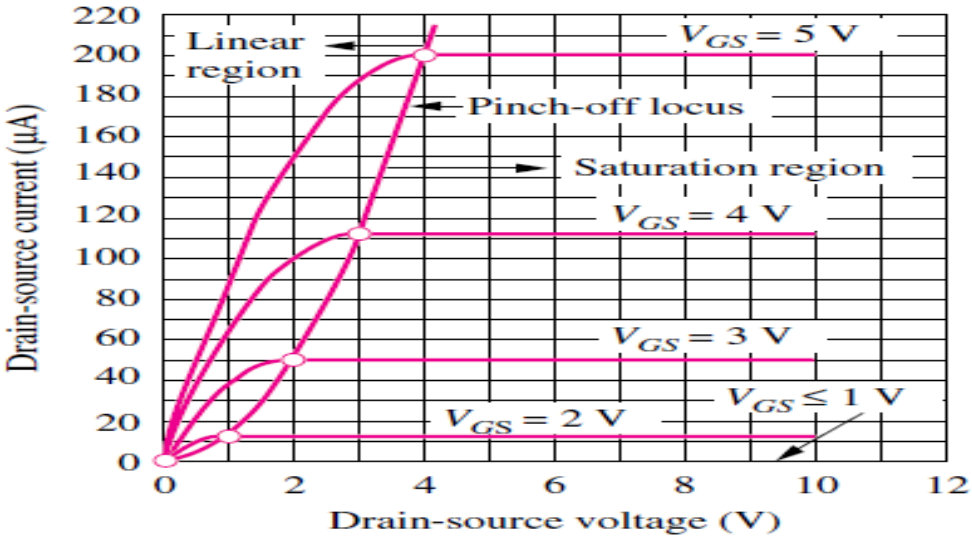


Fig. (2): Circuit symbols for nMOS and pMOS respectively

Output Characteristics

MOSFET output characteristics plot I_D versus V_{DS} for several values of V_{GS}.



The characteristics of an nMOS transistor can be explained as follows. As the voltage on the top electrode increases further, electrons are attracted to the surface. At a particular voltage level, which we will shortly define as the threshold voltage, the electron density at the surface exceeds the hole density. At this voltage, the surface has inverted from the p-type polarity of the original substrate to an n-type inversion layer, or inversion region, directly underneath the top plate as indicated in Fig. 1(b). This inversion region is an extremely shallow layer, existing as a charge sheet directly below the gate. In the MOS capacitor, the high density of electrons in the inversion layer is supplied by the electron–hole generation process within the depletion layer. The positive charge on the gate is balanced by the combination of negative charge in the inversion layer plus negative ionic acceptor charge in the depletion layer. The voltage at which the surface inversion layer just forms plays an extremely important role in field-effect transistors and is called the threshold voltage V_{tn}. The region of output characteristics where $V_{GS}<vtn</v$ and no current flows is called the **cutt-off region**. When the channel forms in the nMOS (pMOS) transistor, a positive (negative) drain voltage with respect to the source creates a horizontal electric field moving the electrons (holes) toward the drain forming a positive (negative) drain current coming into the transistor. The positive current convention is used for electron and hole current, but in both cases electrons are the actual charge carriers. If the channel horizontal electric field is of the same order or smaller than the vertical thin oxide field, then the inversion channel remains almost uniform along the device length. This continuous carrier profile from drain to source puts the transistor in a bias state that is

equivalently called either the **non-saturated, linear, or ohmic bias state**. The drain and source are effectively short-circuited. This happens when *VGS* > *VDS* + *Vtn* for nMOS transistor and *VGS* < *VDS* +*Vtp* for pMOS transistor. Drain current is linearly related to drain-source voltage over small intervals in the linear bias state.

But if the nMOS drain voltage increases beyond the limit, so that *VGS* < *VDS* + *Vtn*, then the horizontal electric field becomes stronger than the vertical field at the drain end, creating an asymmetry of the channel carrier inversion distribution shown in Figure 4.

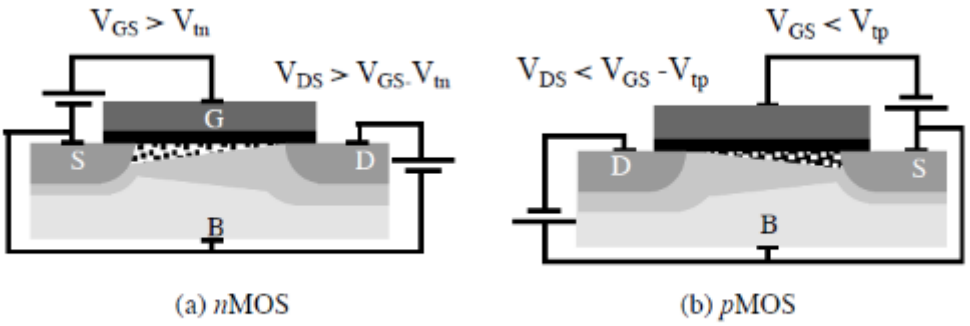


Fig. 4: Channel pinchoff for (a) nMOS and (b) pMOS transistor devices.

If the drain voltage rises while the gate voltage remains the same, then VGD can go below the threshold voltage in the drain region. There can be no carrier inversion at the drain-gate oxide region, so the inverted portion of the channel retracts from the drain, and no longer "touches" this terminal. The pinched-off portion of the channel forms a depletion region with a high electric field. The n-drain and p-bulkform a pn junction. When this happens the inversion channel is said to be "pinched-off" and the device is in the **saturation region**. The characteristics can be loosely modeled by the following equations.

$$I_D = \begin{cases} 0, & \text{Cut-off state} \\ \dfrac{\mu_n \varepsilon_{ox}}{T_{ox}} \dfrac{W}{L}\left[(V_{GS}-V_{tn})V_{DS} - \dfrac{V_{DS}^2}{2}\right], & \text{Ohmic state} \\ \dfrac{\mu_n \varepsilon_{ox}}{2T_{ox}} \dfrac{W}{L}(V_{GS}-V_{tn})^2, & \text{Saturation state} \end{cases}$$

Transfer Characteristics

The transfer characteristic relates drain current (*ID*) response to the input gate-source driving voltage (*VGS*). Since the gate terminal is electrically isolated from the remaining terminals (drain, source, and bulk), the gate current is essentially zero, so that gate current is not part of device characteristics. The transfer characteristic curve can locate the gate voltage at which the transistor passes current and leaves the OFF-state. This is the device threshold voltage (*Vtn*). Figure 5 shows measured input characteristics for an nMOS and pMOS transistor with a small *0.1V* potential across their drain to source terminals.

The transistors are in their non-saturated bias states. As *VGS* increases for the nMOS transistor in Figure 5a, the threshold voltage is reached where drain current elevates.

For *VGS* between *0V* and *0.7V*, *ID* is nearly zero indicating that the equivalent resistance between the drain and source terminals is extremely high. Once *VGS* reaches *0.7V*, the current increases rapidly with *VGS* indicating that the equivalent resistance at the drain decreases with increasing gate-source voltage. Therefore, the threshold voltage of the given nMOS transistor is about $Vtn \approx 0.7V$. The pMOS transistor input characteristic in Figure 5b is analogous to the nMOS transistor except the *ID* and *VGS* polarities are reversed

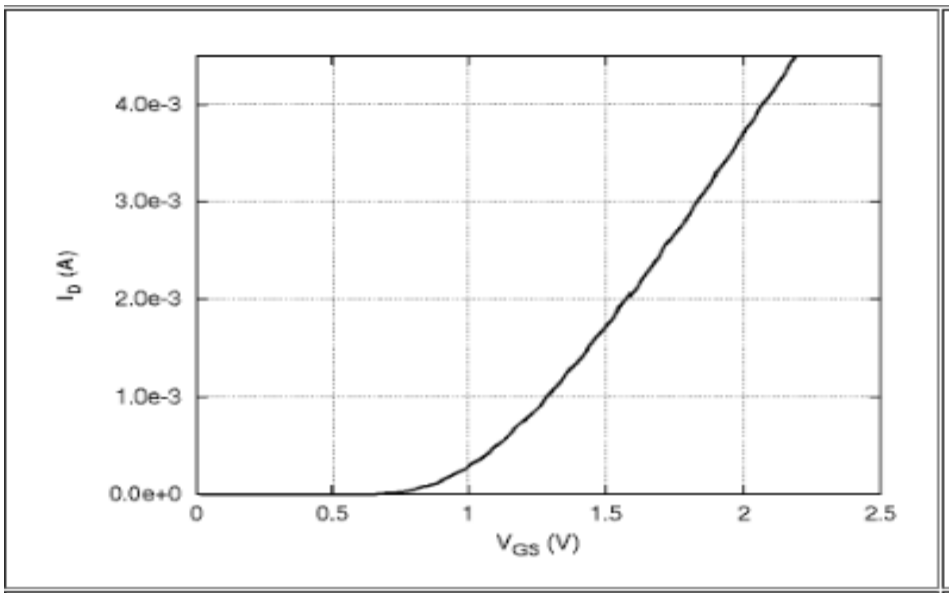
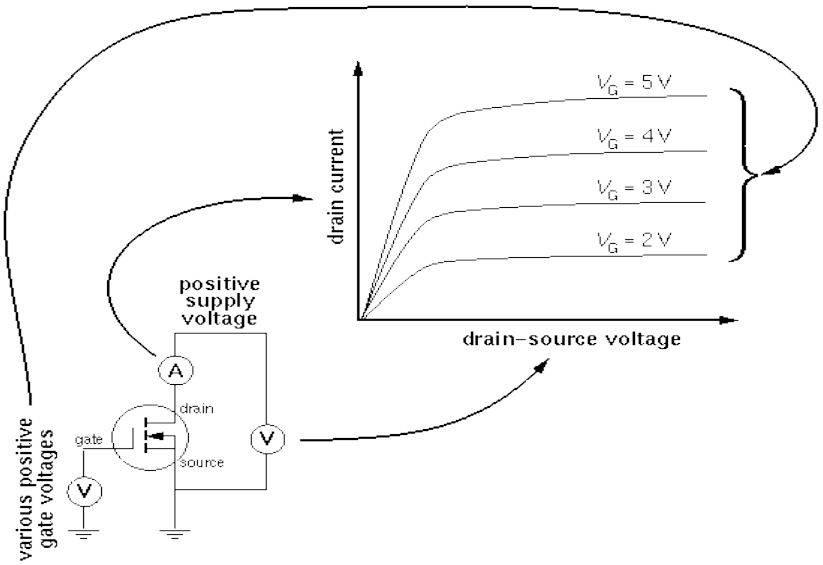
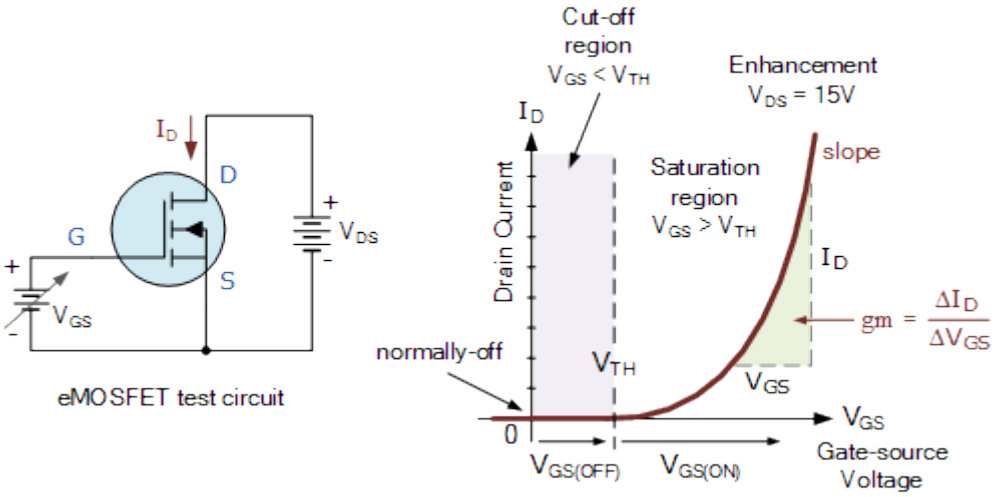


Fig. (5a): Transfer characteristics of nMOS transistor

Diagram:





Procedure: Or-cad P-spice

A) NMOS

Transfer Characteristics

- Go to File → New File → Text File
- Write the code with the nodes specified as shown
- Specify the model specification of MOS
- Save the file with .cir extension
- Open the file from the directory to make it active
- Run the code
- Add trace to axis of ID(M1)
- Observe the output

Output Characteristics

- Make changes to the code with the node of VDD and VGS
- Keep the specifications
- Observe the output for trace ID(M1)
- Save the file with .cir extension
- Open the file from the directory to make it active
- Run the code
- Add trace to axis of ID(M1)
- Observe the output

Procedure: Ng-spice

- Draw the NMOS circuit.
- Mask the nodes and write the corresponding
- P-spice code in text file. Save it with .cir extension.
- Open it from saved directory.
- Simulate the code and observe the output.
- Record the result obtained.

Program:

NMOS OUTPUTCHARACTERISTICS CODE

```
VX 3 2 DC 0v

VDD 3 0 DC 5v

M1 2 1 0 0 NMOS W=1u L=0.35u

.MODEL NMOS NMOS (LEVEL=3

+TOX=7.9E-9 NSUB=1E17 GAMMA=0.5827871

+PHI=0.7 VTO=0.54455489 DELTA=0

+UO=436.256147 ETA=0 THETA=0.1749684

+KP=2.055786E-4 VMAX=8.3099444E4 KAPPA=0.2574081

+RSH=0.0559398 NFS=1E12 TPG=1

+XJ=3E-7 LD=3.162278E-11 WD=7.046724E-8

+CGDO=2.82E-10 CGSO=2.82E-10 CGBO=1E-10

+CJ=1E-3 PB=0.9758533 MJ=0.3448504

+CJSW=3.77852E10 MJSW=0.3508721)

.DC VDD 0 12 0.2 VGS 0 2 0.1

.PLOT DC I(VX)

.PROBE

.END
```

NMOS TRANSFER CHARACTERISTICS

```
VGS 1 0 DC 0V

VX 3 2 DC 0V

VDD 3 0 DC 5V

M1 2 1 0 0 NMOS W=1U L=0.35U

.MODEL NMOS NMOS(LEVEL=3
```

```
+TOX=7.9E-9NSUB=1E17 GAMMA=0.5827871
+PHI=0.7 VTO=0.5445549 DELTA=0
+UO=436.256147 ETA=0 THETA=0.1749648
=KP=2.055786E-4VMAX=8.039444E4 KAPPA=0.2574081
+RSH=0.0559398 NFS=1E12 TPG=1
+XJ=3E-7 LD=3.162278E-11WD=7.04674E-8
=CGDO=2.82E-10 CGSO=2.8E-10 CGBO=1E-10
+CJ=1E-3 PB=0.9758533 MJ=0.3448504
=CJSW=3.77852E-10 MJSW=0.3508721)
.DC VGS 0 2 0.1
.PLOT DCI(VX)
.PROBE
.END
```

Observations:

The explanation provided above can be summarized in the form of a following table

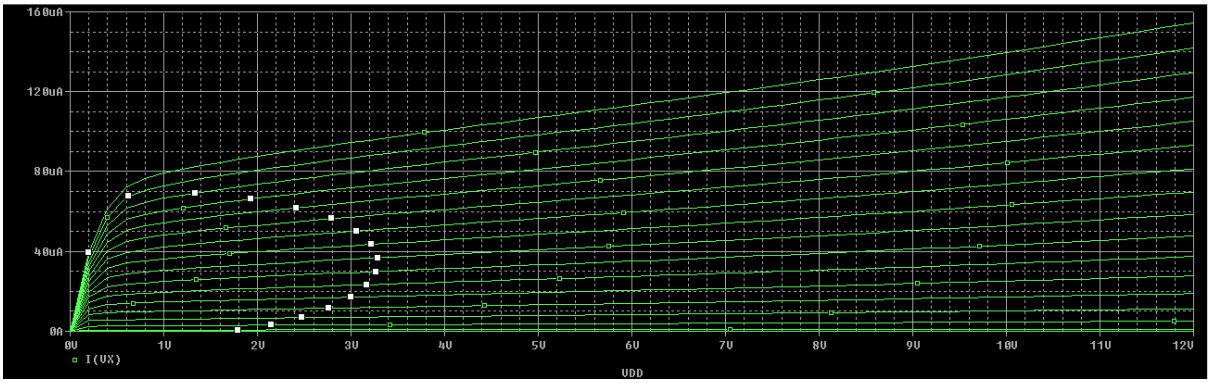
Kind of MOSFET	Region of Operation		
	Cut-Off	**Ohmic/Linear**	**Saturation**
n-channel Enhancement-type	V_{GS}T	$V_{GS} > V_T$ and V_{DS}P	$V_{GS} > V_T$ and $V_{DS} > V_P$
p-channel Enhancement-type	$V_{GS} > -V_T$	V_{GS}T and $V_{DS} > -V_P$	V_{GS}T and V_{DS}P
n-channel Depletion-type	V_{GS}T	$V_{GS} > -V_T$ and V_{DS}P	$V_{GS} > -V_T$ and $V_{DS} > V_P$
p-channel Depletion-type	$V_{GS} > V_T$	V_{GS}T and $V_{DS} > -V_P$	V_{GS}T and V_{DS}P

Calculations:

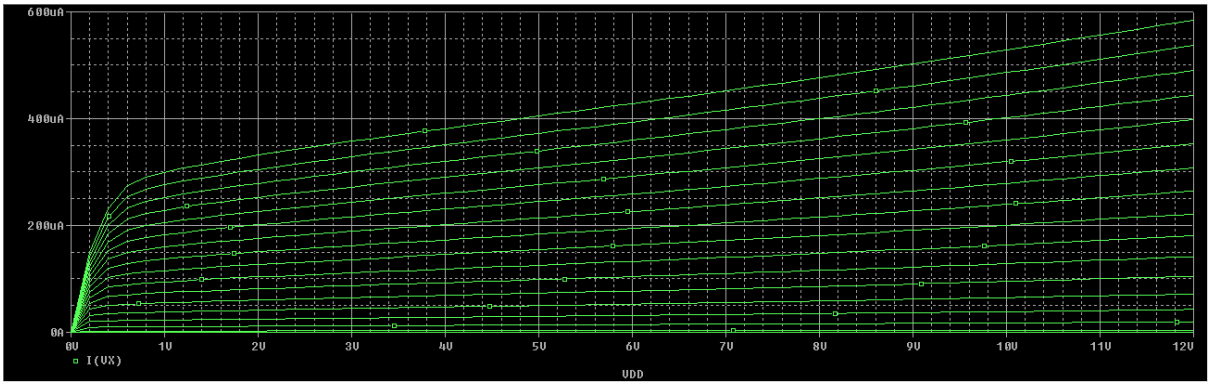
$$I_D = \begin{cases} 0, & \text{Cut-off state} \\ \dfrac{\mu_n \varepsilon_{ox}}{T_{ox}} \dfrac{W}{L}\left[(V_{GS} - V_{tn})V_{DS} - \dfrac{V_{DS}^2}{2}\right], & \text{Ohmic state} \\ \dfrac{\mu_n \varepsilon_{ox}}{2T_{ox}} \dfrac{W}{L}(V_{GS} - V_{tn})^2, & \text{Saturation state} \end{cases}$$

Output (Output Characteristics):

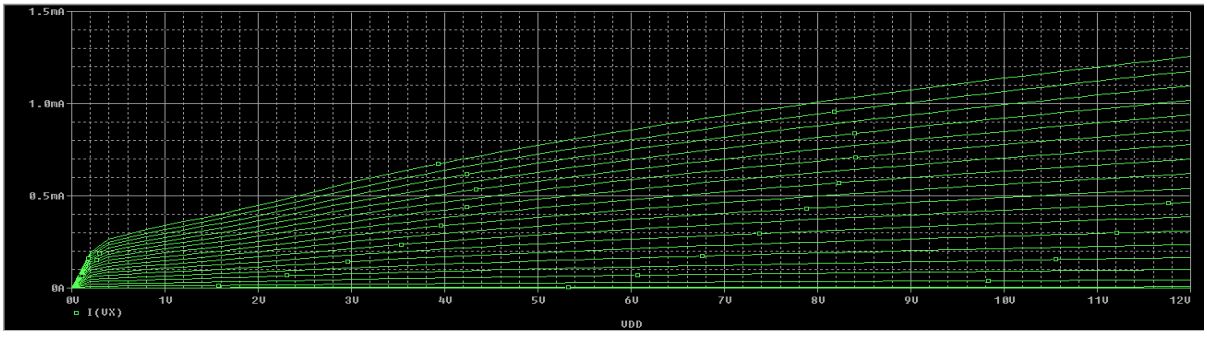
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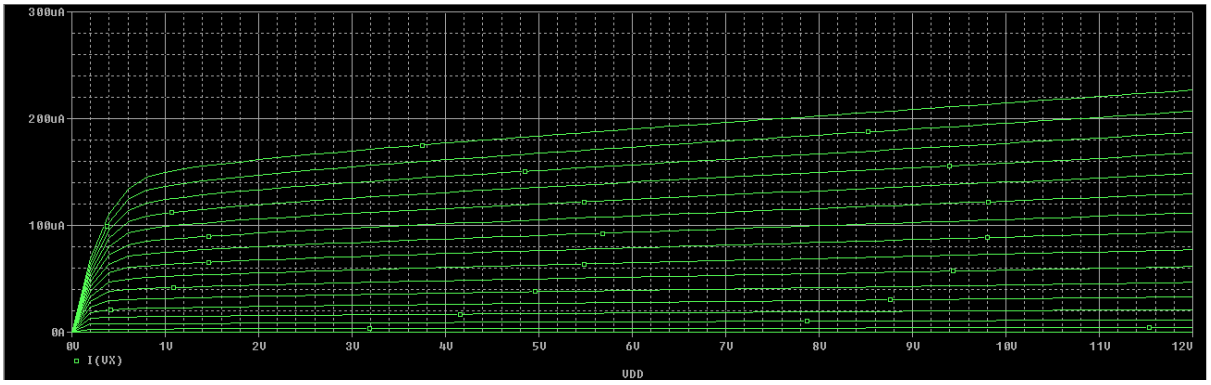
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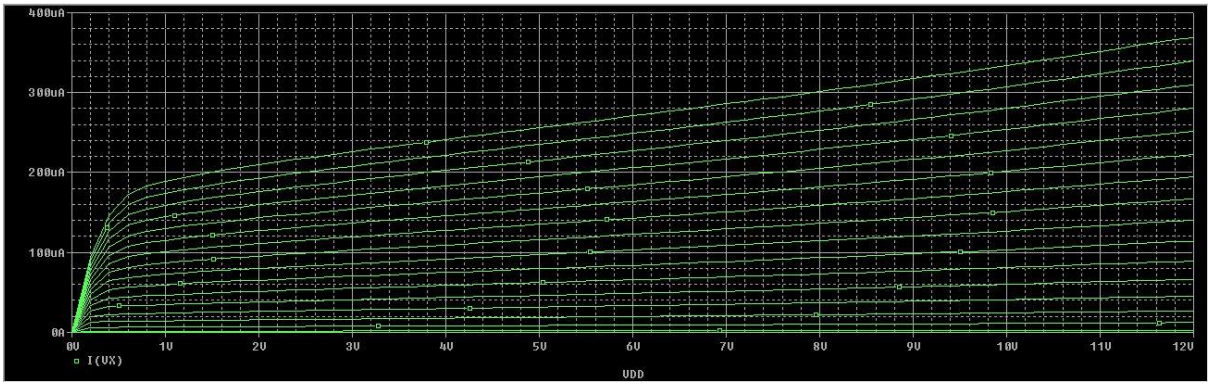
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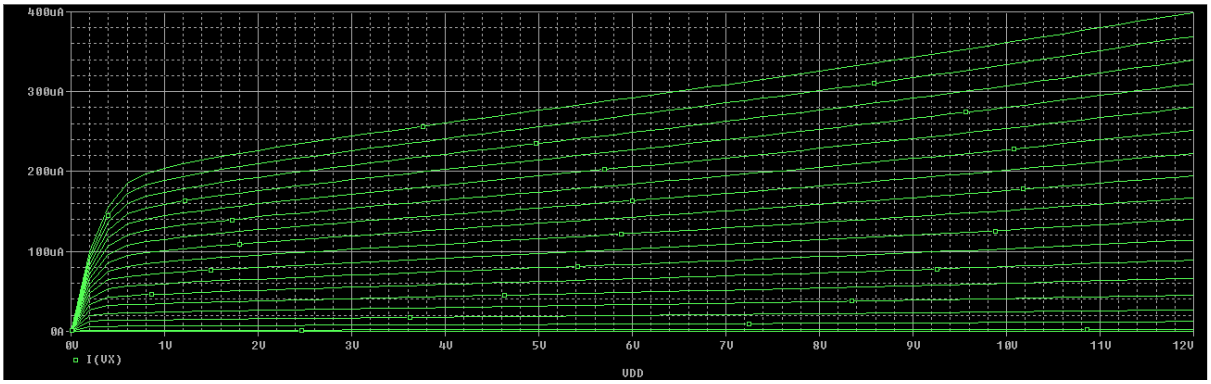
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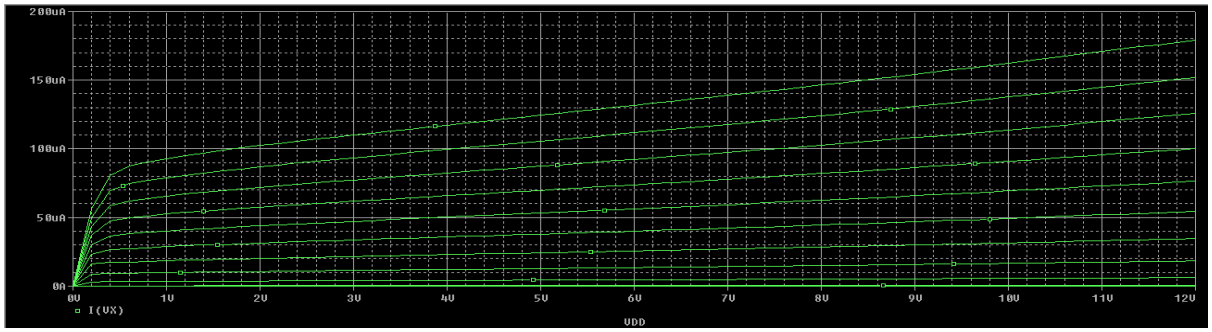
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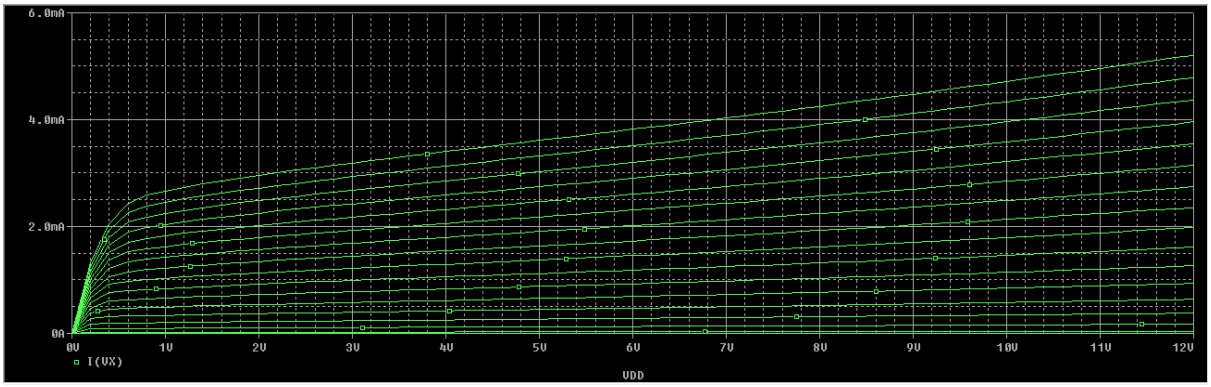
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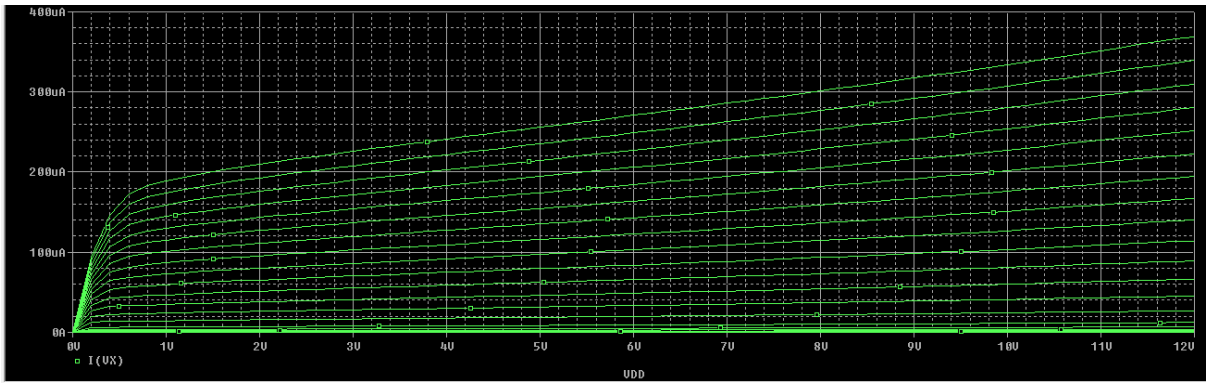
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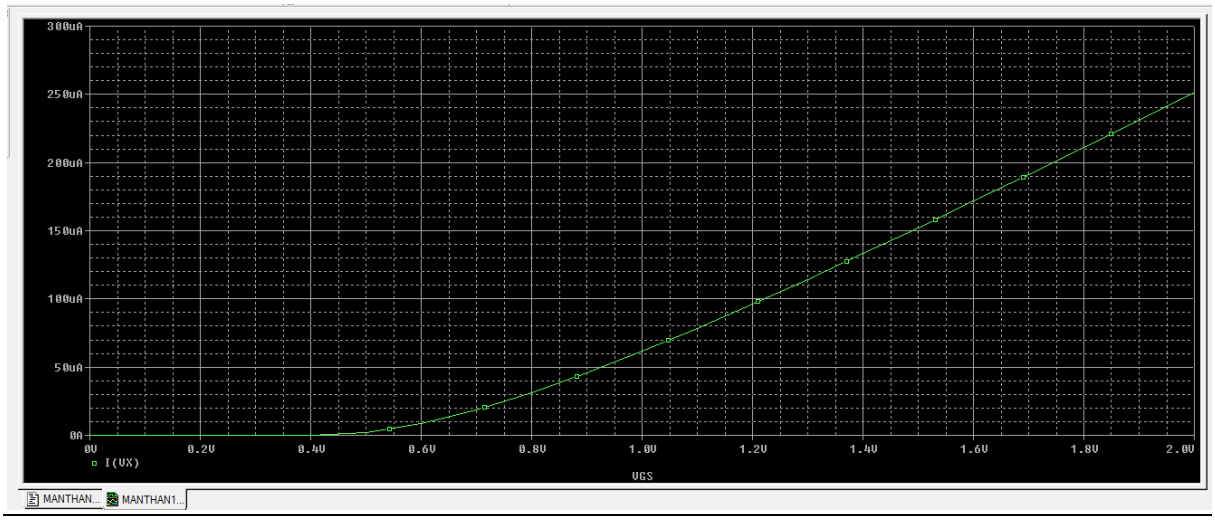
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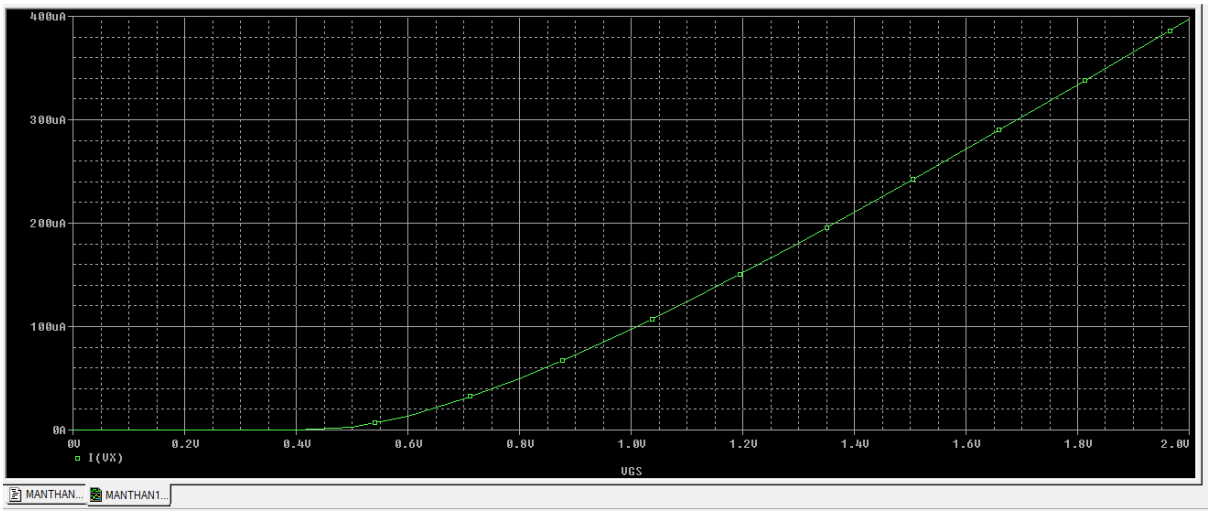
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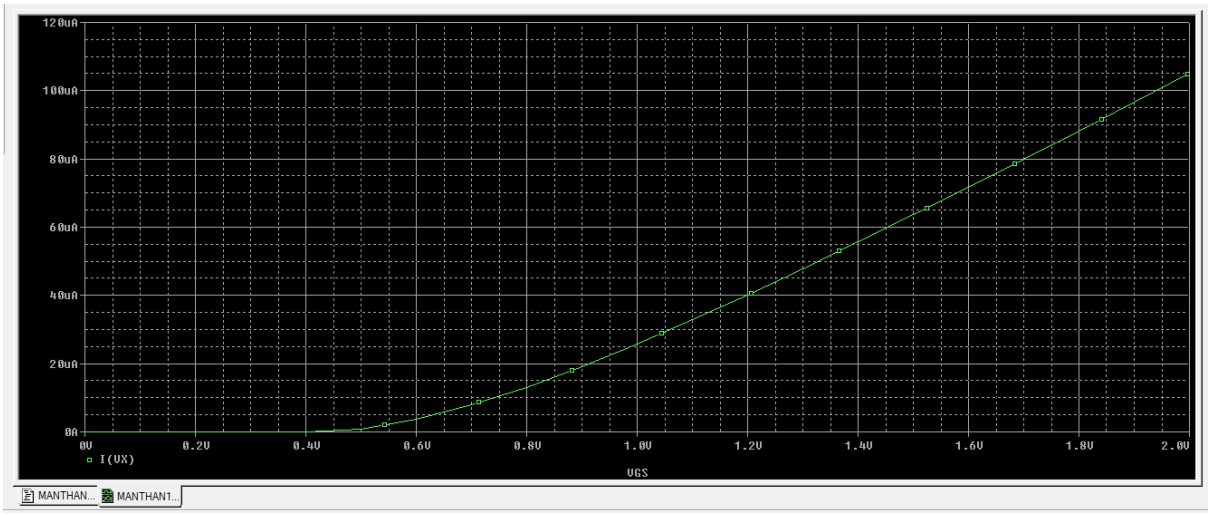
Output (Input Characteristics):



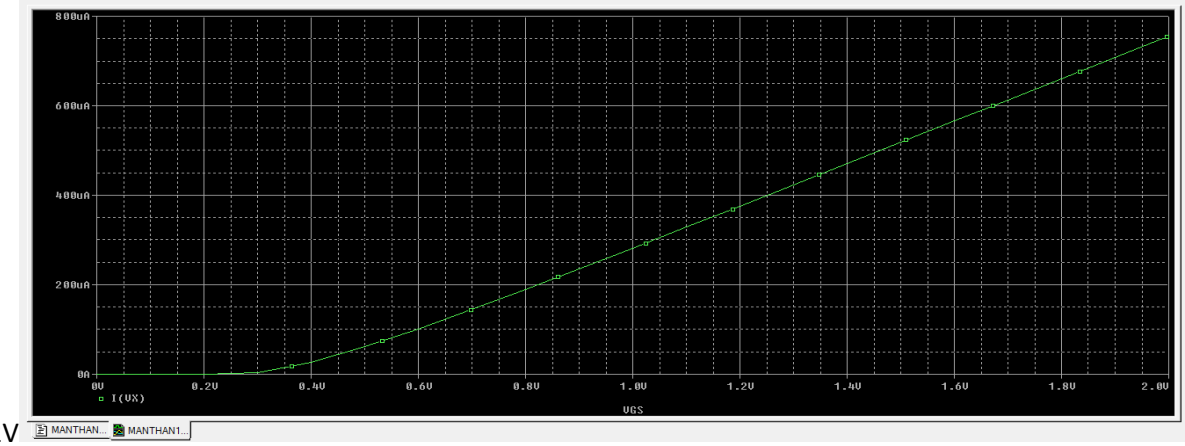
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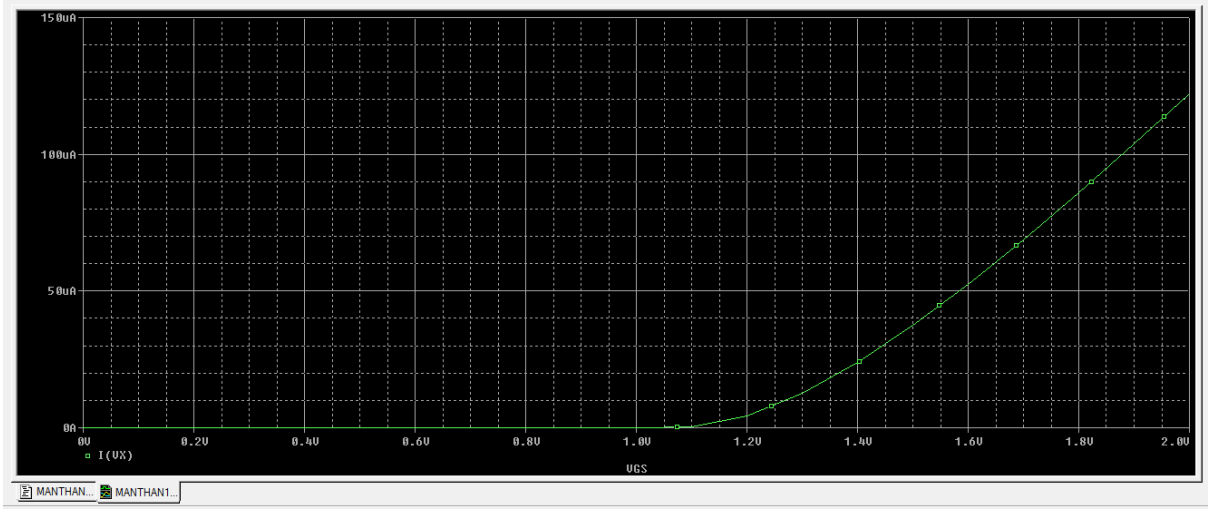
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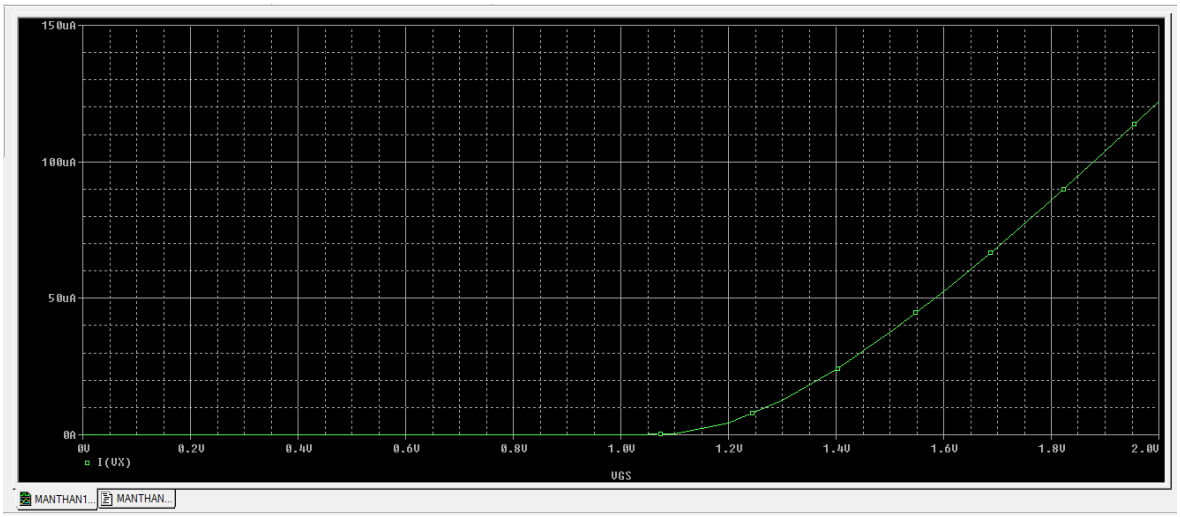
L=0.15

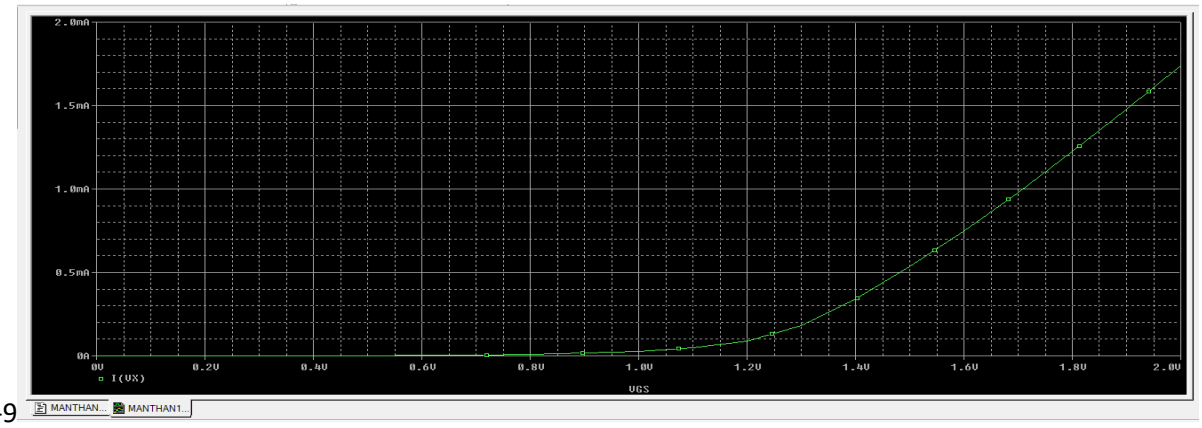


VTO=1.2V



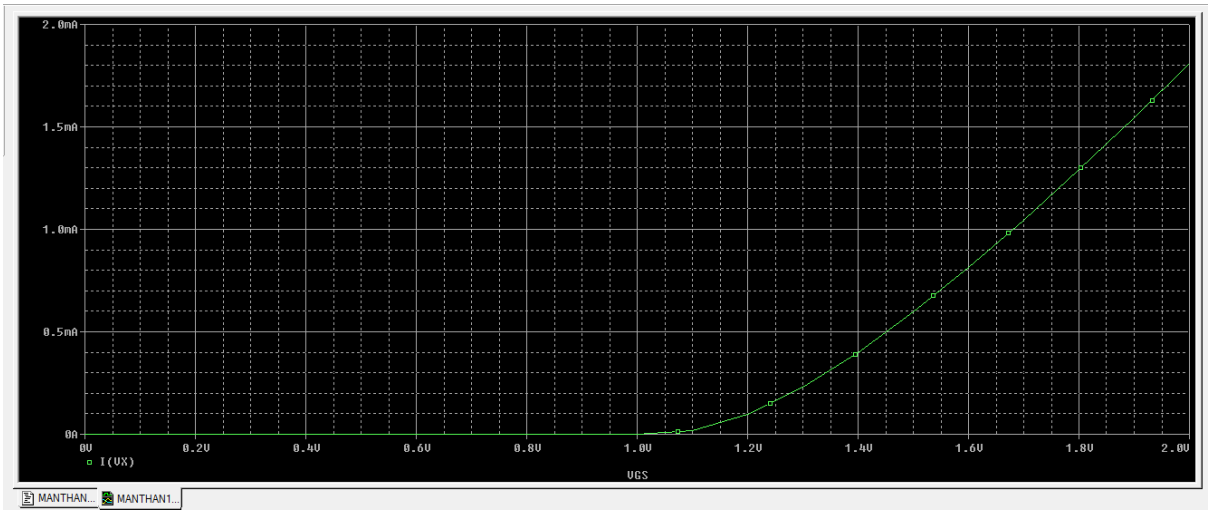
KP=29.30





TOX=120E-9

GAMMA=0.744



Pre Lab Questions:

1. The n-MOS invertor is better than BJT in terms of:
2. The n-MOS invertor consists of n-MOS transistor as driven and

Post Lab Questions:

1. The average power dissipated in resistive load n-MOS inverter is:
2. The depletion mode n-MOS as an active load is better than enhancement load n-MOS in

Conclusion: Thus we have studied and experimentally verified the operation of NMOS inverter circuit and recorded the results of output and transfer characteristics.

CO's Covered:

…….....
……..
.

Expt. No. 2

Title: To study and verification of output and transfer characteristics of PMOS transistor.

Rubric	*Score (1 to 5)*
Timely Submission	
Result Interpretation and Output	
Program Logic and Concept Understanding	
Total	

Performed On: _____

Sign:_____

Experiment No. 2

Aim: To study and verification of Output and Transfer characteristics of PMOS transistor.

Apparatus/Software: Computer with P-spice

Link for virtual lab: http://www.vlsi-iitg.vlabs.ac.in

Theory:

Introduction

The metal–oxide–semiconductor field-effect transistor (MOSFET) is a transistor used for amplifying or switching electronic signals. In MOSFETs, a voltage on the oxide-insulated gate electrode can induce a conducting channel between the two other contacts called source and drain. The channel can be of n-type or p-type, and is accordingly called an nMOSFET or a pMOSFET. Figure 1 shows the schematic diagram of the structure of an nMOS device before and after channel formation.

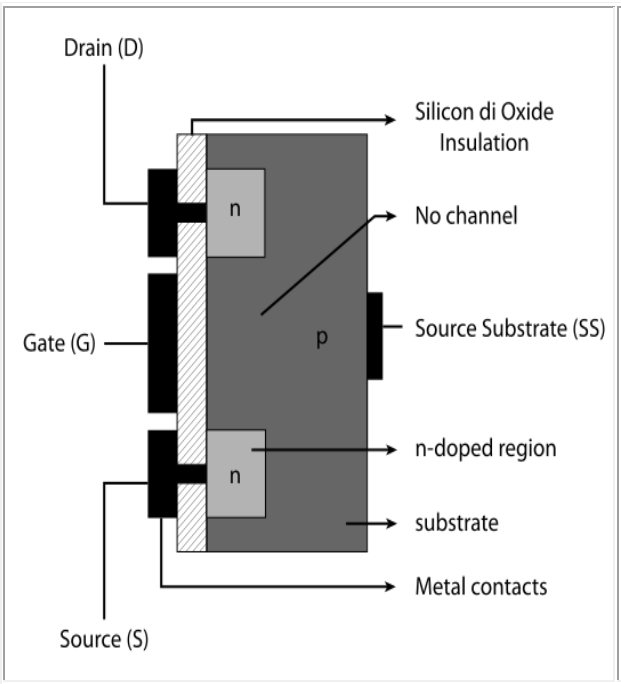


Fig. (1a): nMOSFET before channel formation

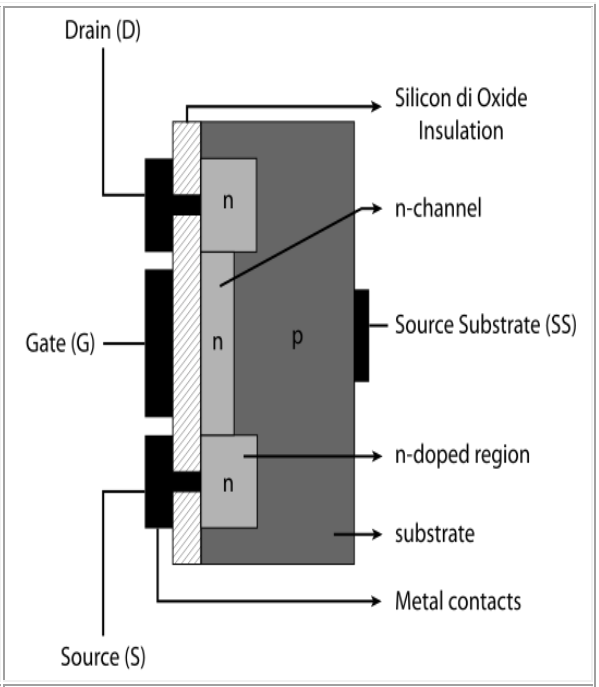


Fig. (1b): nMOSFET structure after channel formation

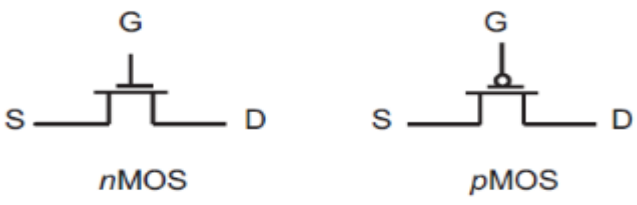
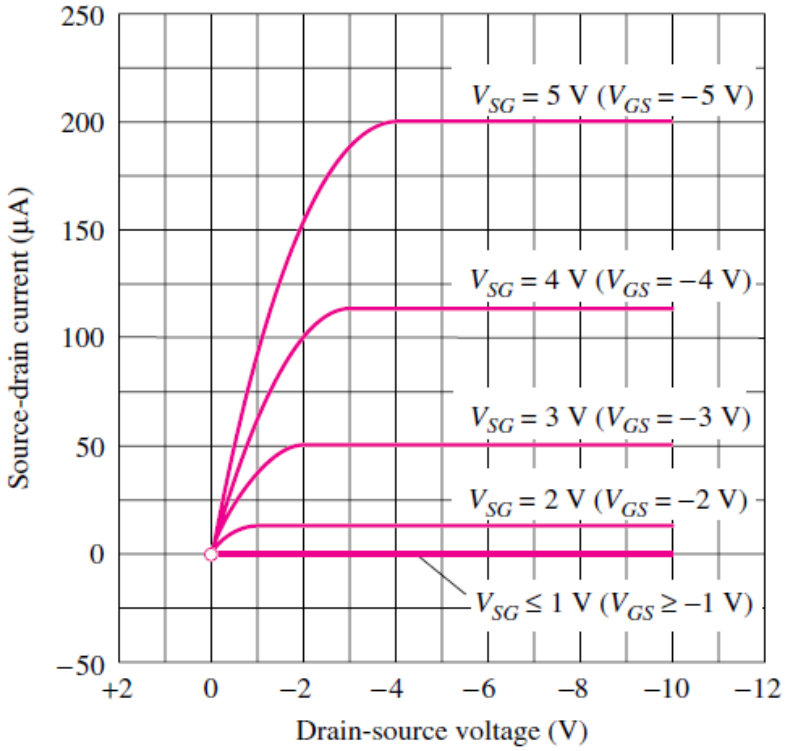


Fig. (2): Circuit symbols for nMOS and pMOS respectively

Output Characteristics

MOSFET output characteristics plot I_D versus V_{DS} for several values of V_{GS}.



P-Channel MOSFET:

The P- Channel MOSFET has a P- Channel region between source and drain. It is a four terminal device such as gate, drain, source, body. The drain and source are heavily doped p+ region and the body or substrate is n-type. The flow of current is positively charged holes. When we apply the negative gate voltage, the electrons present under the oxide layer with are pushed downward into the substrate with a repulsive force. The depletion region populated by the bound positive charges which are associated with the donor atoms. The negative gate voltage also attracts holes from p+ source and drain region into the channel region.

The characteristics of an nMOS transistor can be explained as follows. As the voltage on the top electrode increases further, electrons are attracted to the surface. At a particular voltage level, which we will shortly define as the threshold voltage, the electron density at the surface exceeds the hole density. At this voltage, the surface has inverted from the p-type polarity of the original substrate to an n-type inversion layer, or inversion region, directly underneath the top plate as indicated in Fig. 1(b). This inversion region is an extremely shallow layer, existing as a charge sheet directly below the gate. In the MOS capacitor, the high density of electrons in the inversion layer is supplied by the electron–hole generation process within the depletion layer. The positive charge on the gate is balanced by the combination of negative charge in the inversion layer plus negative ionic acceptor charge in the depletion layer. The voltage at which the surface inversion layer just forms plays an extremely important role in field-effect transistors and is called the threshold voltage V_{tn}. The region of output characteristics where $V_{GS}<vtn</v$ and no

current flows is called the **cutt-off region**. When the channel forms in the nMOS (pMOS) transistor, a positive (negative) drain voltage with respect to the source creates a horizontal electric field moving the electrons (holes) toward the drain forming a positive (negative) drain current coming into the transistor. The positive current convention is used for electron and hole current, but in both cases electrons are the actual charge carriers. If the channel horizontal electric field is of the same order or smaller than the vertical thin oxide field, then the inversion channel remains almost uniform along the device length. This continuous carrier profile from drain to source puts the transistor in a bias state that is equivalently called either the **non-saturated, linear, or ohmic bias state**. The drain and source are effectively short-circuited. This happens when *VGS* > *VDS* + *Vtn* for nMOS transistor and *VGS* < *VDS* +*Vtp* for pMOS transistor. Drain current is linearly related to drain-source voltage over small intervals in the linear bias state.

But if the nMOS drain voltage increases beyond the limit, so that *VGS* < *VDS* + *Vtn*, then the horizontal electric field becomes stronger than the vertical field at the drain end, creating an asymmetry of the channel carrier inversion distribution shown in Figure 4.

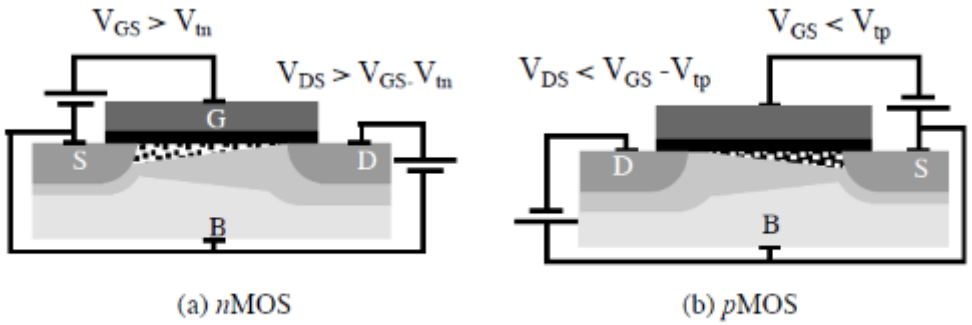


Fig. 4: Channel pinchoff for (a) nMOS and (b) pMOS transistor devices.

If the drain voltage rises while the gate voltage remains the same, then VGD can go below the threshold voltage in the drain region. There can be no carrier inversion at the drain-gate oxide region, so the inverted portion of the channel retracts from the drain, and no longer "touches" this terminal. The pinched-off portion of the channel forms a depletion region with a high electric field. The n-drain and p-bulk form a pn junction. When this happens the inversion channel is said to be "pinched-off" and the device is in the **saturation region**. The characteristics can be loosely modeled by the following equations.

$$I_D = \begin{cases} 0, & \text{Cut-off state} \\ \dfrac{\mu_n \varepsilon_{ox}}{T_{ox}} \dfrac{W}{L}\left[(V_{GS} - V_{tn})V_{DS} - \dfrac{V_{DS}^2}{2}\right], & \text{Ohmic state} \\ \dfrac{\mu_n \varepsilon_{ox}}{2T_{ox}} \dfrac{W}{L}(V_{GS} - V_{tn})^2, & \text{Saturation state} \end{cases}$$

Transfer Characteristics

The transfer characteristic relates drain current (*ID*) response to the input gate-source driving voltage (*VGS*). Since the gate terminal is electrically isolated from the remaining

terminals (drain, source, and bulk), the gate current is essentially zero, so that gate current is not part of device characteristics. The transfer characteristic curve can locate the gate voltage at which the transistor passes current and leaves the OFF-state. This is the device threshold voltage (*Vtn*). Figure 5 shows measured input characteristics for an nMOS and pMOS transistor with a small *0.1V* potential across their drain to source terminals.

The transistors are in their non-saturated bias states. As *VGS* increases for the nMOS transistor in Figure 5a, the threshold voltage is reached where drain current elevates. For *VGS* between *0V* and *0.7V*, *ID* is nearly zero indicating that the equivalent resistance between the drain and source terminals is extremely high. Once *VGS* reaches *0.7V*, the current increases rapidly with *VGS* indicating that the equivalent resistance at the drain decreases with increasing gate-source voltage. Therefore, the threshold voltage of the given nMOS transistor is about $Vtn \approx 0.7V$. The pMOS transistor input characteristic in Figure 5b is analogous to the nMOS transistor except the *ID* and *VGS* polarities are reversed

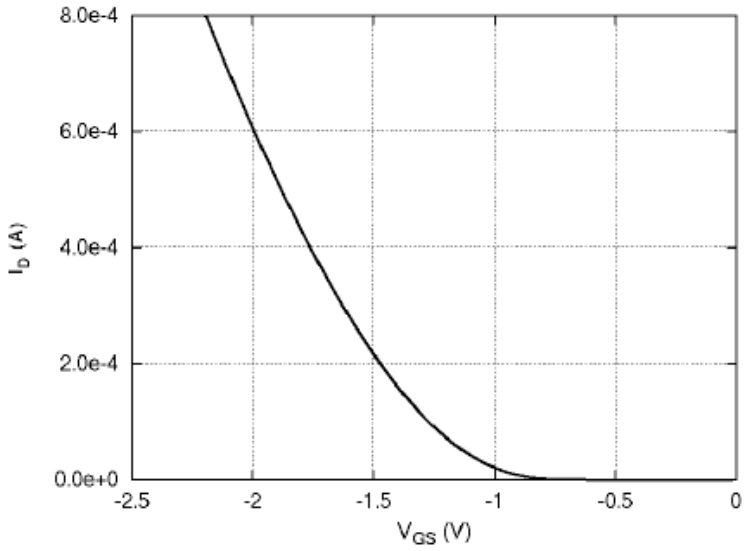
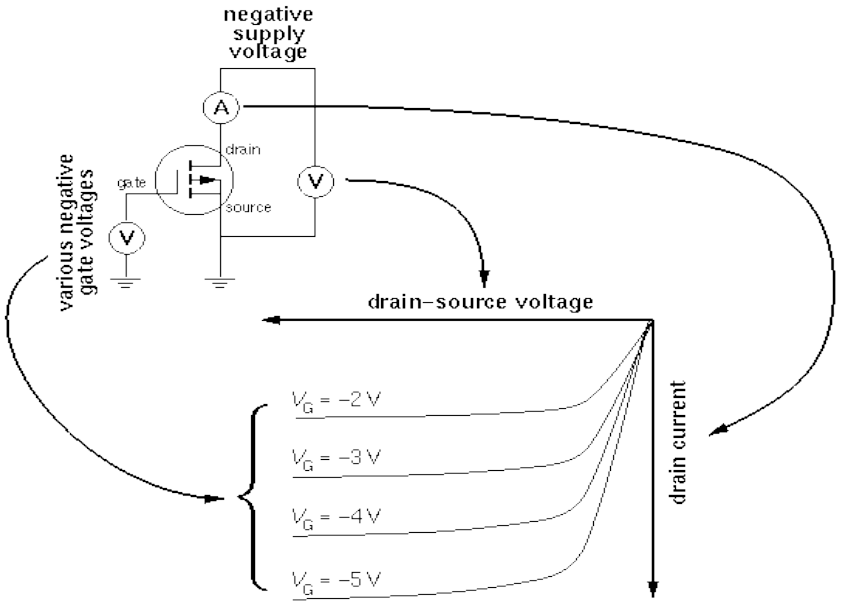


Diagram:



Procedure: Ng-spice

- 1Draw the PMOS circuit.
- Mask the nodes and write the corresponding
- P-spice code in text file. Save it with .cir extension.
- Open it from saved directory.
- Simulate the code and observe the output.
- Record the result obtained.

Procedure: P-spice

PMOS Transfer Characteristics

- Go to File → New File → Text File
- Write the code with the nodes specified as shown
- Specify the model specification of MOS
- Save the file with .cir extension
- Open the file from the directory to make it active
- Run the code
- Add trace to axis of ID(M1)
- Observe the output

PMOS Output Characteristics

- Make changes to the code with the node of VDD and VGS
- Keep the specifications
- Observe the output for trace ID(M1)
- Save the file with .cir extension
- Open the file from the directory to make it active
- Run the code
- Add trace to axis of ID(M1)
- Observe the output.

Program:

PMOS OUTPUT CHARACTERISTICS CODE

```
VGS 1 0 DC 0V

VX 3 2 DC 5v

VDD 3 0 DC 5v

M1 2 1 0 0 PMOS W=1u L=0.35u

.MODEL PMOS PMOS (LEVEL=3

+TOX=7.9E-9 NSUB=1E17 GAMMA=0.4083894

+PHI=0.7 VTO=0.7140674 DELTA=0

+UO=212.23198801 ETA=9.999762E-4 THETA=0.2020774

+KP=6.733755E-5 VMAX=1.181551E5 KAPPA=1.5

+RSH=30.0712458 NFS=1E12 TPG=-1

+XJ=2E-7 LD=5.000001E-13 WD=1.249872E-7

+CGDO=3.09E-10 CGSO=3.09E-10 CGBO=1E-10

+CJ=1.419508E-3 PB=0.8152753 MJ=0.5

+CJSW=4.813504E-10 MJSW=0.5)

.DC VDD 5 -15 -0.01 VGS 5 -5 -1

.PLOT DC I(VX)

.PROBE

.END
```

PMOS TRANSFER CHARACTERISTICS CODE

```
//TRANSFER CHARACTERICTICS OF PMOS

VGS 1 0 DC 0V

VX 3 2 DC 5V

VDD 3 0 DC -15V

M1 2 1 0 0 PMOS W=1u L=0.35u

.MODEL PMOS PMOS (LEVEL=3

+TOX =120E-9 NSUB=1E17 GAMMA=0.40

+PHI=0.7 VTO=0.7 DELTA=0

+UO=212.23198801 ETA=9.999762E-4 THETA=0.2020744

+KP=6.733755E-5 VMAX=1.181551E5 KAPPA=1.5

+RSH=30.0712458 NFS=1E12 TPG=-1

+XJ=2E-7 LD=5.000001E-13 WD=1.249872E-7

+CGDO=3.09E-10 CGSO=3.09E10 CGBO=1E-10

+CJ=1.419508E-3 PB=0.8152753 MJ=0.5

+CJSW=4.813504E-10 MJSW=0.5)

.DC VGS 0 -5 -1

.PLOT DC i(VX)

.PROBE

.END
```

Observations:

The explanation provided above can be summarized in the form of a following table

Kind of MOSFET	Region of Operation		
	Cut-Off	**Ohmic/Linear**	**Saturation**
n-channel	V_{GS}T	$V_{GS} > V_T$ and V_{DS}P	$V_{GS} > V_T$ and $V_{DS} > V_P$

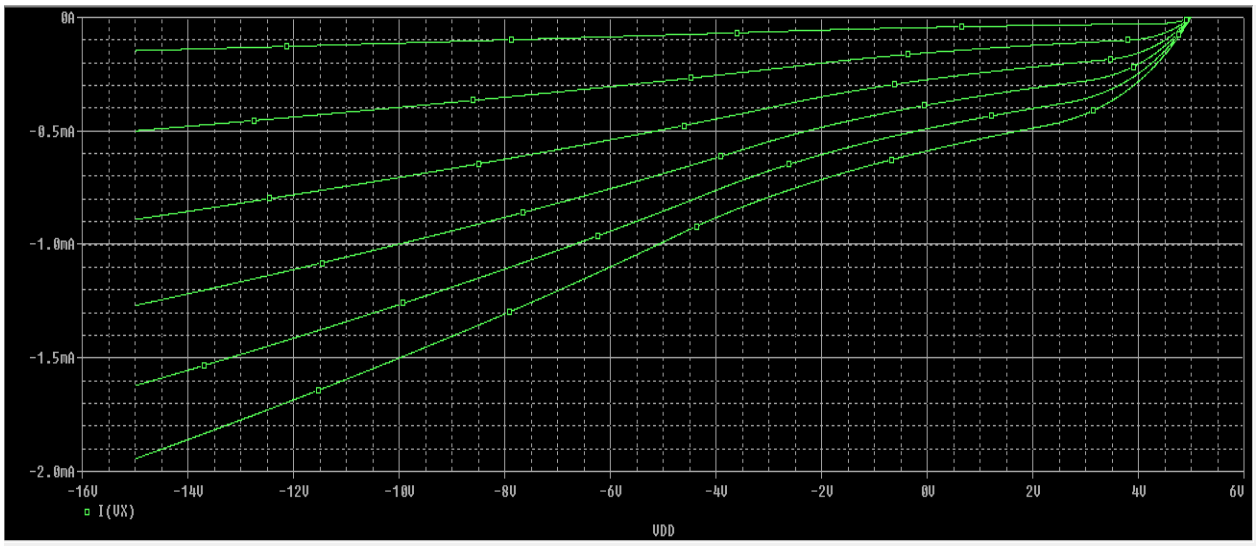
Enhancement-type			
p-channel Enhancement-type	$V_{GS} > -V_T$	$V_{GS}T$ and $V_{DS} > -V_P$	$V_{GS}T$ and $V_{DS}P$
n-channel Depletion-type	$V_{GS}T$	$V_{GS} > -V_T$ and $V_{DS}P$	$V_{GS} > -V_T$ and $V_{DS} > V_P$
p-channel Depletion-type	$V_{GS} > V_T$	$V_{GS}T$ and $V_{DS} > -V_P$	$V_{GS}T$ and $V_{DS}P$

Calculations:

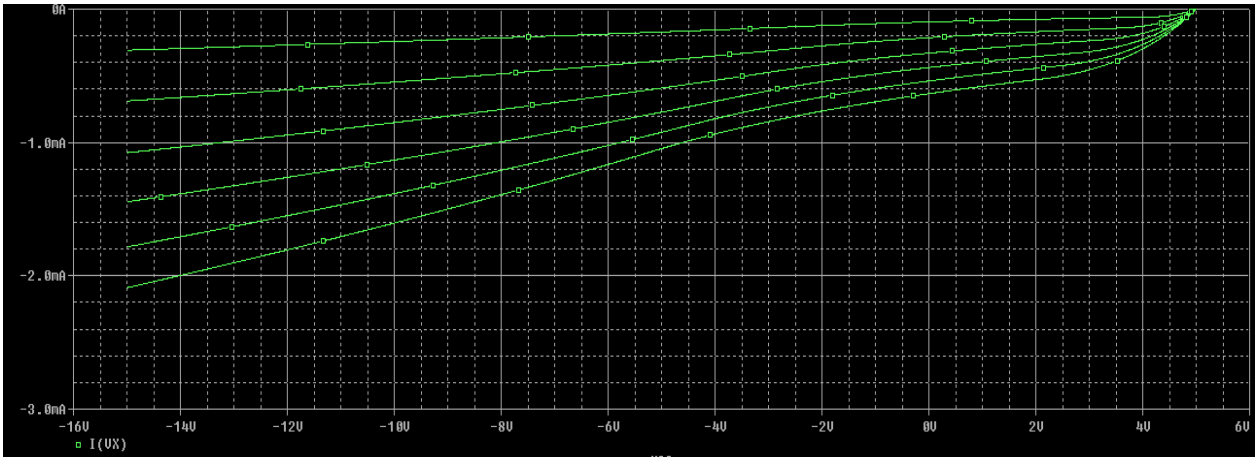
$$I_D = \begin{cases} 0, & \text{Cut-off state} \\ \frac{\mu_n \varepsilon_{ox}}{T_{ox}} \frac{W}{L} \left[(V_{GS} - V_{tn}) V_{DS} - \frac{V_{DS}^2}{2} \right], & \text{Ohmic state} \\ \frac{\mu_n \varepsilon_{ox}}{2T_{ox}} \frac{W}{L} (V_{GS} - V_{tn})^2, & \text{Saturation state} \end{cases}$$

Output (Output Characteristics):

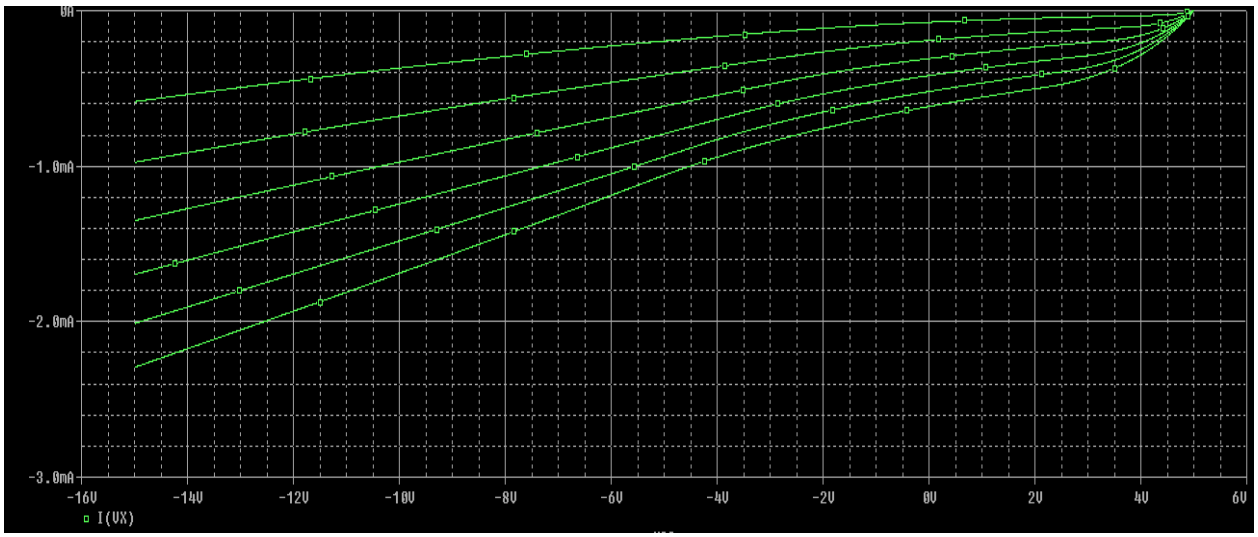
ID=-1.95mA



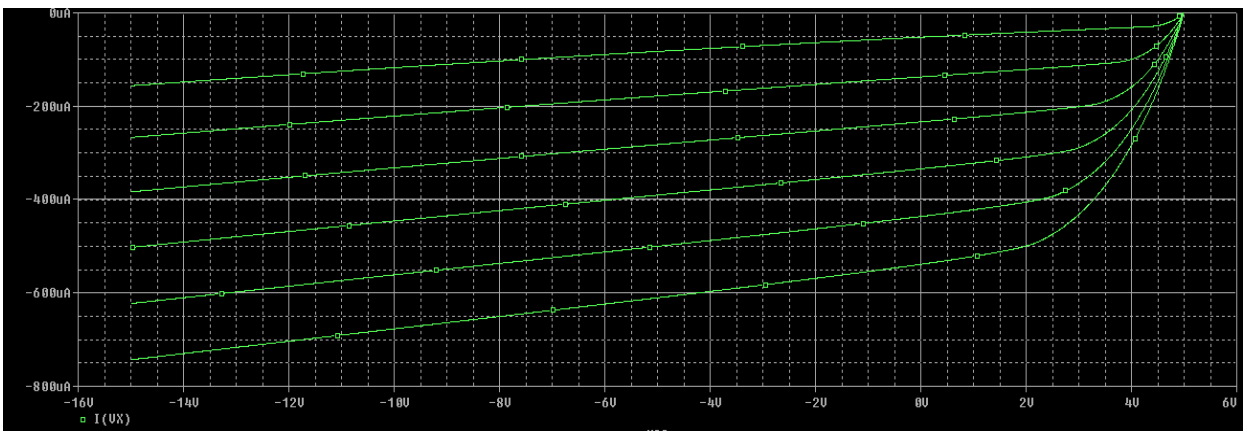
VTO=1.2V ID=-2.1mA



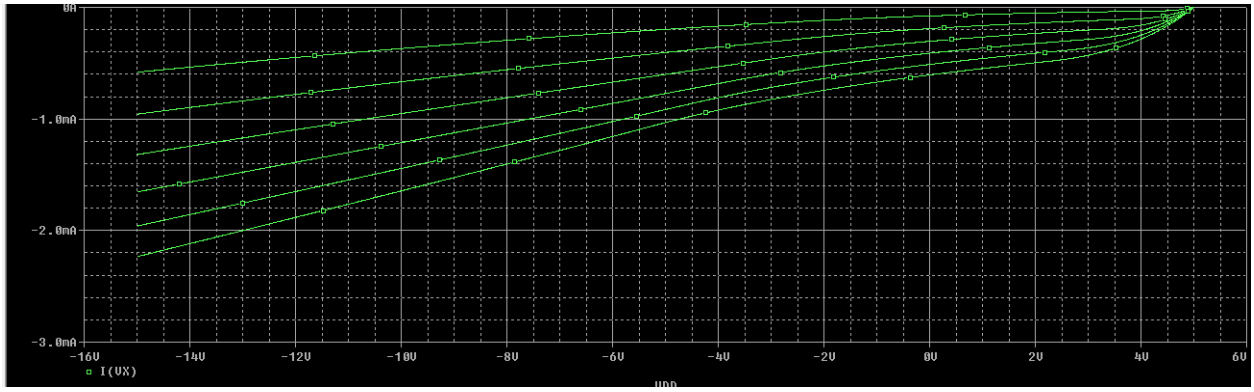
TOX=120E-9 ID=-2.2mA



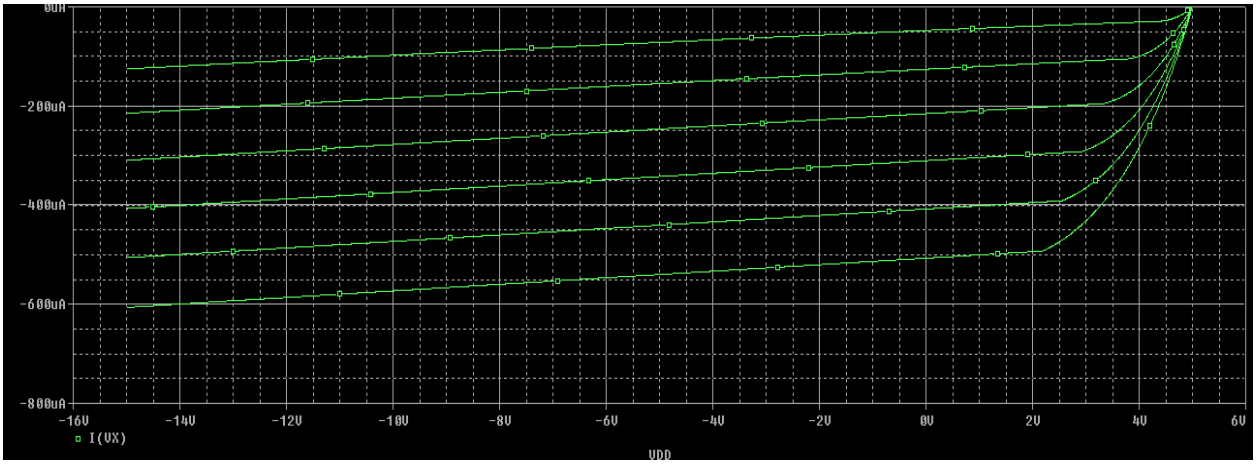
LAMBDA=0.02V-1



GAMMA=0.744 ID=-2.2mA

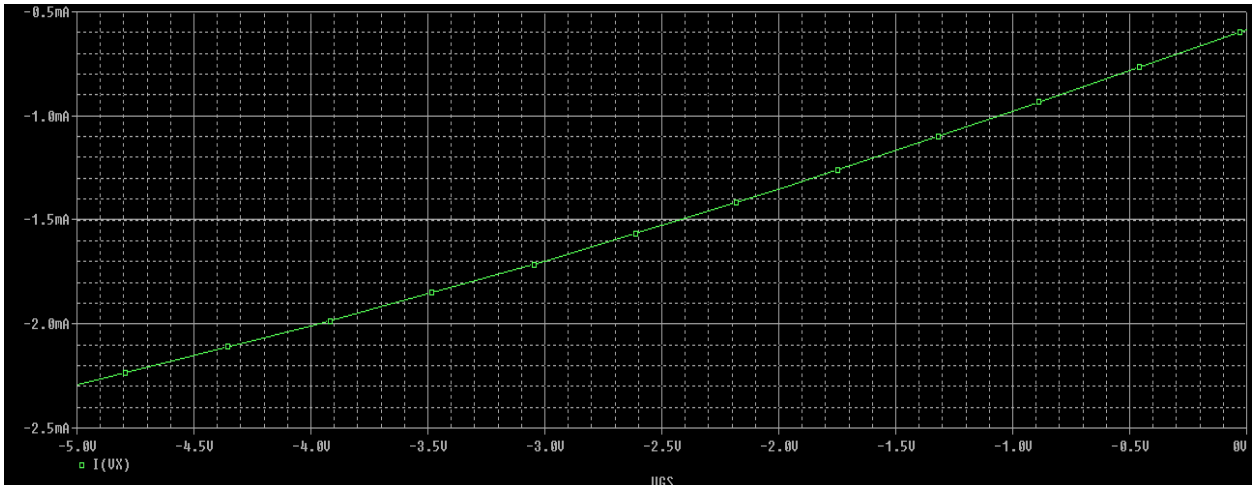


LAMBDA=0V-1 ID=-600mA

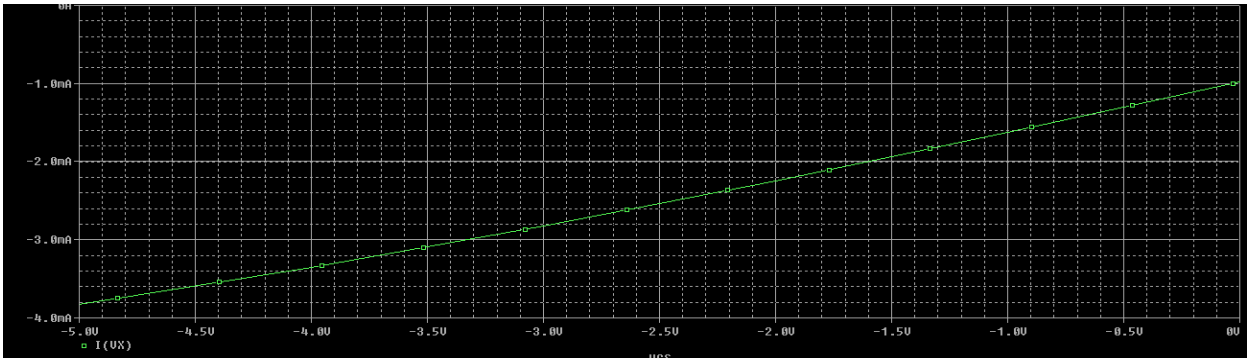


Output (Transfer Characteristics):

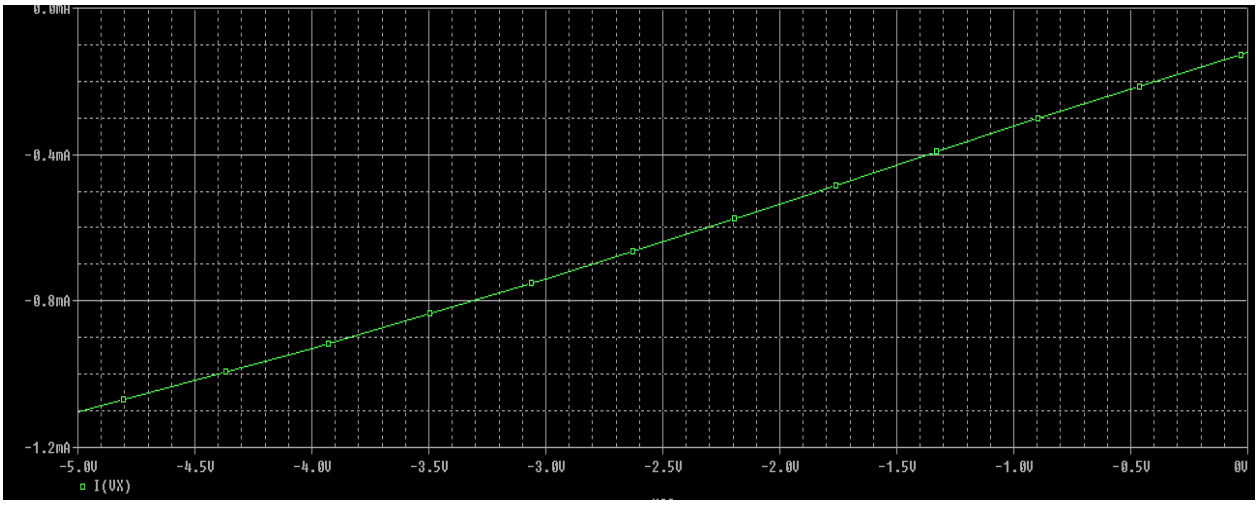
ID=-2.2mA



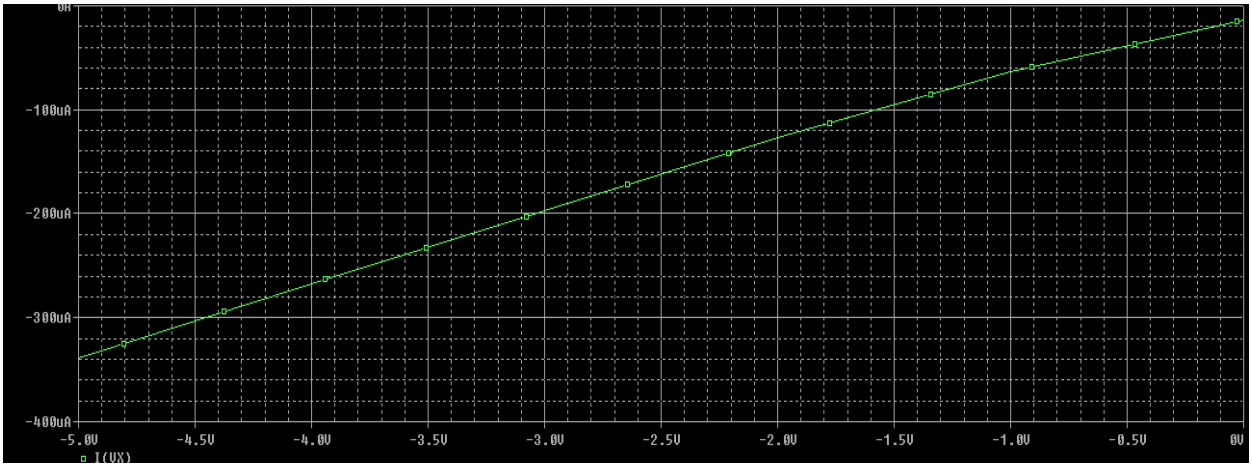
W=1.5u



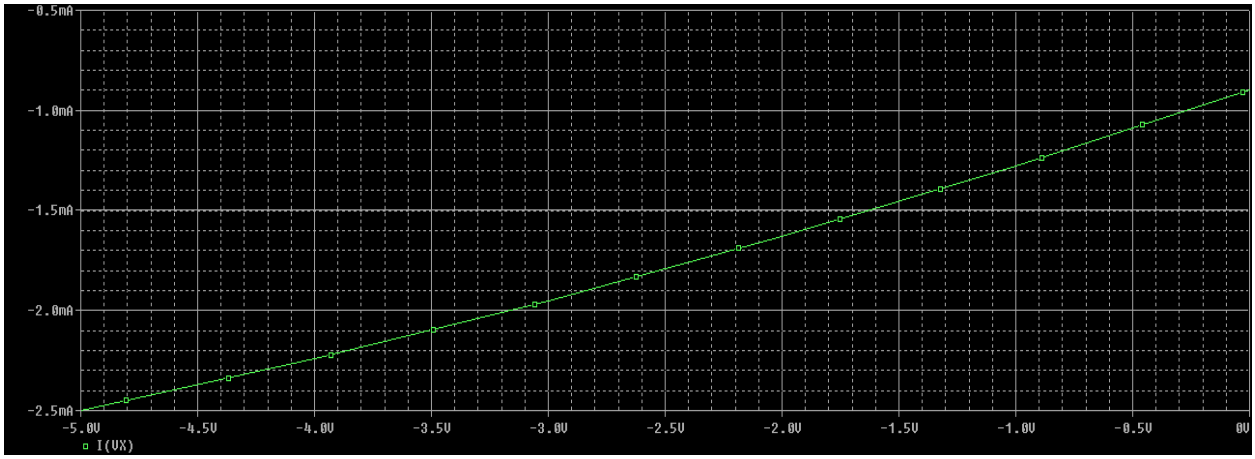
L=0.5u



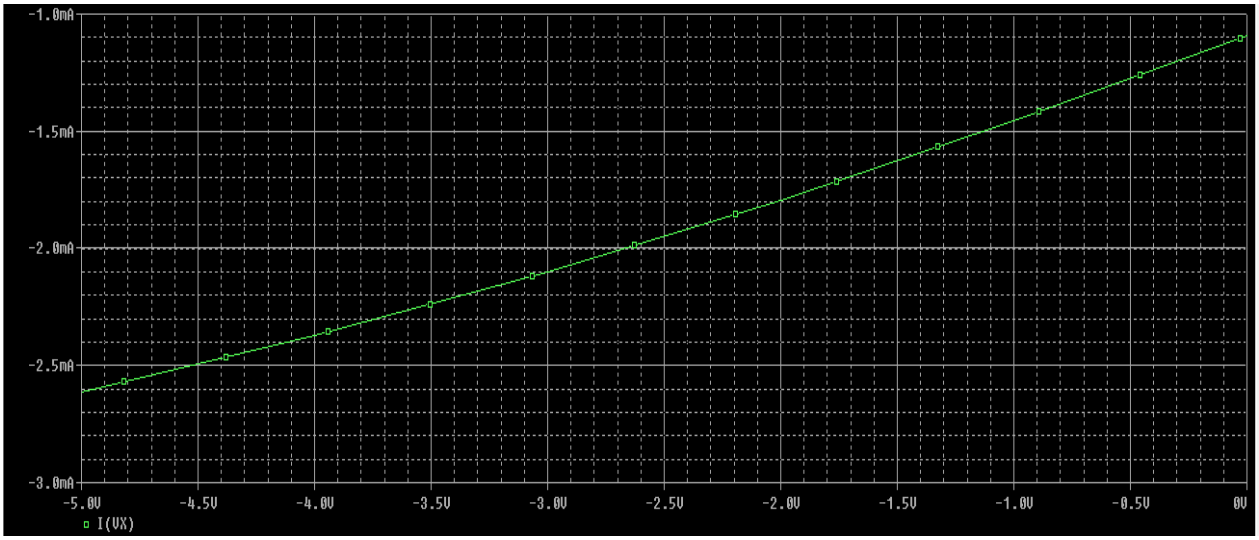
L=1u



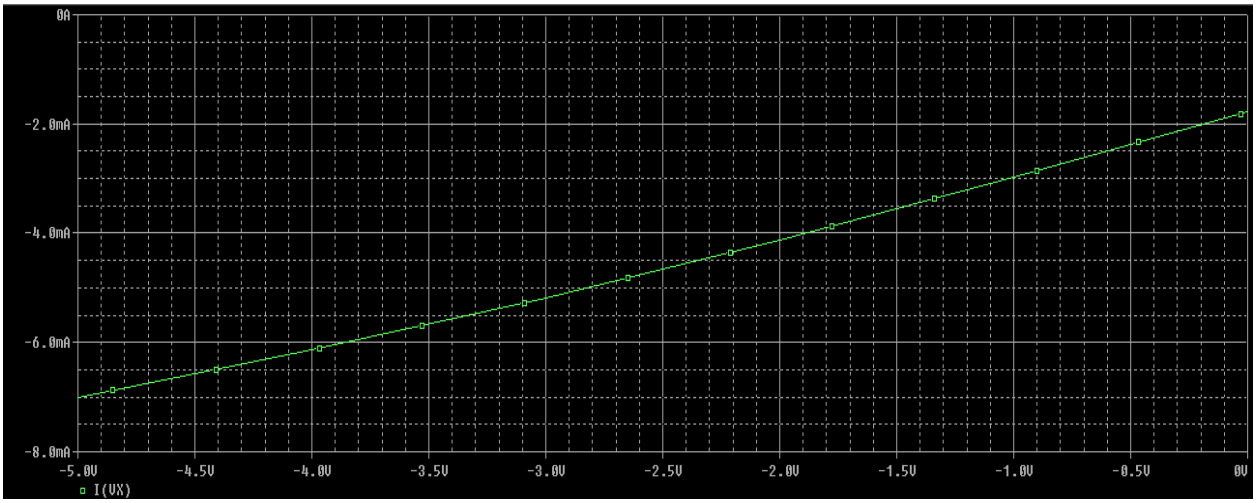
Vto=1.5


-0.5mA
-1.0mA
-1.5mA
-2.0mA
-2.5mA
-5.0V
-4.5V
-4.0V
-3.5V
-3.0V
-2.5V
-2.0V
-1.5V
-1.0V
-0.5V
0V
I(VX)

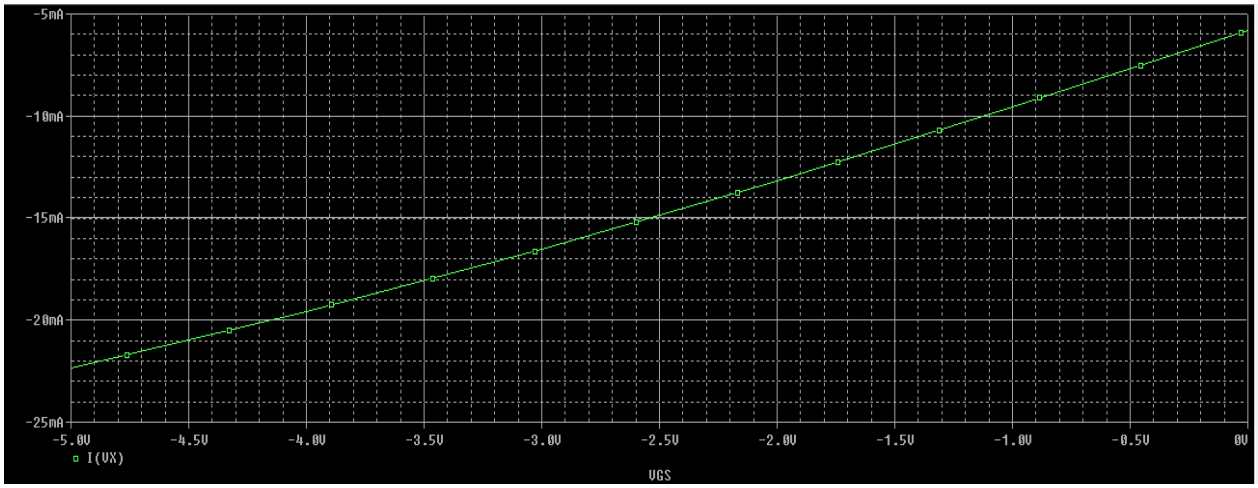
Vto=2


-1.0mA
-1.5mA
-2.0mA
-2.5mA
-3.0mA
-5.0V
-4.5V
-4.0V
-3.5V
-3.0V
-2.5V
-2.0V
-1.5V
-1.0V
-0.5V
0V
I(VX)

Kp=2.055


0A
-2.0mA
-4.0mA
-6.0mA
-8.0mA
-5.0V
-4.5V
-4.0V
-3.5V
-3.0V
-2.5V
-2.0V
-1.5V
-1.0V
-0.5V
0V
I(VX)

Gamma=0.744



Pre Lab Questions:

1. The gate region consists of _____
2. In P channel MOSFET _____ is constant

Post Lab Questions:

. 1. PMOS fabrication process is carried out in _____

Conclusion: Thus, we have studied and experimentally verified the operation of PMOS inverter circuit and recorded the results of output and transfer characteristics

CO's Covered:

……
……

Expt. No. 3

Title: Design and verify the layout of NMOS and PMOS devices in micro-wind.

Rubric	*Score (1 to 5)*
Timely Submission	
Result Interpretation and Output	
Program Logic and Concept Understanding	
Total	

Performed On: _____

Sign:_____

Experiment No. 3

Aim: Design and verify the layout of NMOS and PMOS devices in micro-wind.

Software Required: Micro-wind software

Hardware: computer

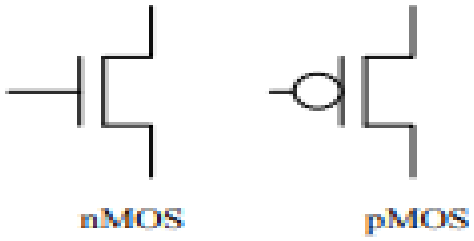
Theory:

NMOS

NMOS uses N-type FET to implement logic gates and other digital circuits. This NMOS transistor operates by creating an inversion layer in an P-type transistor body. The inversion layer called the n channel can conduct electrons between n and p-type source and drain.

PMOS

P-type MOS logic uses p channel metal oxide semiconductor FET to implement logic gates and other digital circuits. A PMOS transistor operates by creating an inversion layer in an N-type transistor body. This inversion layer called the P-channel, can conduct holes between P-type source and drain terminals. The strong logic is logic 1.



Lay Out Design Rules:

The design rules are usually described in two ways:

Micron rules, in which the layout constraints such as minimum feature sizes and minimum allowable feature separations, are stated in terms of absolute dimensions in micrometres or Lambda rules, which specify the layout constraints in terms of a single parameter (λ) and thus, allow linear, proportional scaling of all geometrical constraints.

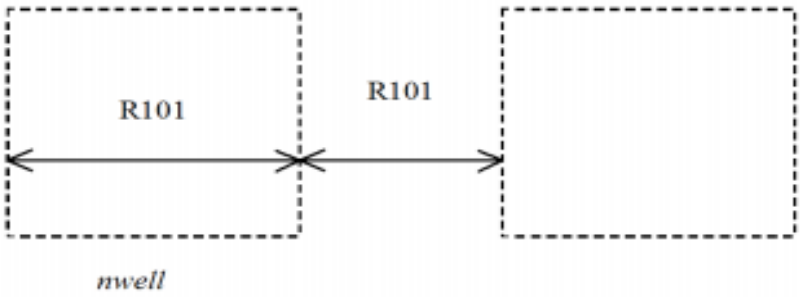
Lambda-based layout design rules were originally devised to simplify the industry-standard micron-based design rules and to allow scaling capability for various processes. It must be emphasized, however, that most of the submicron CMOS process design rules do not lend themselves to straightforward linear scaling. The use of lambda-based design rules must therefore be handled with caution in submicron geometries

N-Well

r101 Minimum well size 12 λ

r102 Between wells 12 λ

r110 Minimum well area 144 λ 2



Diffusion

r201 Minimum N+ and P+ diffusion width 4 λ

r202 Between two P+ and N+ diffusions 4 λ

r203 Extra n-well after P+ diffusion : 6 λ

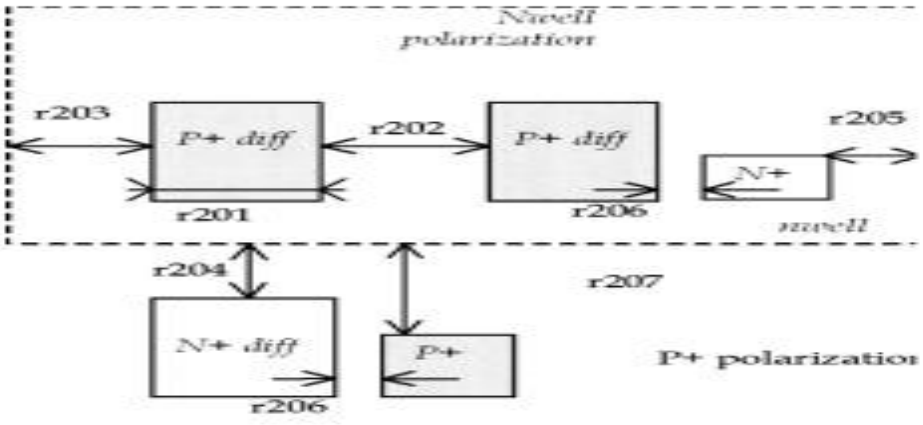
r204: Between N+ diffusion and n-well 6 λ

r205 Border of well after N+ polarization 2 λ

r206 Between N+ and P+ polarization 0 λ

r207 Border of N-well for P+ polarization 6 λ

r210 Minimum diffusion area 24 λ 2



Polysilicon

r301 Polysilicon Width 2 λ

r302 Polysilicon gate on diffusion 2 λ

r303 Polysilicon gate on diffusion for high voltage MOS 4 λ

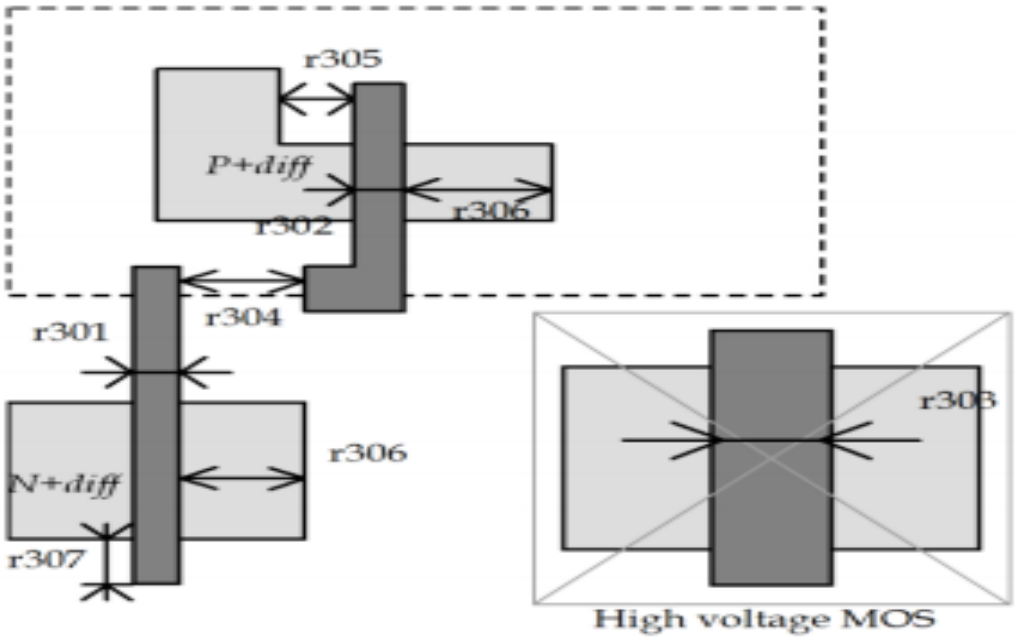
r304 Between two polysilicon boxes 3 λ

r305 Polysilicon Vs. other diffusion 2 λ

r306 Diffusion after polysilicon 4 λ

r307 Extra gate after polysilicon 3 λ

r310 Minimum surface 8 λ 2

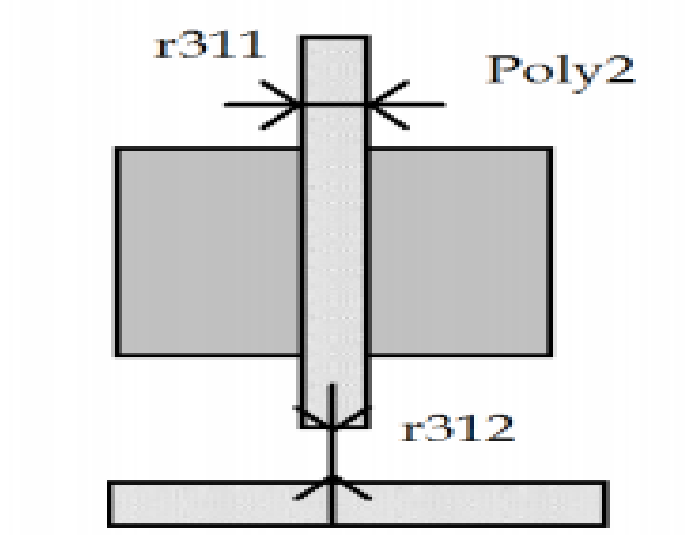


2nd Polysilicon Design Rules

r311 Polysilicon2 with 2 λ

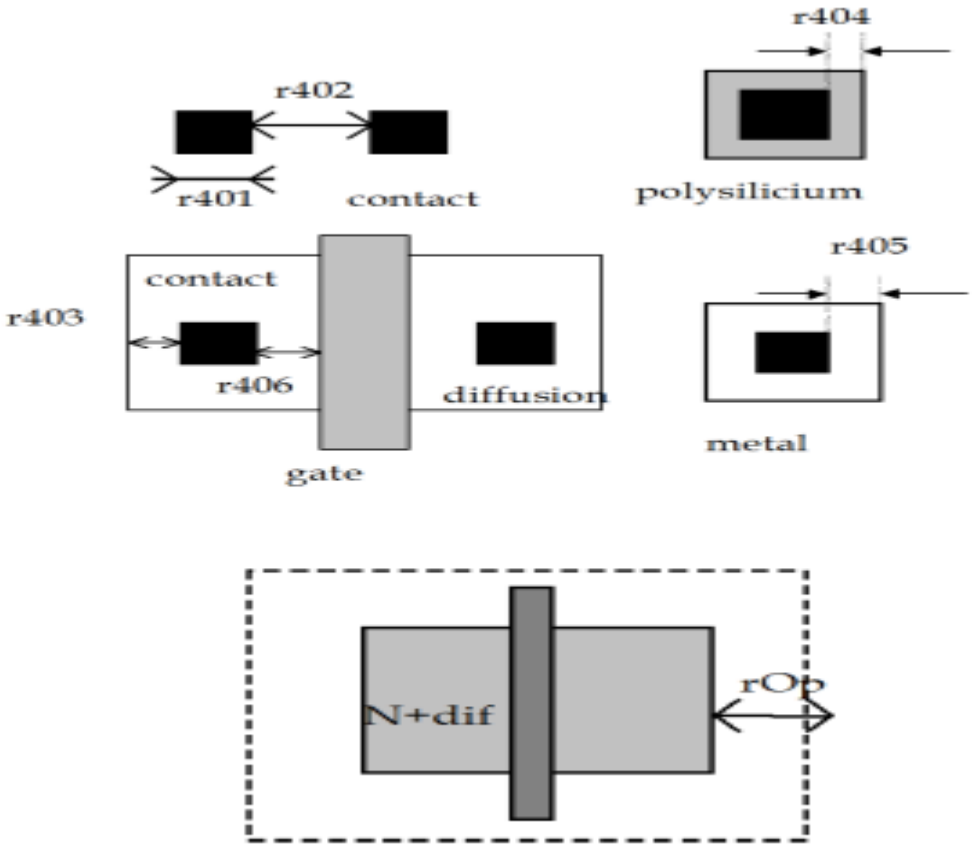
r312 Polysilicon gate on diffusion 2 λ

r320 Polysilicon2 minimum surface 8 λ



MOS Option

rOp Border of "option" layer over diff 7 N+ and diff P+



Contact

r401 Contact width 2λ

r402 Between two contacts 5λ

r403 Extra diffusion over contact 2λ

r404 Extra poly over contact 2λ

r405 Extra metal over contact 2λ

r406 Distance between contact and poly gate 3λ

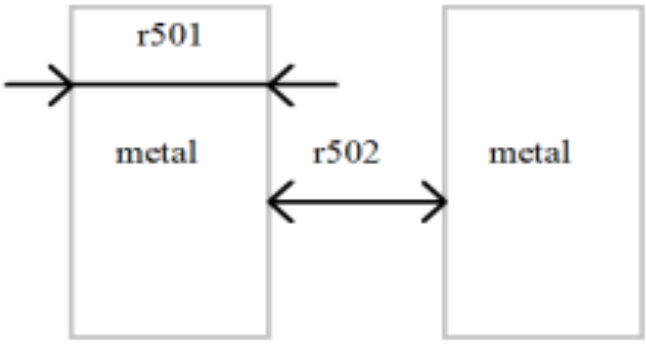
r407 Extra poly2 over contact 2λ

Metal 1

r501 Metal width 2λ

r502 Between two metals 2λ

r510 Minimum surface 16λ2



Via

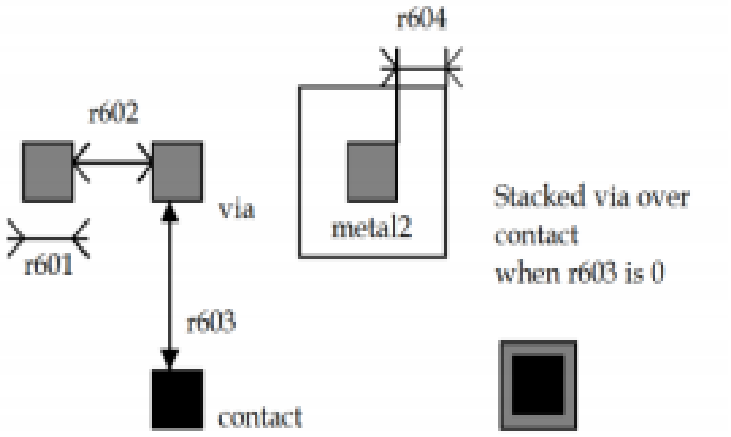
r601 Via width 2λ

r602 Between two Via 5λ

r603 Between Via and contact 0λ

r604 Extra metal over via 2λ

r605 Extra metal2 over via: 2λ

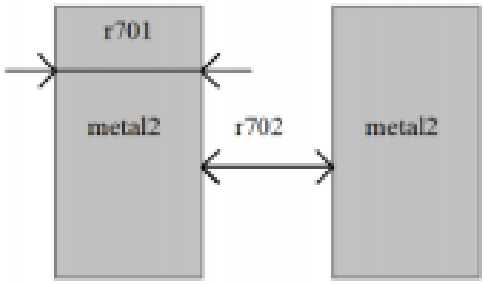


Metal 2

r701 Metal width: 4λ

r702 Between two metal2 4λ

r710 Minimum surface 16λ2



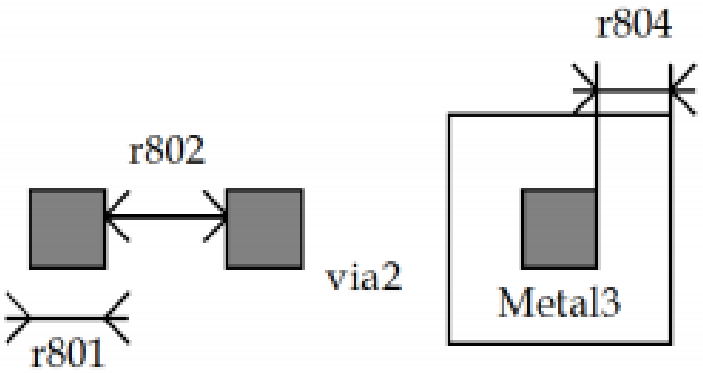
Via 2

r801 Via 2 width: 2λ

r802 Between two Via2: 5λ

r804 Extra metal2 over via2: 2λ

r805 Extra mental3 over via2: 2λ

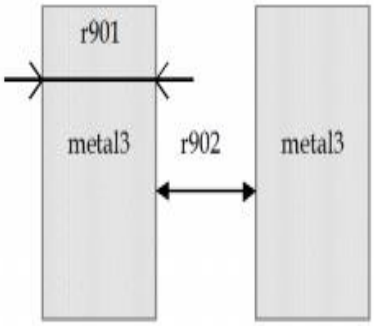


Metal 3

r901 Metal3 width: 4λ

r902 Between two metal3: 4λ

r910 Minimum surface: 32λ 2



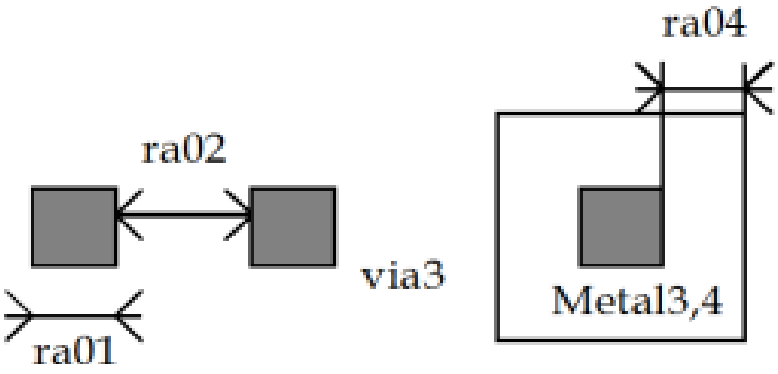
Via 3

ra01 Via3 width : 2λ

ra02 Between two Via3 : 5λ

ra04 Extra metal3 over via3 : 2λ

ra05 Extra metal4 over via3 : 2λ



Pads

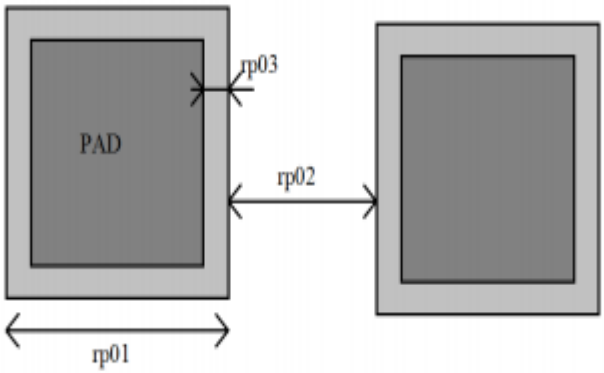
rp01 Pad width : 10µm

rp02 Between two pads 100µm

rp03 opening in passivation v.s via 5µm

rp04 opening in passivation v.s metals : 5µm

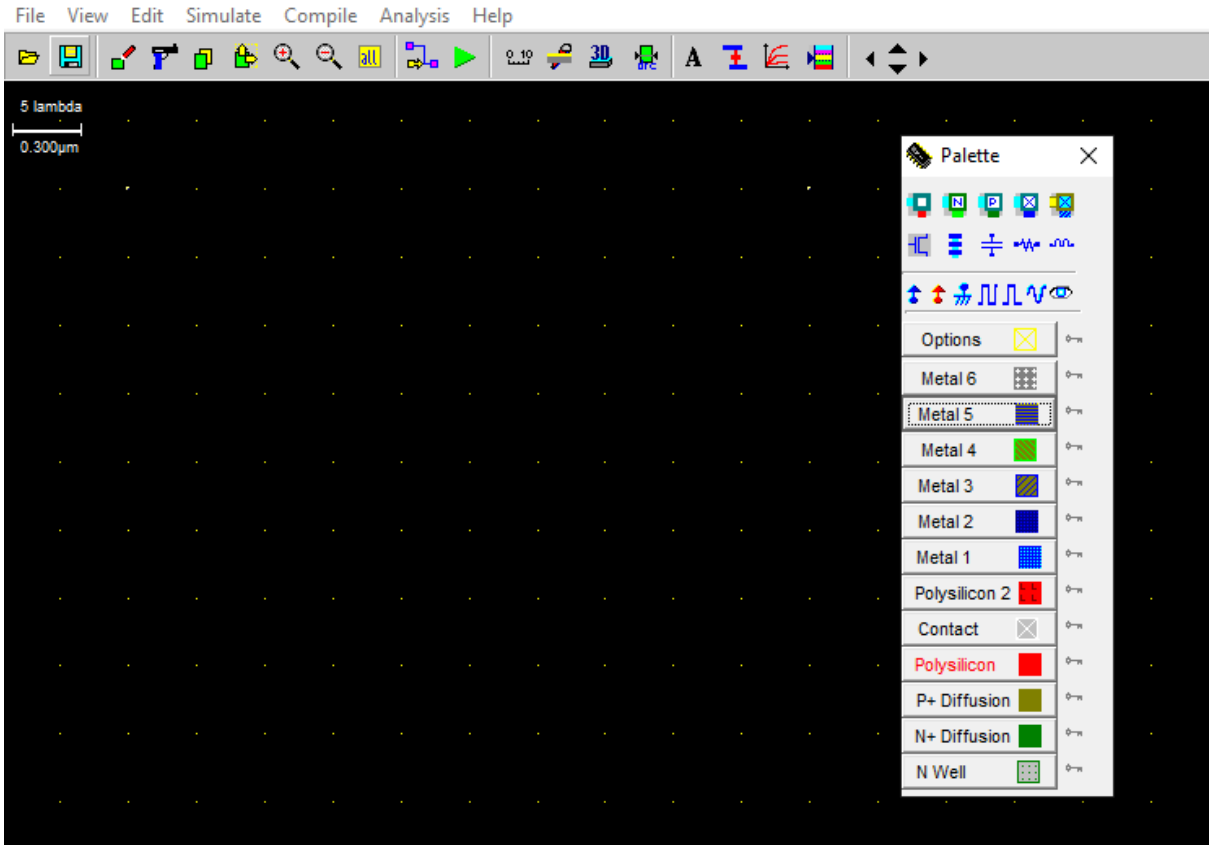
rp05 Between pad and unrelated active area : 20µm

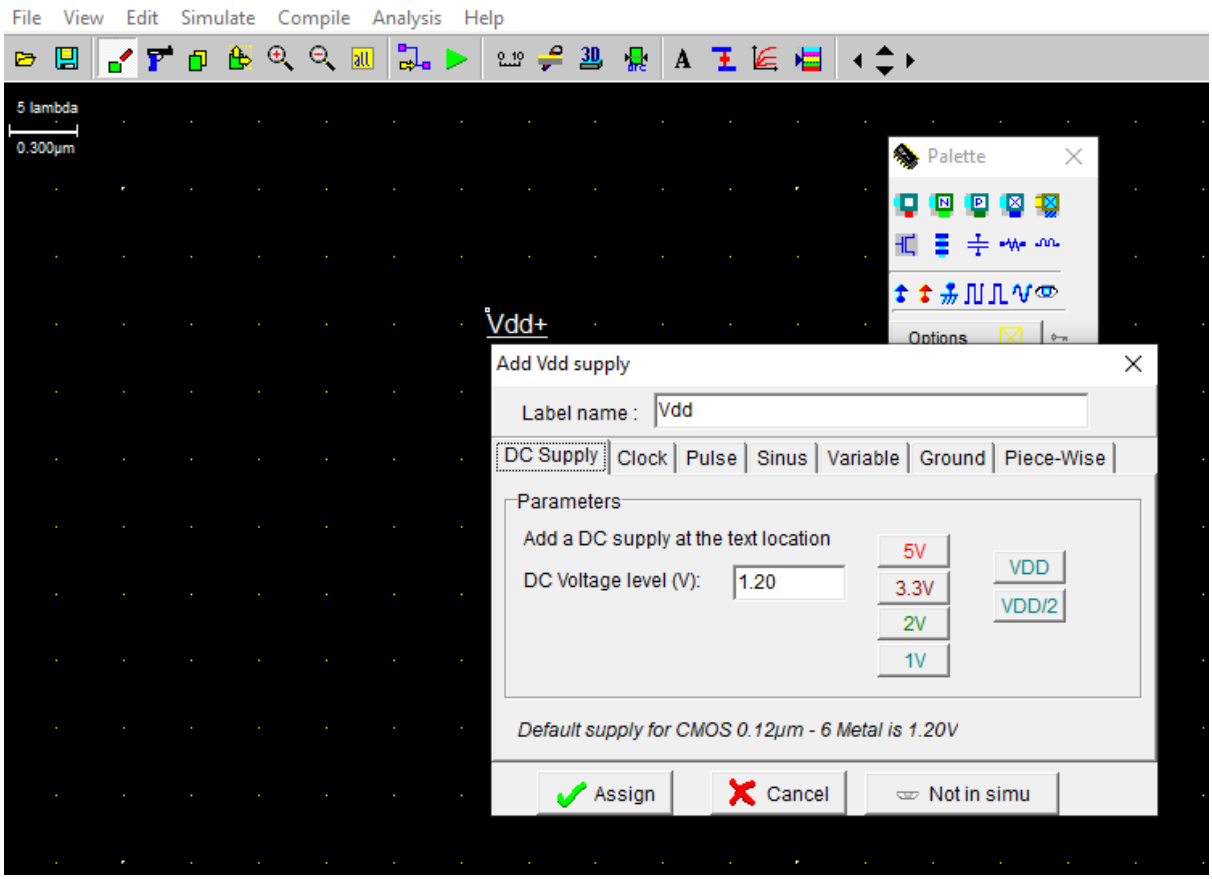


MOS layout

We use Microwind3.1 to draw the MOS layout and simulate its behavior. Go to the directory in which the software has been copied (By default microwind3). Double-click on the Microwind3.1 icon. The Microwind3.1 display window includes four main windows: the main menu, the lay out display window, the icon menu and the layer palette. The lay out window features a grid, scaled in lambda units. The lambda unit is fixed to half of the

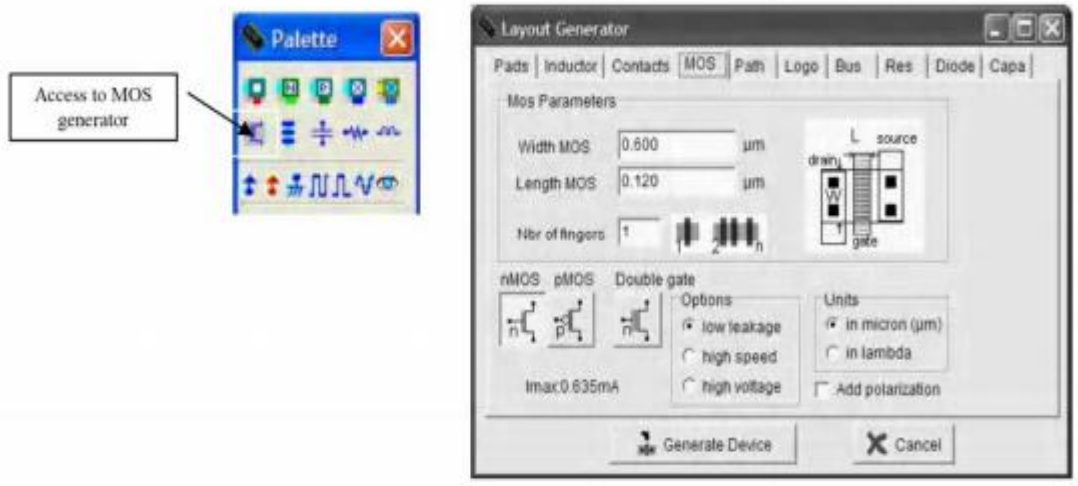
minimum available lithography of the technology. The default technology is a CMOS 6-metal layers 0.12µm technology, consequently lambda is 0.06µm (60nm).





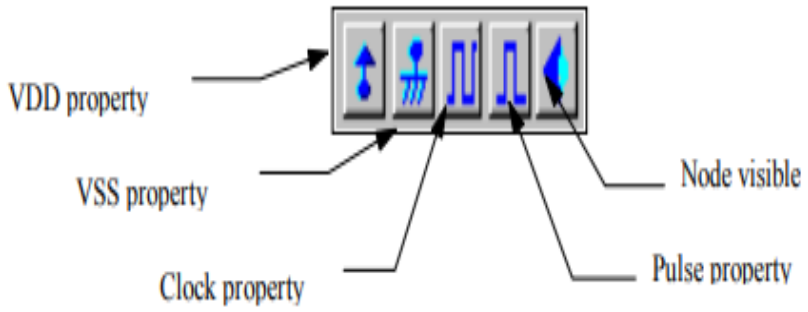
Layout considerations

The safest way to create a MOS device is to use the MOS generator. In the palette, click the MOS generator icon. A window appears as reported below. The programmable parameters are the MOS width, length, the number of gates in parallel and the type of device (n-channel or p-channel). By default metal interconnects and contacts are added to the drain and source of the MOS. We may add a supplementary metal2 interconnect on the top of metal 1 for drain and source

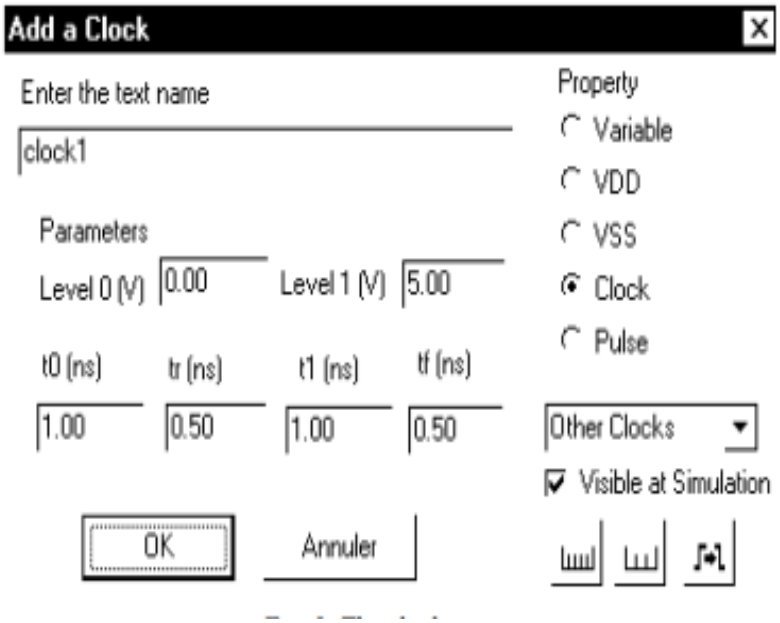


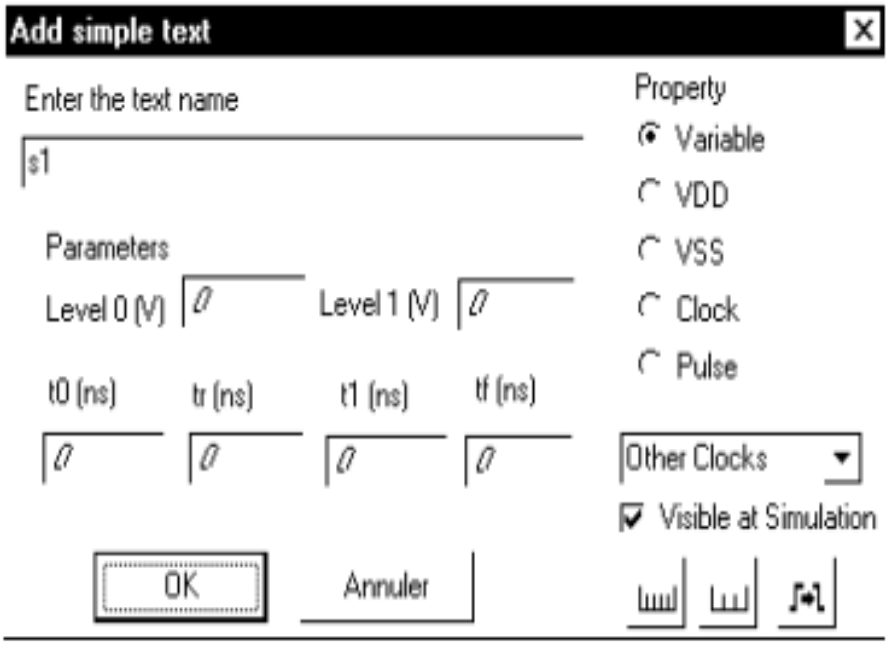
Add Properties for Simulation

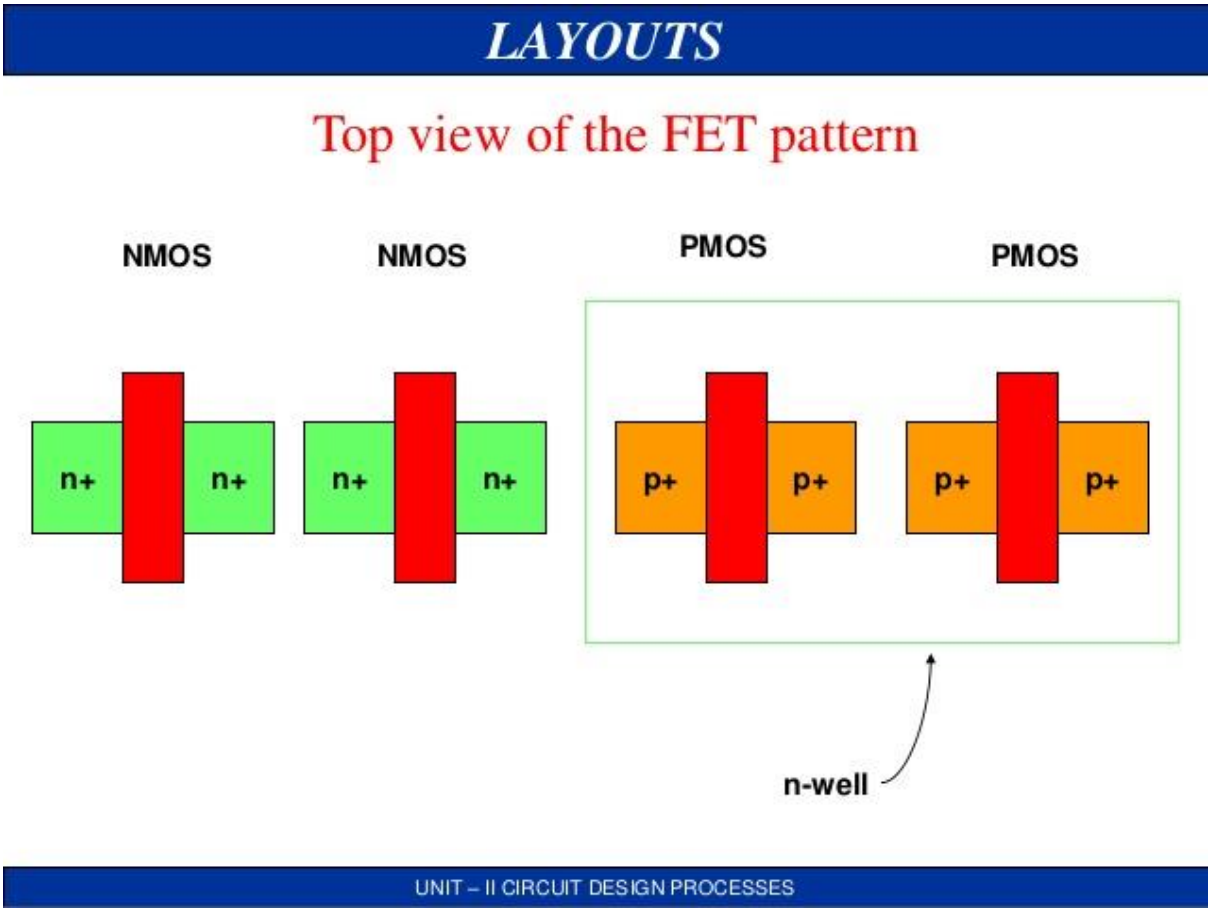
Properties must be added to the layout to fix the ground, the supply, the input and the outputs. The summary of available properties is reported below.



Stuck at Vdd: Click on the Vdd icon and click on the upper p-diffusion box. The Vdd property is sent to the node. Page 6 o Also click in the N-Well region inside which the pMOS is located. The NWell must always be at VDD voltage to keep the PMOS junctions inverted. p Hold at Vss: Click on the Vss icon, click on the lower n-diffusion box. The Vss property is sent to the node. q Apply a clock to node IN: Click on the Clock icon and on the polysilicon gate. The Clock menu appears (See below). Click on OK. A default clock with 3 ns period is generated. The Clock property is sent to the node and appears at the right hand side of the desired location with the name « clock1 ».


Add a Clock
Enter the text name
clock1
Parameters
Level 0 (V)
0.00
Level 1 (V)
5.00
t0 (ns)
tr (ns)
t1 (ns)
tf (ns)
1.00
0.50
1.00
0.50
Property
Variable
VDD
VSS
Clock
Pulse
Other Clocks
Visible at Simulation
OK
Annuler


Add simple text
Enter the text name
$1
Parameters
Level 0 (V)
0
Level 1 (V)
0
t0 (ns)
tr (ns)
t1 (ns)
tf (ns)
0
0
0
0
Property
Variable
VDD
VSS
Clock
Pulse
Other Clocks
Visible at Simulation
OK
Annuler



Procedure:

1) Start micro-wind software.

2) Design the PMOS and NMOS devices following the layout design.

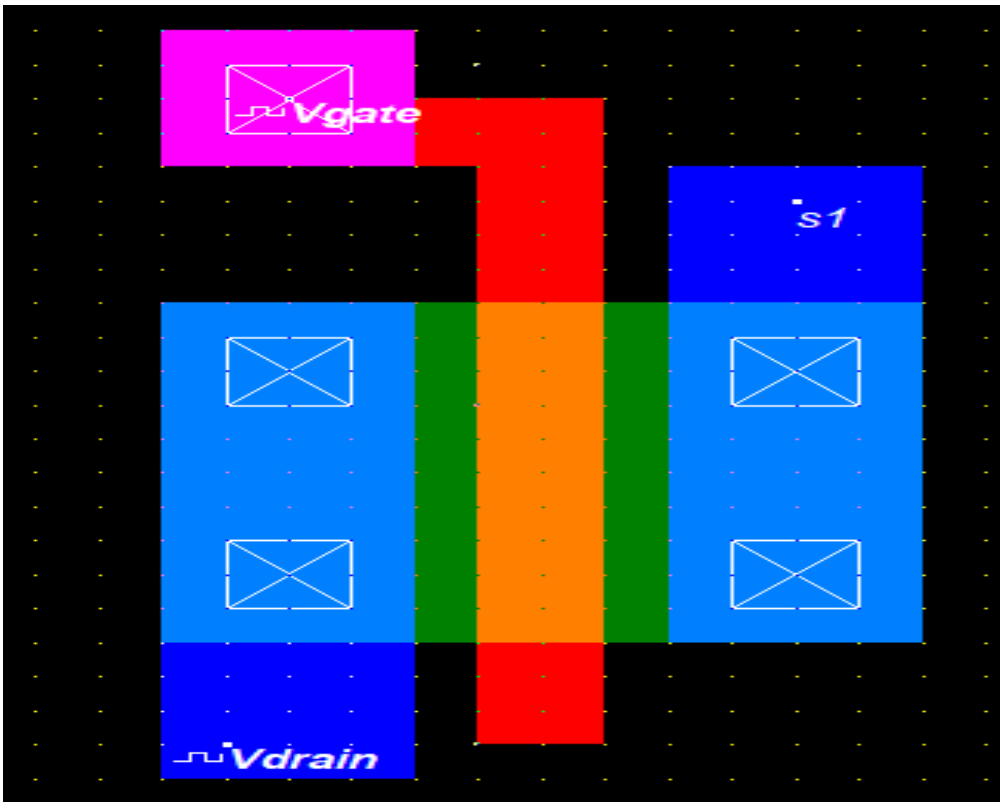
3) Click on DRC at every stage of design.

4) Save your design and finally run DRC again.

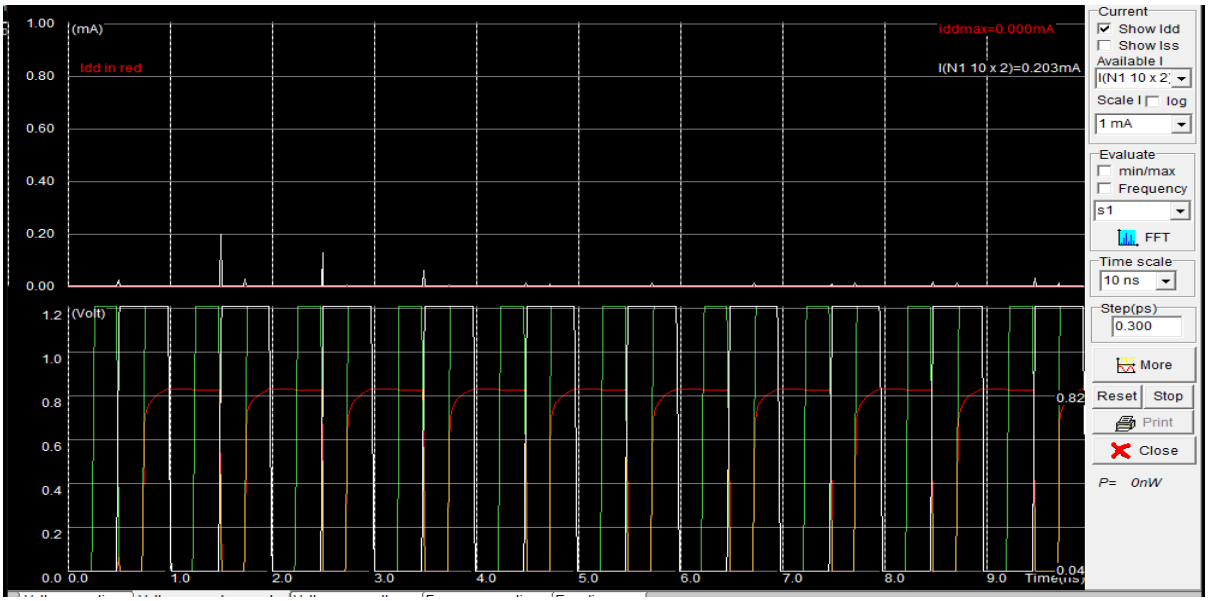
5) Observe various outputs using the stimulation option

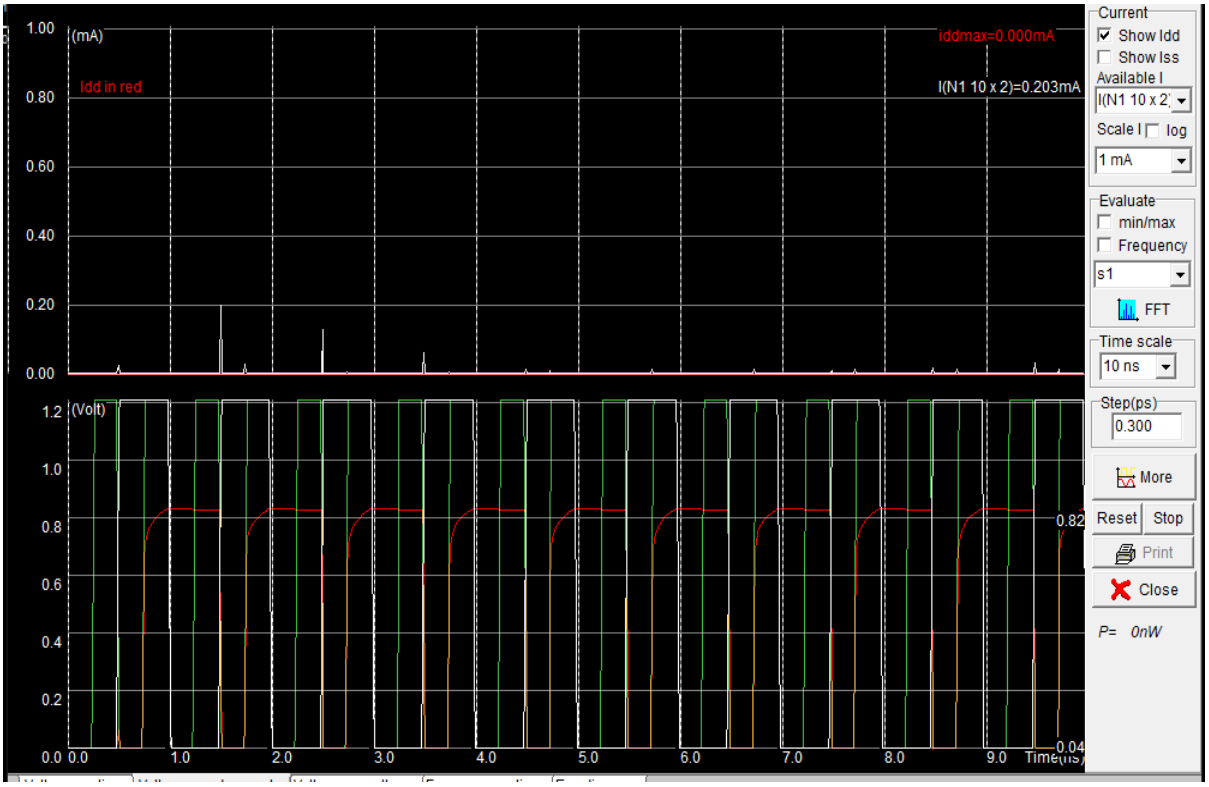
Layout:

N-MOS

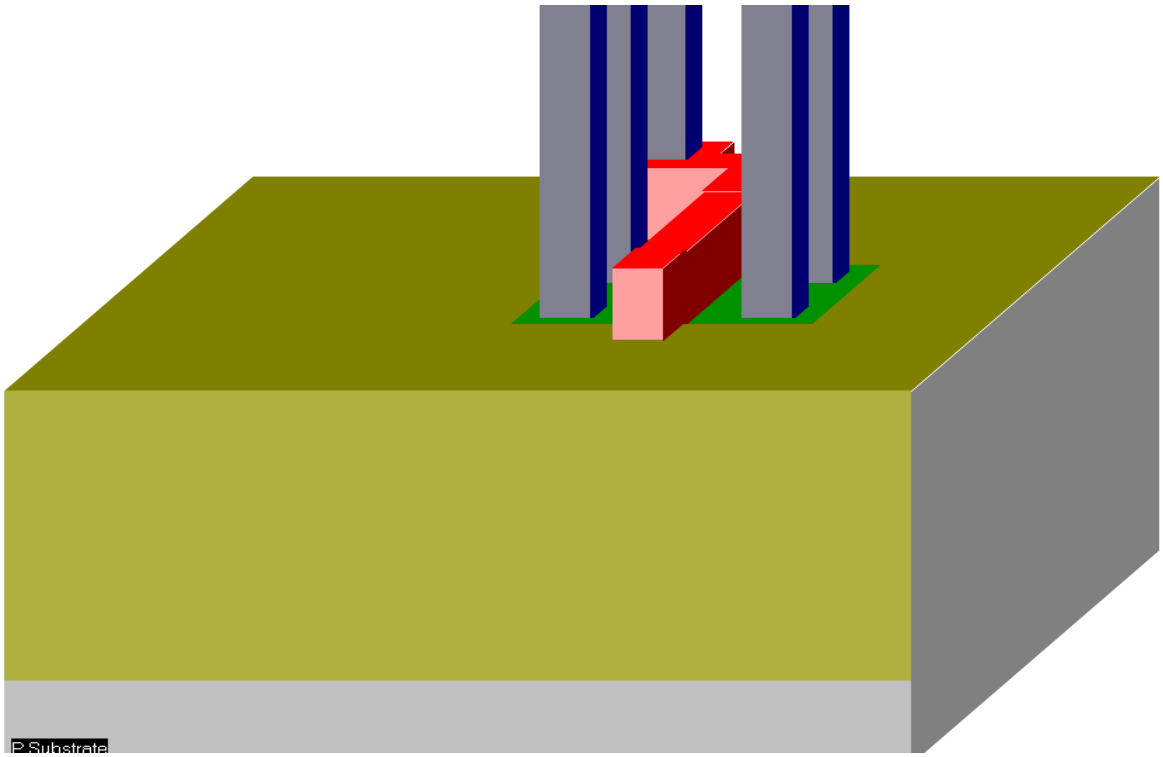


Output:



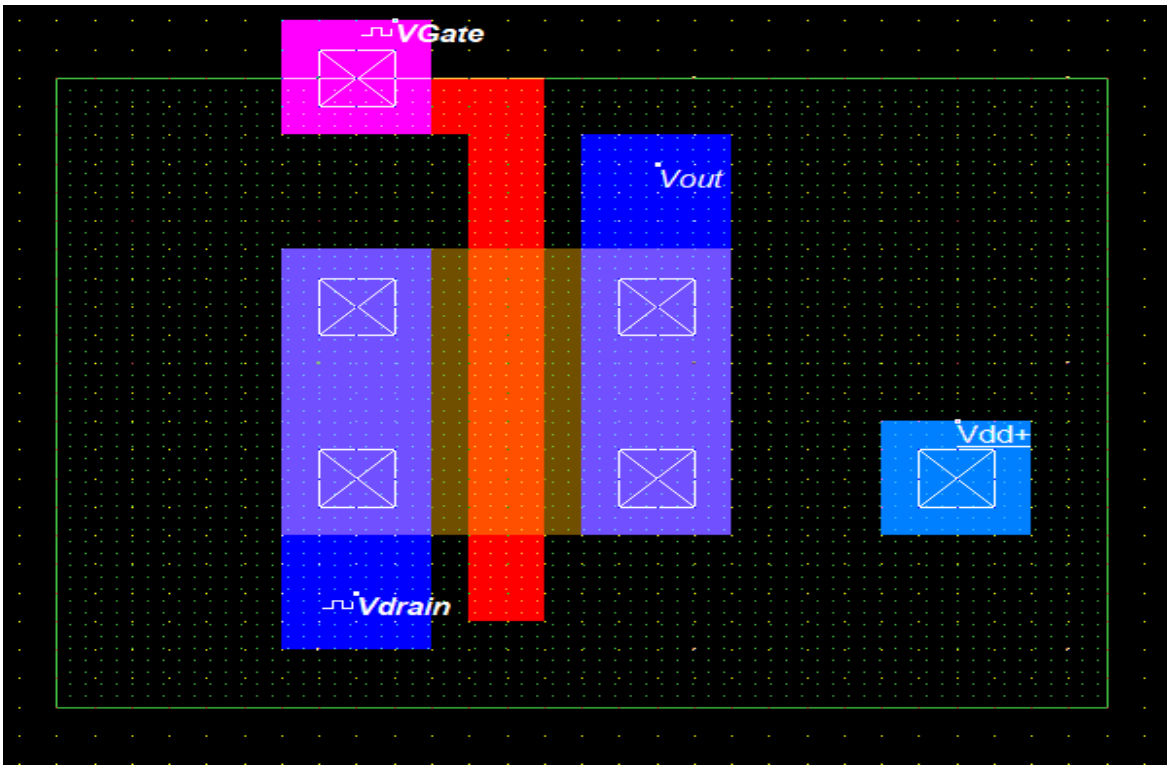


NMOS 3D Process View

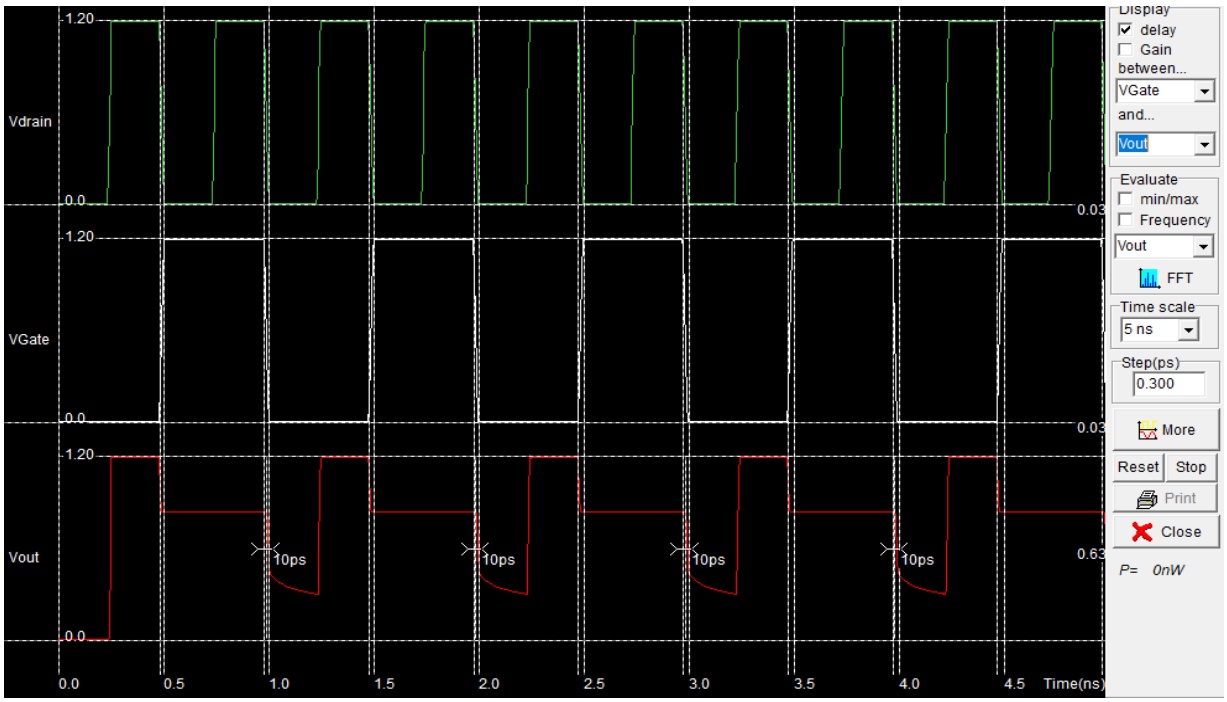


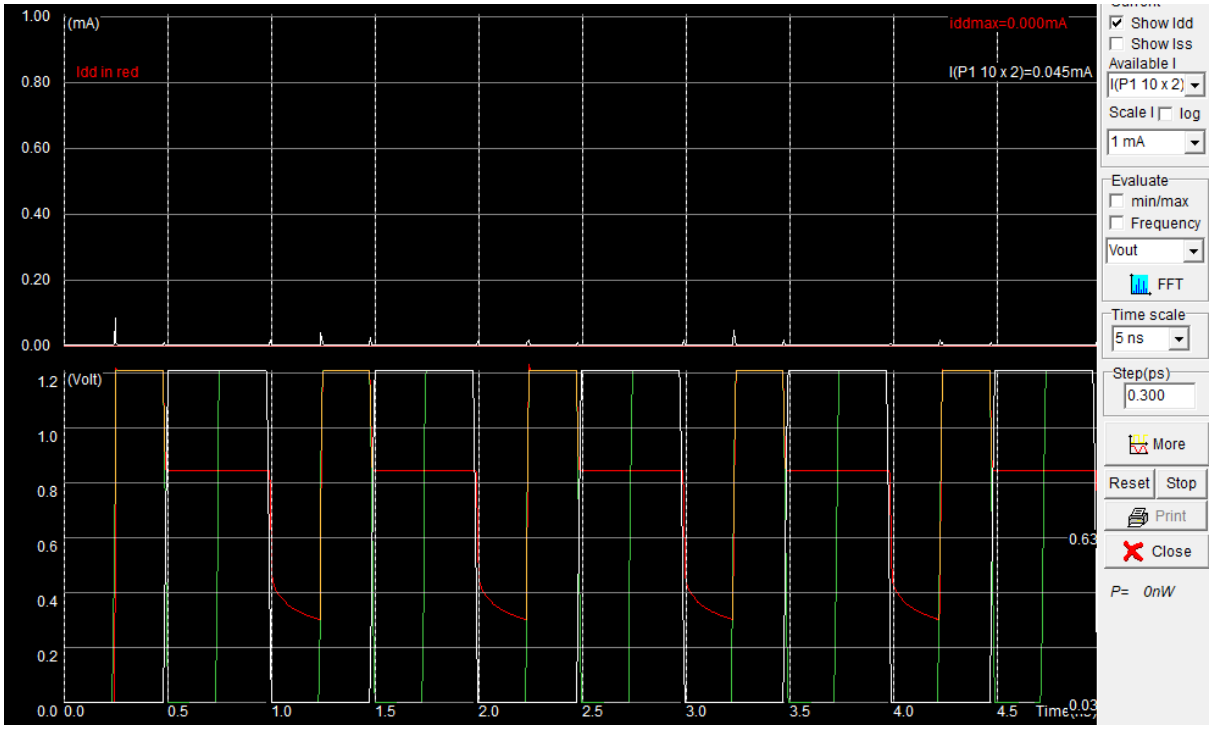
Layout:

P-MOS

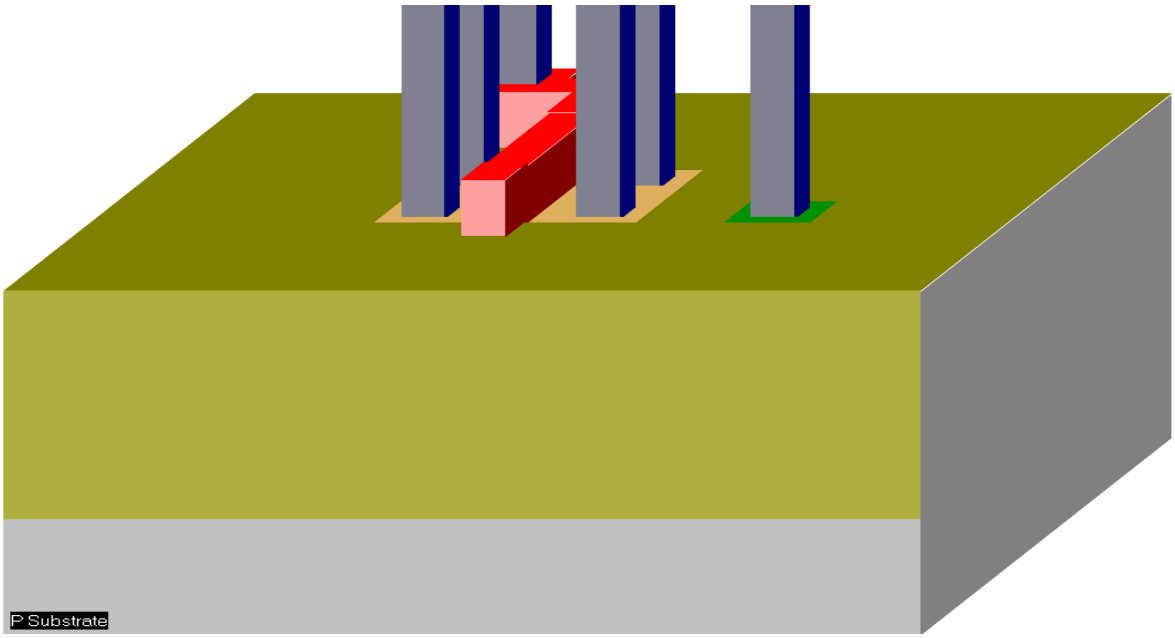


Output:





PMOS 3D Process View



Pre Lab Questions:

1. What is the LAMBDA width of N-mos Diffusion
2. What is the poly-silicon width of N-mos and P-mos

Post Lab Questions:

3. Why do ICs need small power for their operation?
4. Why are ICs more reliable than discrete circuits?

Conclusion: Thus, we have designed and simulated Layout of NMOS and PMOS devices in micro-wind.

CO's Covered:

……
……

Expt. No. 4

Title: To design and verify the layout of CMOS inverter.

Rubric	*Score (1 to 5)*
Timely Submission	
Result Interpretation and Output	
Program Logic and Concept Understanding	
Total	

Performed On: _____

Sign:_____

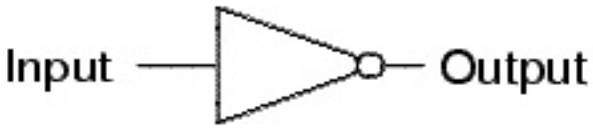
Experiment No. 4

Aim: To design and verify the layout of CMOS inverter.

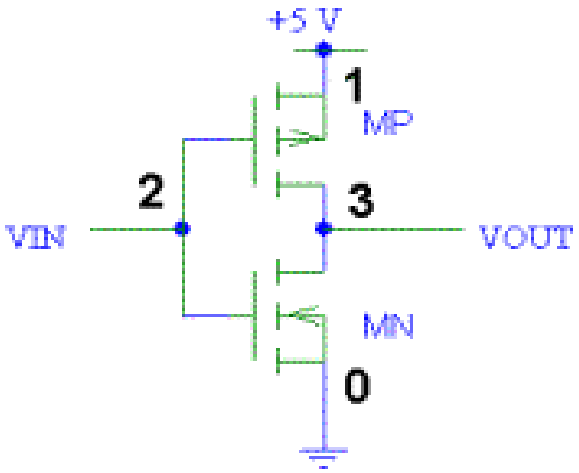
Software Required: Micro-wind software

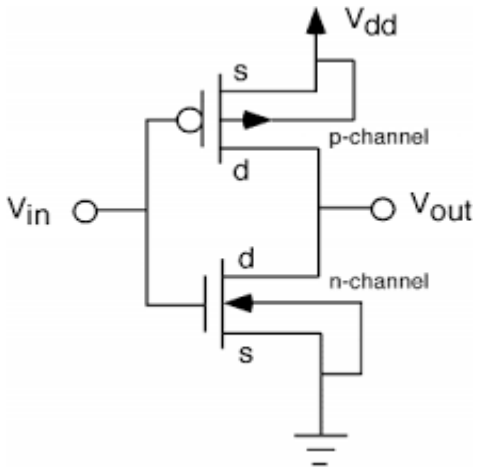
Hardware: computer

Circuit Diagram:



Input	Output
1	0
0	1





Theory:

The inverter is universally accepted as the most basic logic gate doing a Boolean operation on a single input variable. Fig.1 depicts the symbol, truth table and a general structure of a CMOS inverter. As shown, the simple structure consists of a combination of an pMOS transistor at the top and a nMOS transistor at the bottom.

CMOS is also sometimes referred to as **complementary-symmetry metal–oxide–semiconductor**. The words "complementary-symmetry" refer to the fact that the typical digital design style with CMOS uses complementary and symmetrical pairs of p-type and n-type metal oxide semiconductor field effect transistors (MOSFETs) for logic functions. Two important characteristics of CMOS devices are high noise immunity and low static power consumption. Significant power is only drawn while the transistors in the

CMOS device are switching between on and off states. Consequently, CMOS devices do not produce as much waste heat as other forms of logic, for example transistor-transistor logic (TTL) or NMOS logic, which uses all n-channel devices without p-channel devices. Some important condition to be kept in doing layout are

1] CMOS logic is based on a pair of NMOS and PMOS transistor, two different transistors have to be in layer, so that it will be a good for reading software of both transistors.
2] There must be a single nail for VDD as well as VDS.
3] Width of PMOS to be kept at a value of double for as compared as to NMOS.
4] Also follow the layout rules, it keeps checking DRC at every step so as to a void error in output at the end of simulator.

Procedure:

1] Start micro-wind software for layout.

2] Draw the CMOS inverter following the design.

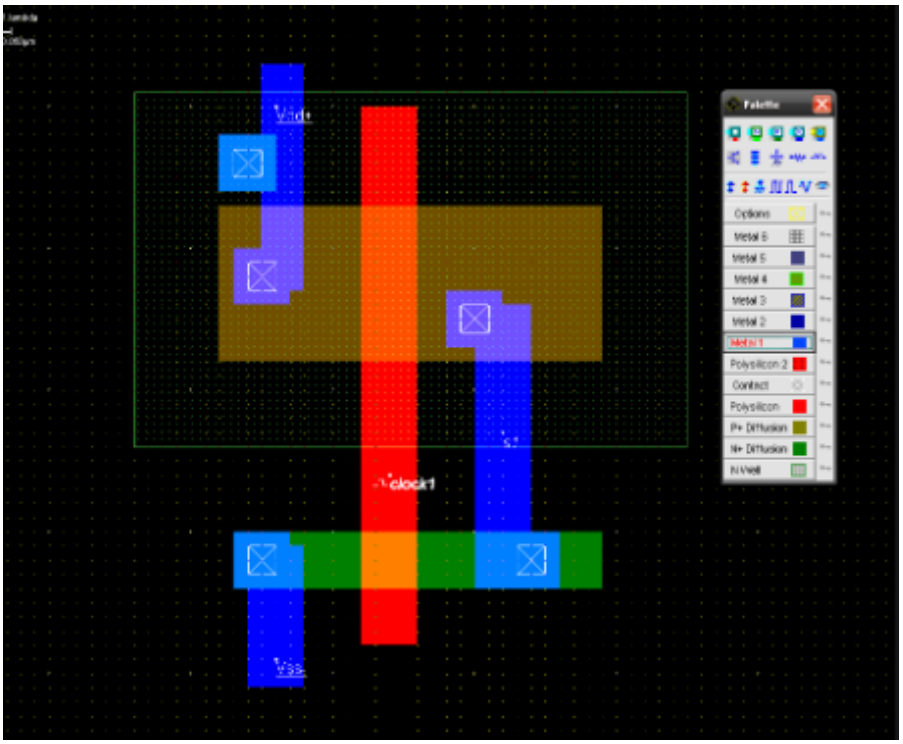
3] Click on DRC at every stage of design to avoid error.

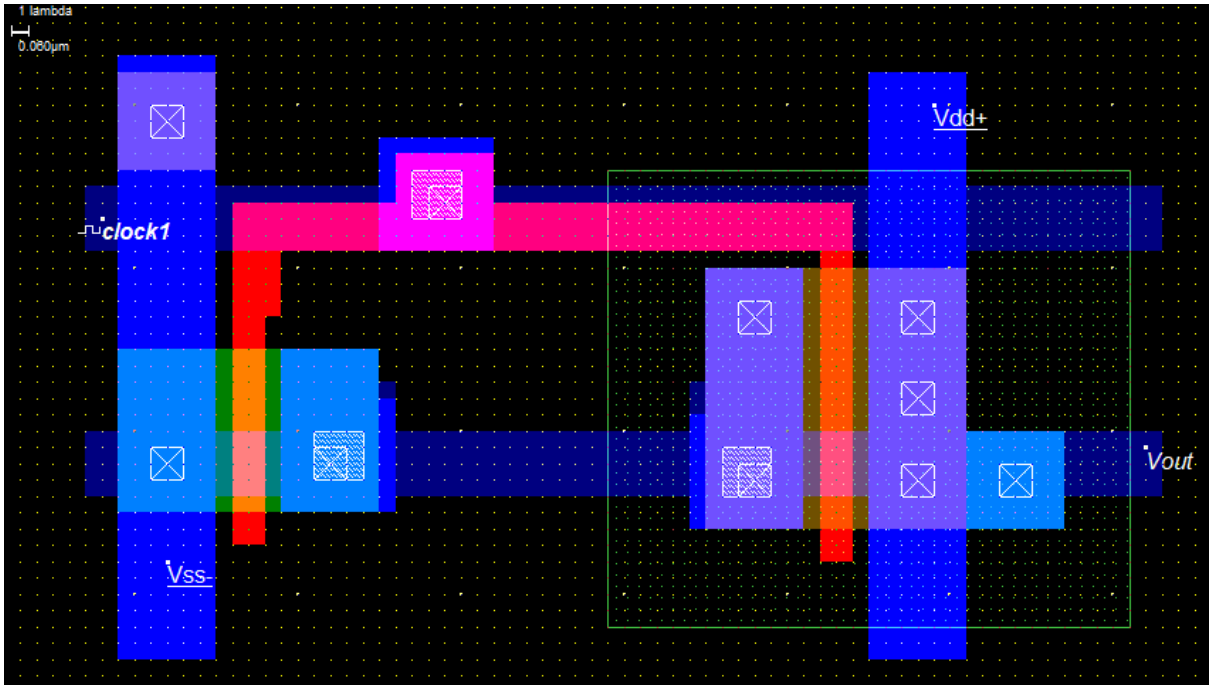
4] Observe various output while using simulation output.

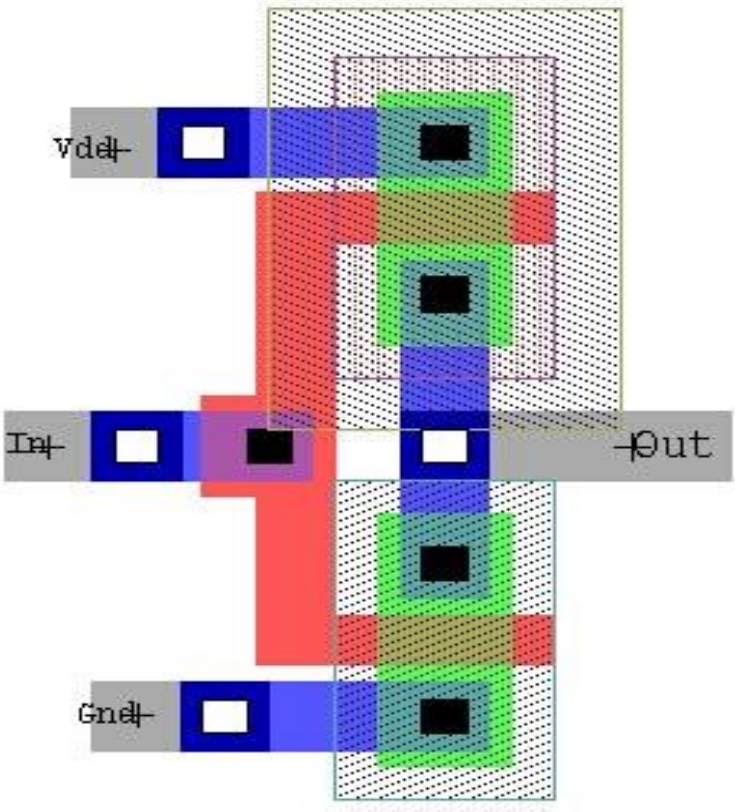
5] Software design and finally run a DRC.

6] Record the output waveform observed

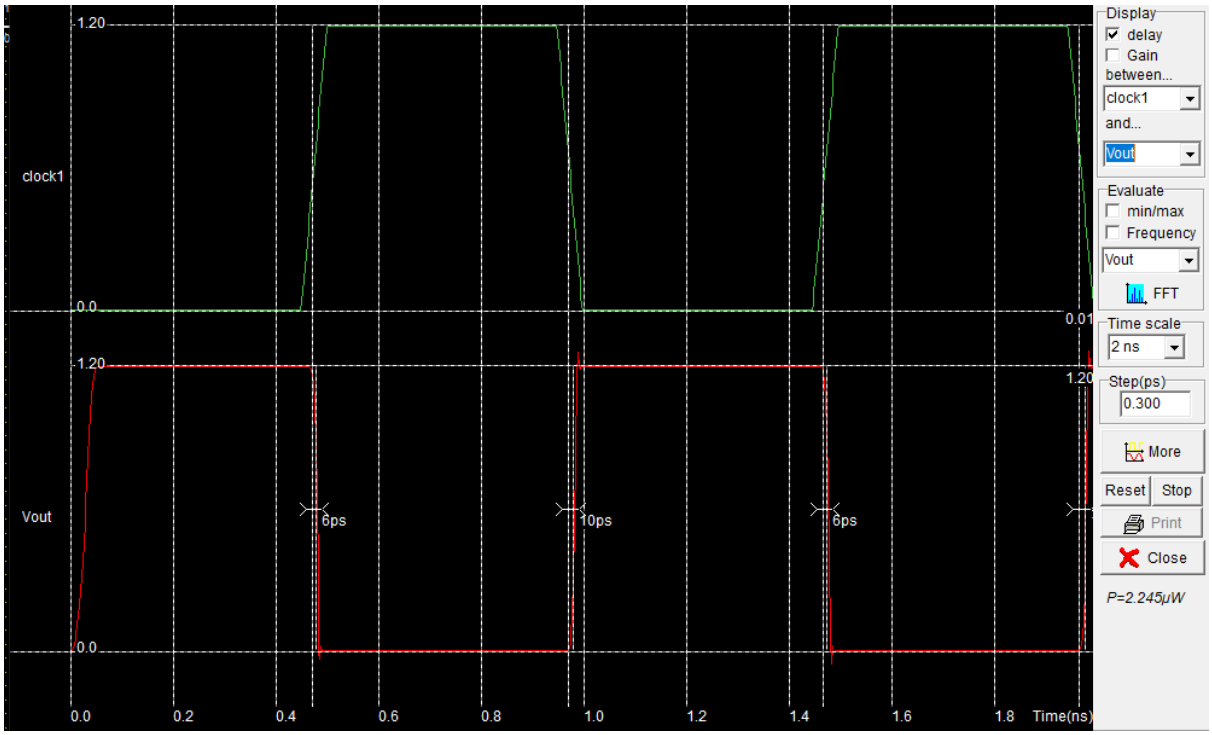
Layout:

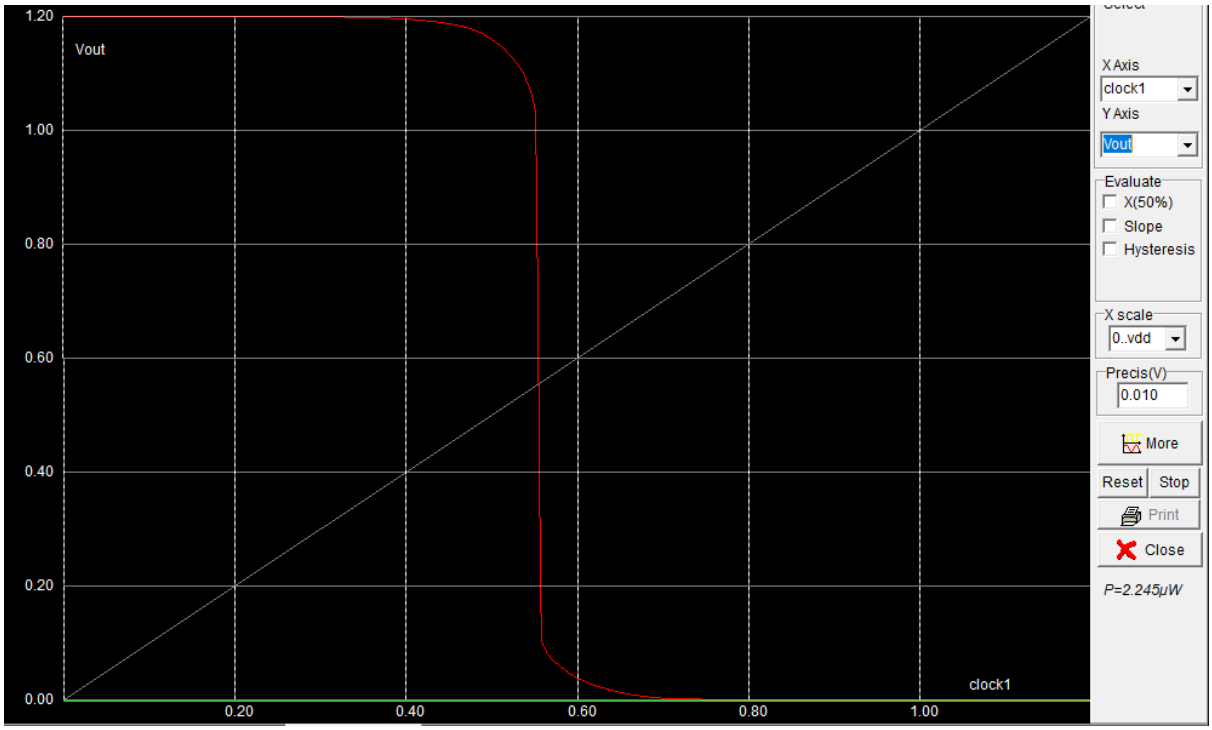


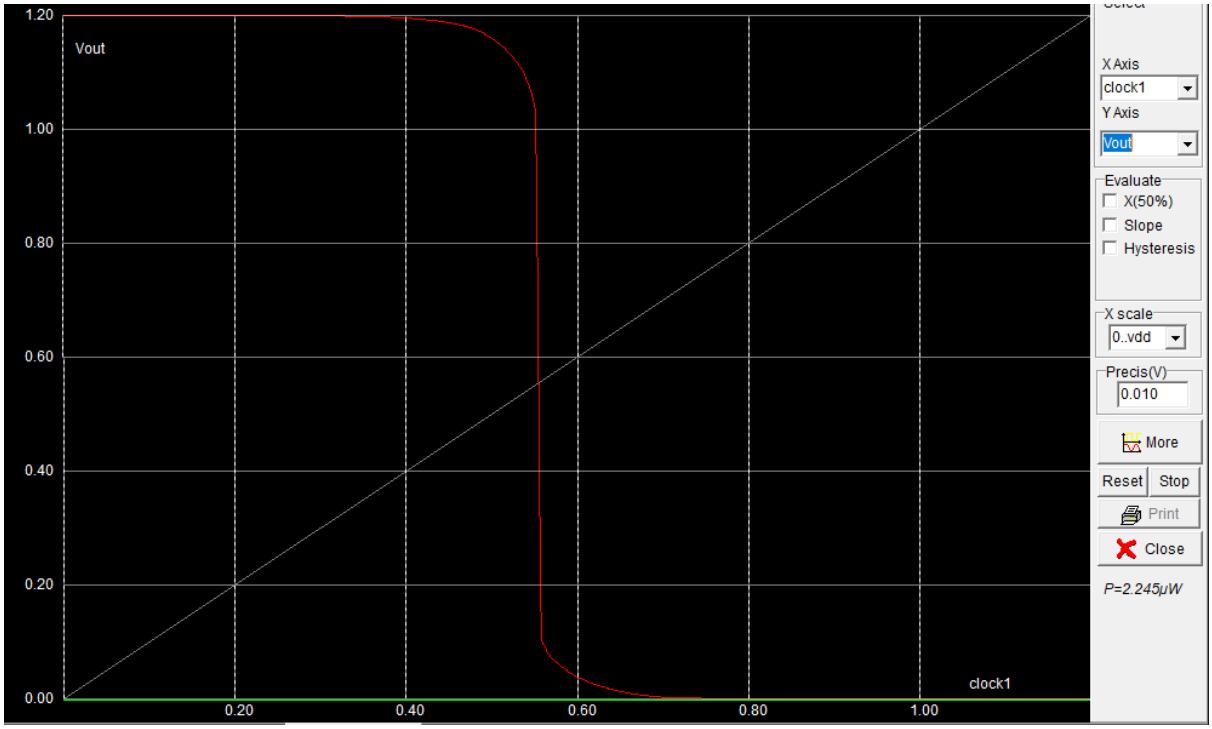

1 lambda
0.060µm
Vdd+
clock1
Vout
Vss-

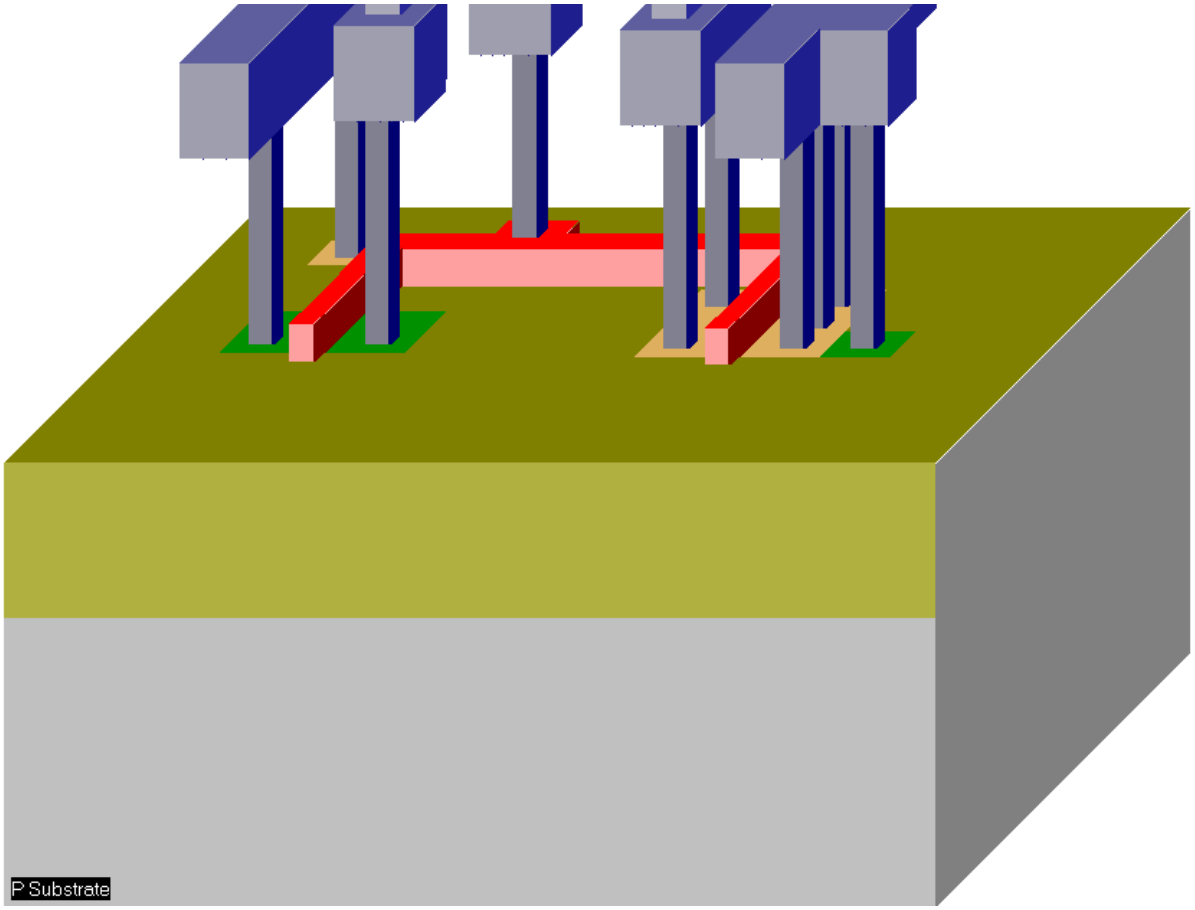

Vdd
In
Out
Gnd

Output:


1.20
clock1
0.0
1.20
Vout
0.0
6ps
10ps
6ps
0.01
1.20
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
Time(ns)
Display
delay
Gain
between...
clock1
and...
Vout
Evaluate
min/max
Frequency
Vout
FFT
Time scale
2 ns
Step(ps)
0.300
More
Reset
Stop
Print
Close
P=2.245µW


1.20
Vout
1.00
0.80
0.60
0.40
0.20
0.00
0.20
0.40
0.60
0.80
1.00
clock1
X Axis
clock1
Y Axis
Vout
Evaluate
X(50%)
Slope
Hysteresis
X scale
0..vdd
Precis(V)
0.010
More
Reset
Stop
Print
Close
P=2.245µW


1.20
1.00
0.80
0.60
0.40
0.20
0.00
Vout
0.20
0.40
0.60
0.80
1.00
clock1
X Axis
clock1
Y Axis
Vout
Evaluate
X(50%)
Slope
Hysteresis
X scale
0..vdd
Precis(V)
0.010
More
Reset
Stop
Print
Close
P=2.245µW


P Substrate

Pre Lab Questions:

1 For Cmos Logic, Give The Various Techniques You Know To Minimize Power Consumption?
2 Why Do We Gradually Increase The Size Of Inverters In Buffer Design? Why Not Give The Output Of A Circuit To One Large Inverter**?**

Post Lab Questions:

1 Why Is The Substrate In Nmos Connected To Ground And In Pmos To Vdd?
2 What Is The Fundamental Difference Between A Mosfet And Bjt?

Conclusion: Thus, we have designed and verified the layout of CMOS inverter circuit using micro-wind waveform.

CO's Covered:

……
……

Expt. No. 5

Title: To design layout of NAND and NOR gate using micro-wind.

Rubric	*Score (1 to 5)*
Timely Submission	
Result Interpretation and Output	
Program Logic and Concept Understanding	
Total	

Performed On: _____

Sign:_____

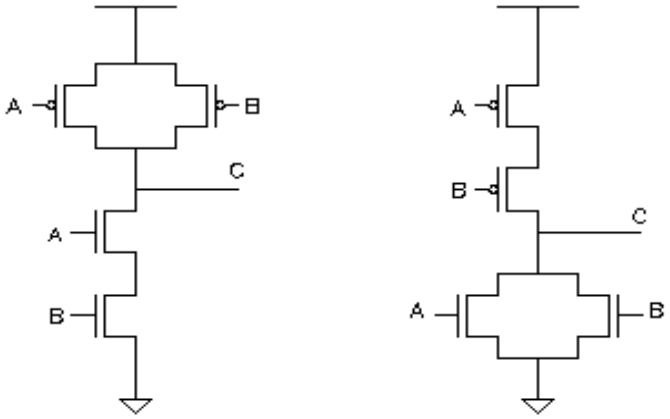
Experiment No. 5

Aim: To design layout of NAND and NOR gate using micro-wind.

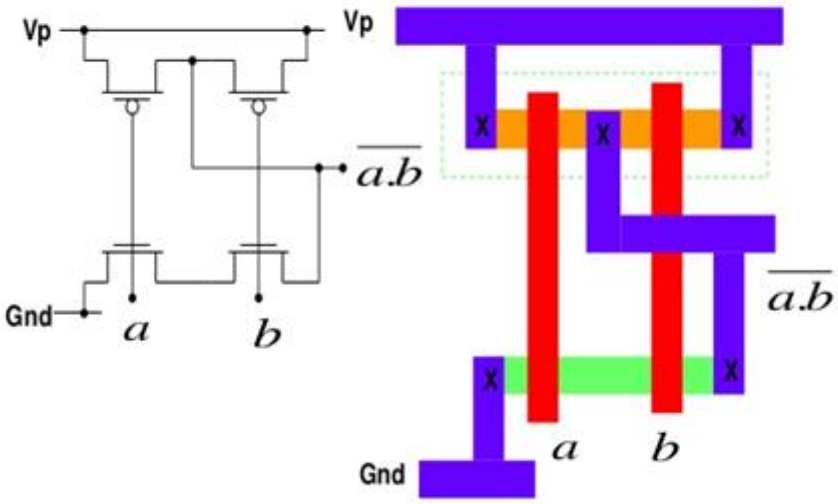
Software Required: Micro-wind software, DSCH software

Hardware: computer

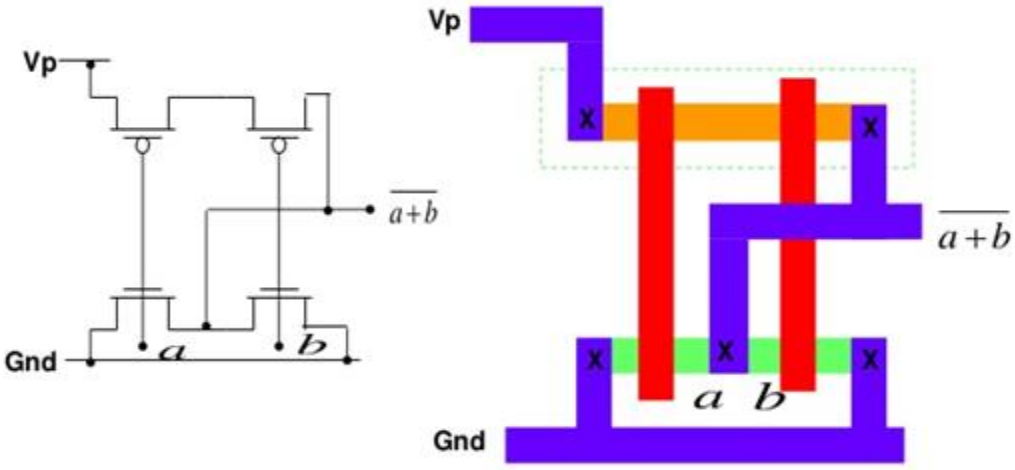
Circuit Diagram:



The CMOS NAND Gate



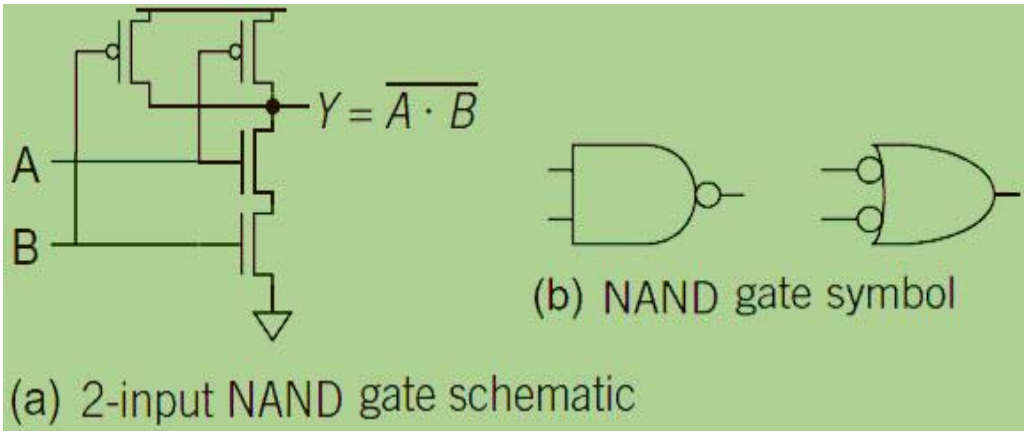
The CMOS NOR Gate



Theory:

CMOS NAND Gate

The below figure shows a 2-input Complementary MOS NAND gate. It consists of two series NMOS transistors between Y and Ground and two parallel PMOS transistors between Y and VDD.



CMOS NAND Gate

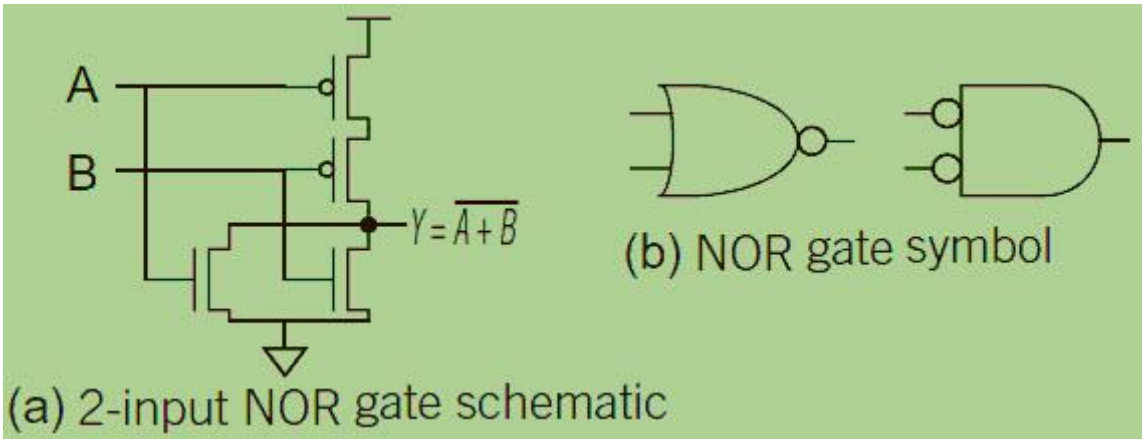
If either input A or B is logic 0, at least one of the NMOS transistors will be OFF, breaking the path from Y to Ground. But at least one of the pMOS transistors will be ON, creating a path from Y to VDD.

Hence, the output Y will be high. If both inputs are high, both of the NMOS transistors will be ON and both of the PMOS transistors will be OFF. Hence, the output will be logic low. The truth table of NAND logic gate given in below table.

A	B	Pull-Down Network	Pull-up Network	OUTPUT Y
0	0	OFF	ON	1
0	1	OFF	ON	1
1	0	OFF	ON	1
1	1	ON	OFF	0

CMOS NOR Gate

A 2-input NOR gate is shown in the figure below. The NMOS transistors are in parallel to pull the output low when either input is high. The PMOS transistors are in series to pull the output high when both inputs are low, as given in below table. The output is never left floating.



The truth table of NOR logic gate given in below table.

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

NAND and NOR gate can be implemented using CMOS inverter. It consists of NMOS and PMOS two transistors and they have to be proved in a layout such that we get device output according to the logic. Some important conditions to be kept in mind during layout design are

1. N well/ P well spacing =6λ
2. N well/ P well width =6λ
3. Active spacing = 3λ
4. Poly width = 3λ
5. Metal width = 2λ
6. Metal to metal = 2λ
7. Mask = 3λ

Procedure: DSCH software

1. Open DSCH software.
2. Do the connections of circuit in software as per circuit diagram.
3. Check the connections before running it. If not connected, do the connections without any open circuit.
4. Save & Run the software.
5. See the simulated waveforms.
6. Check whether the waveforms follow the desired results
7. Record the waveform obtained.

Procedure: Micro-wind software

1] Start micro-wind software for layout**.**

2] Draw the CMOS inverter following the design.

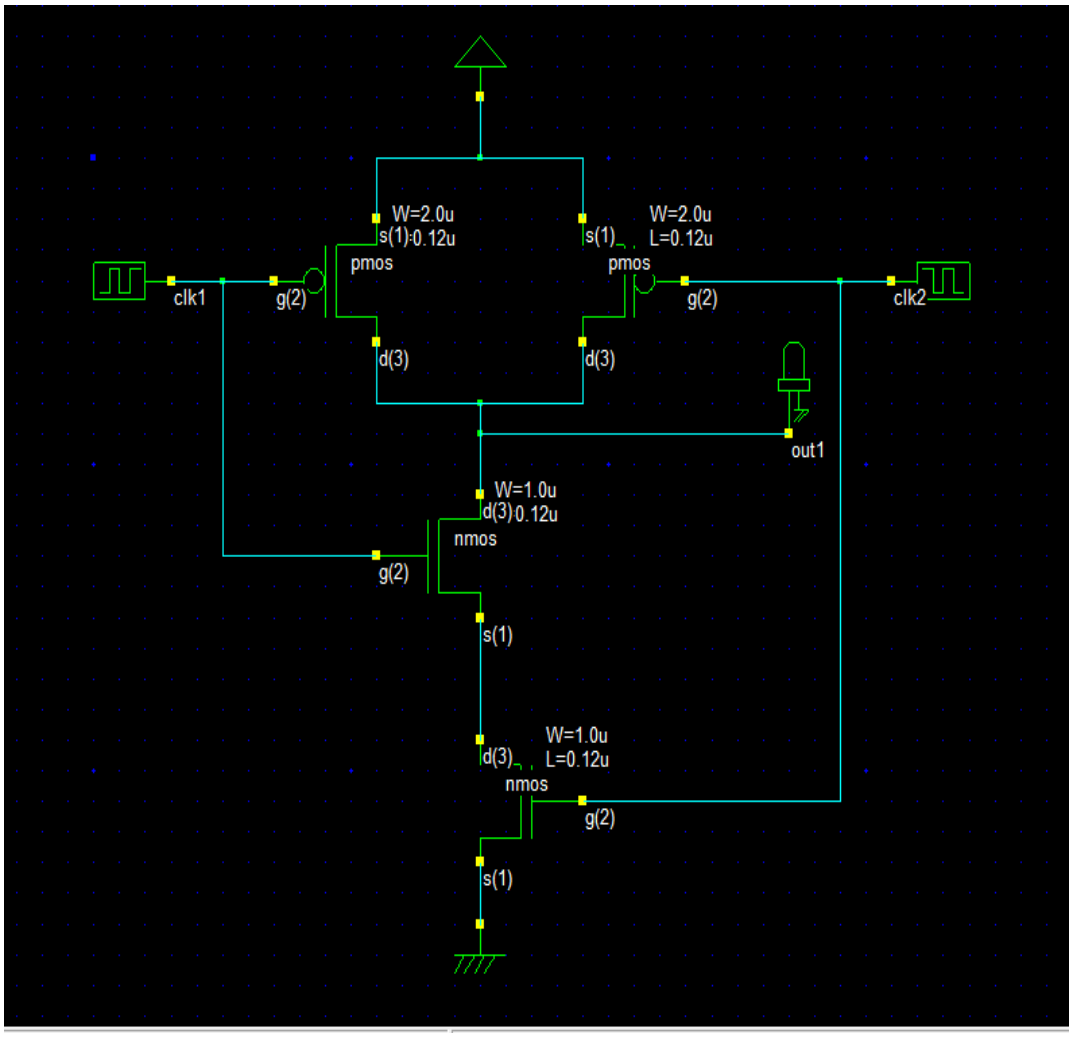
3] Click on DRC at every stage of design to avoid error.

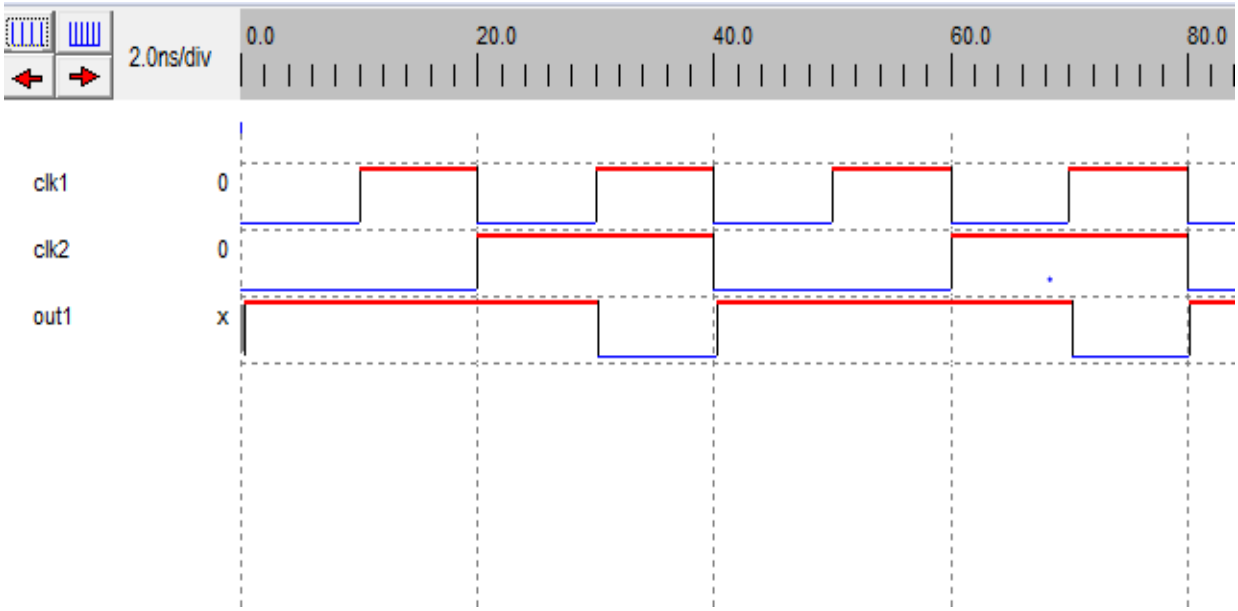
4] Observe various output while using simulation output.

5] Software design and finally run a DRC.

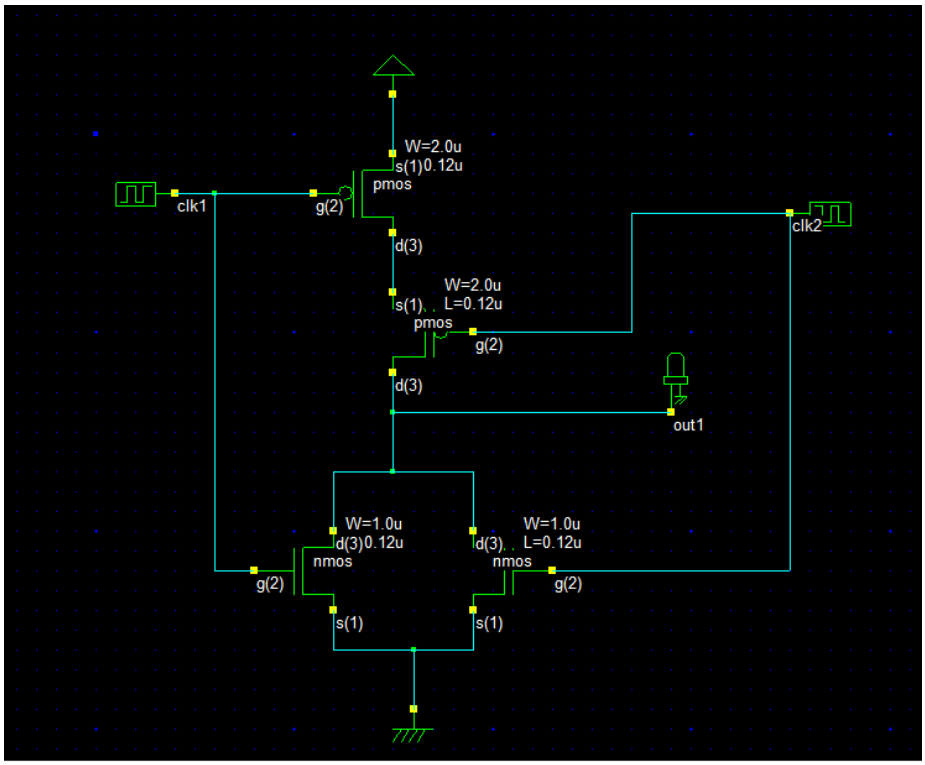
6] Record the output waveform observed

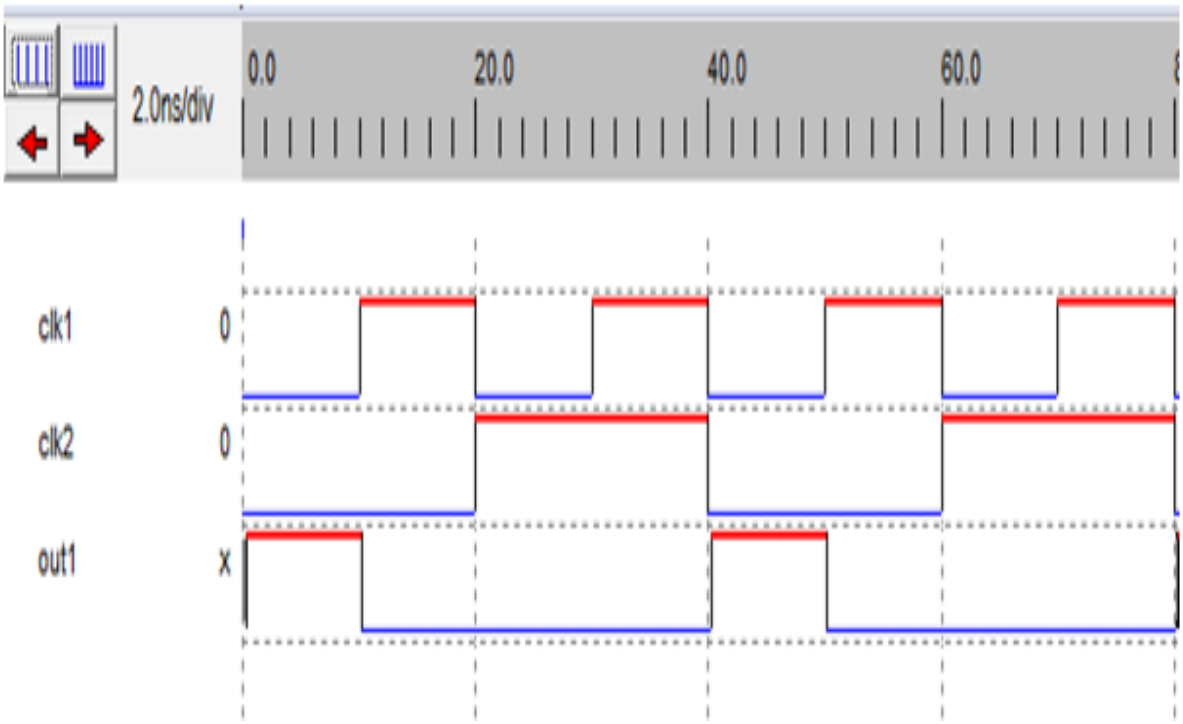
Output (NAND circuit):



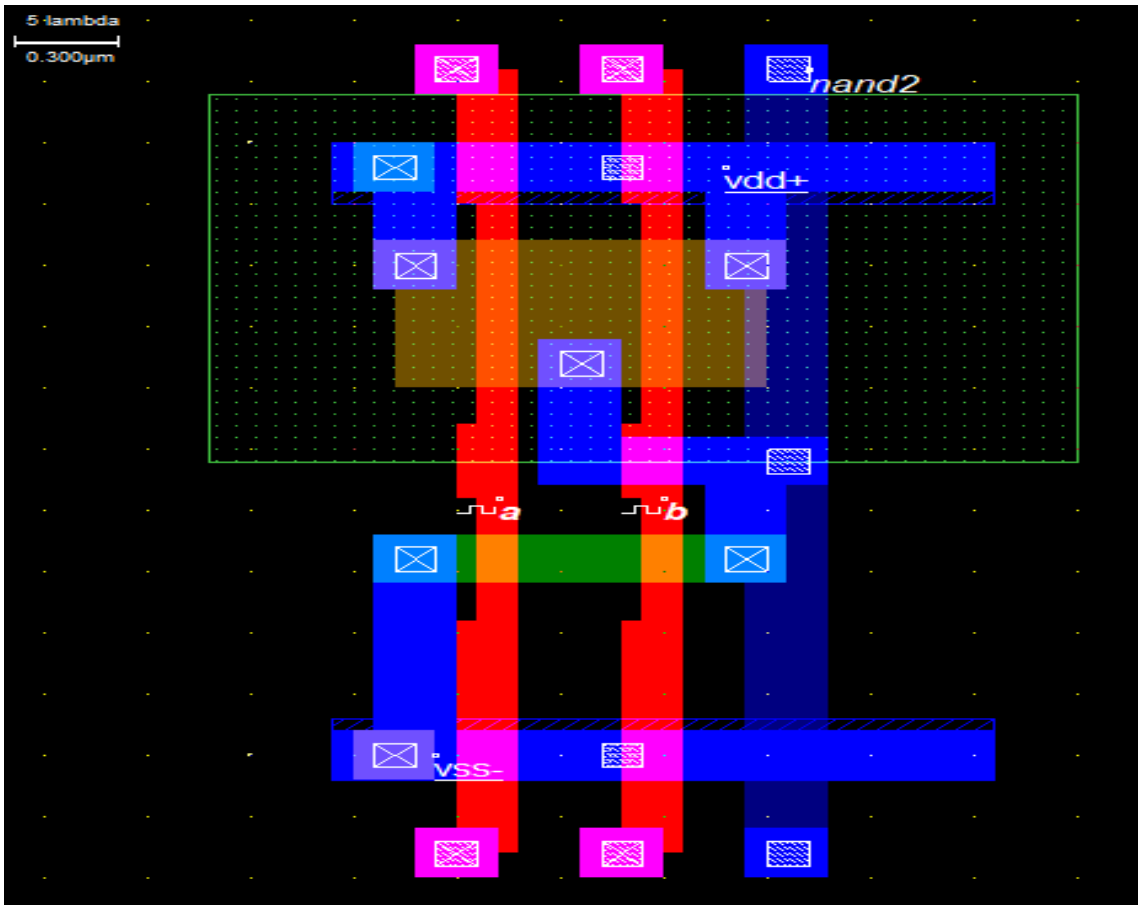


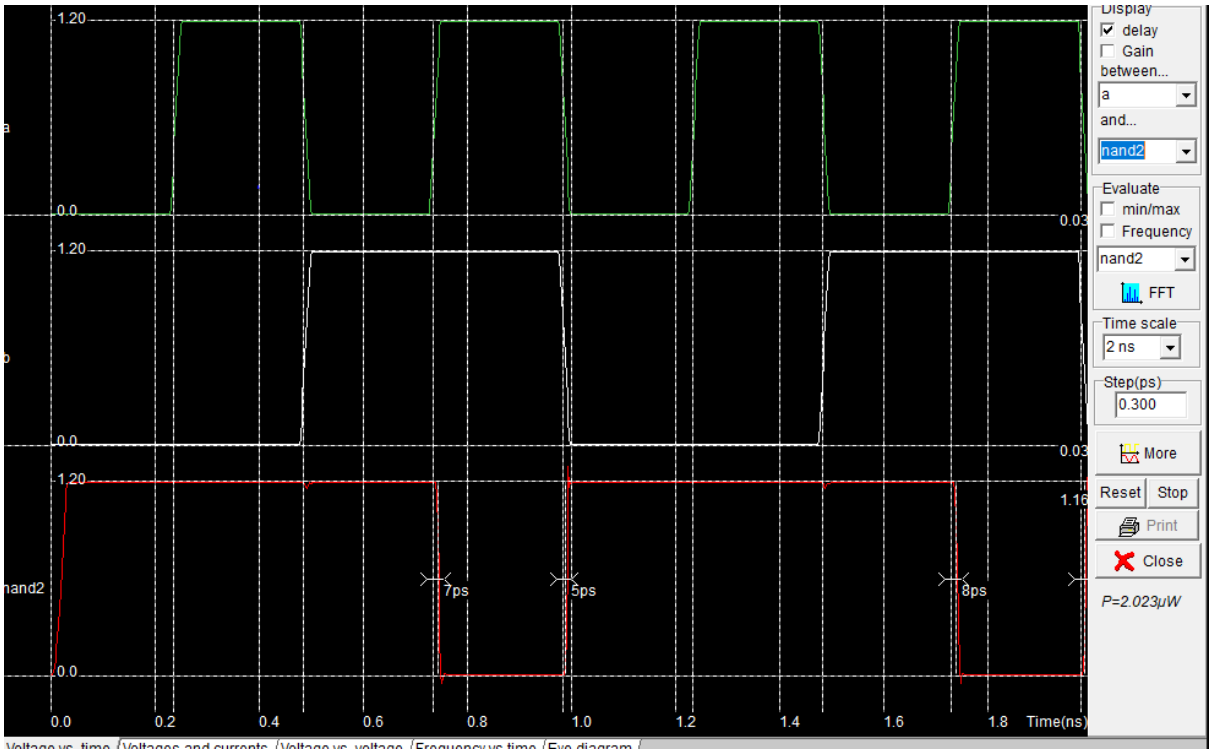
Output (NOR circuit):

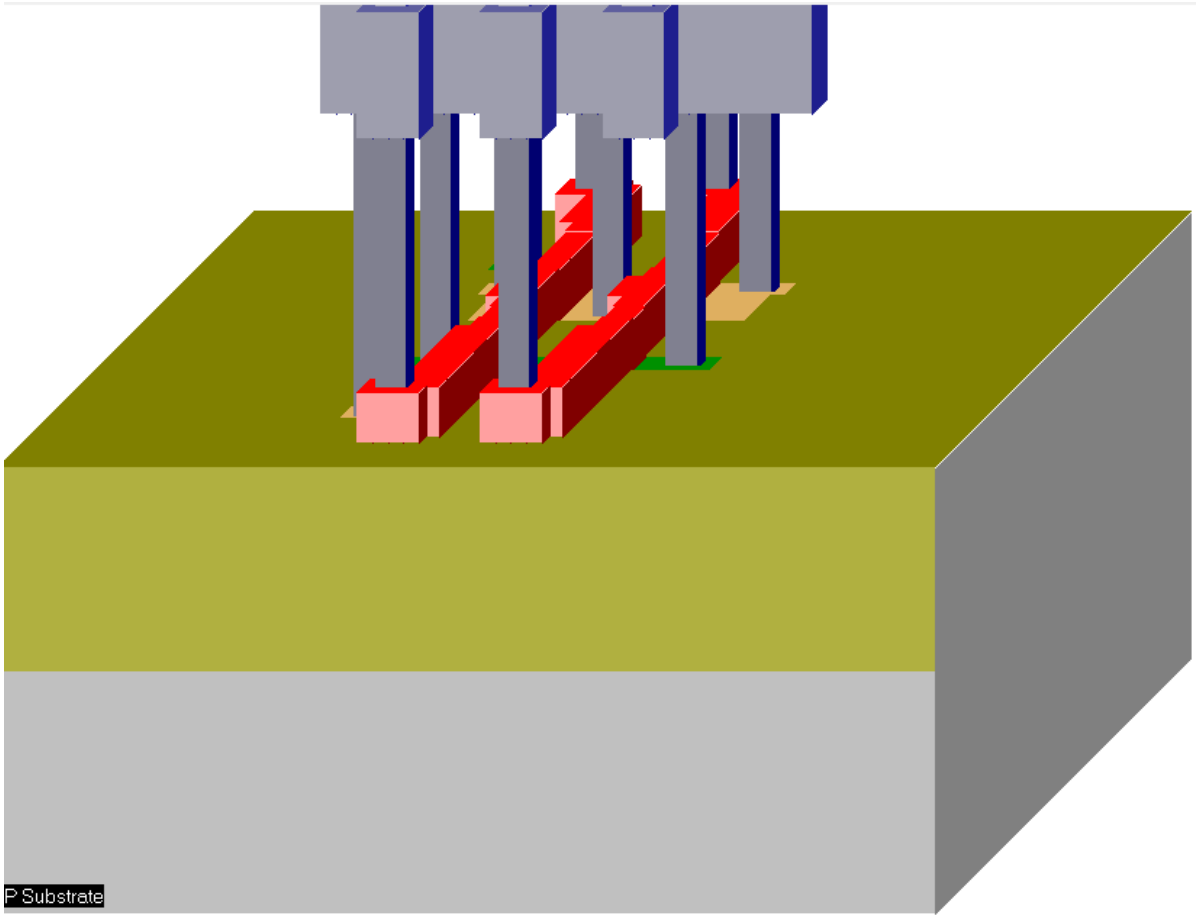




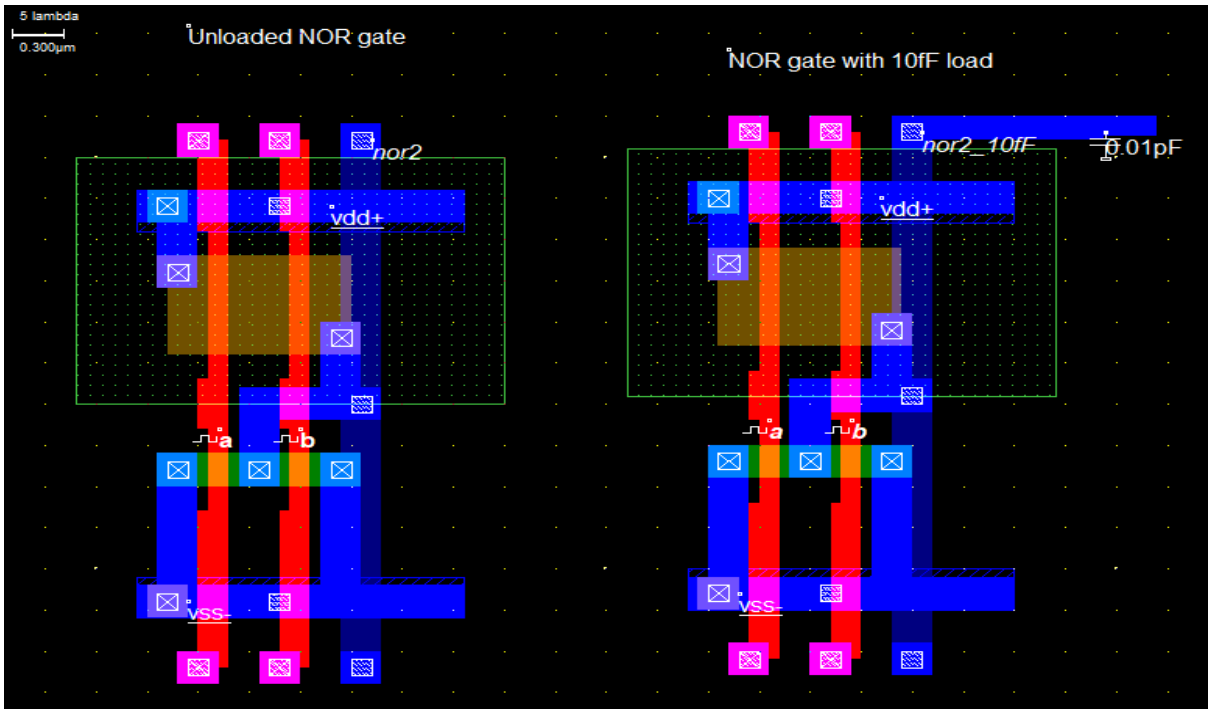
Layout NAND circuit:

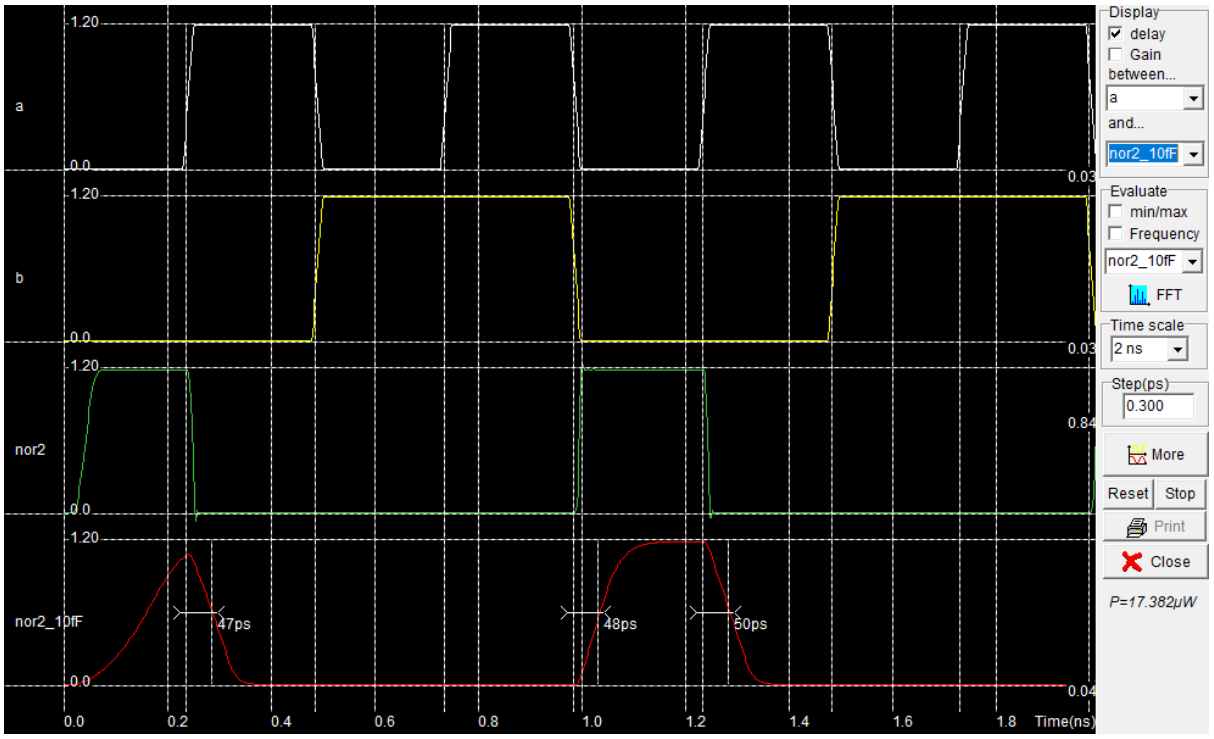


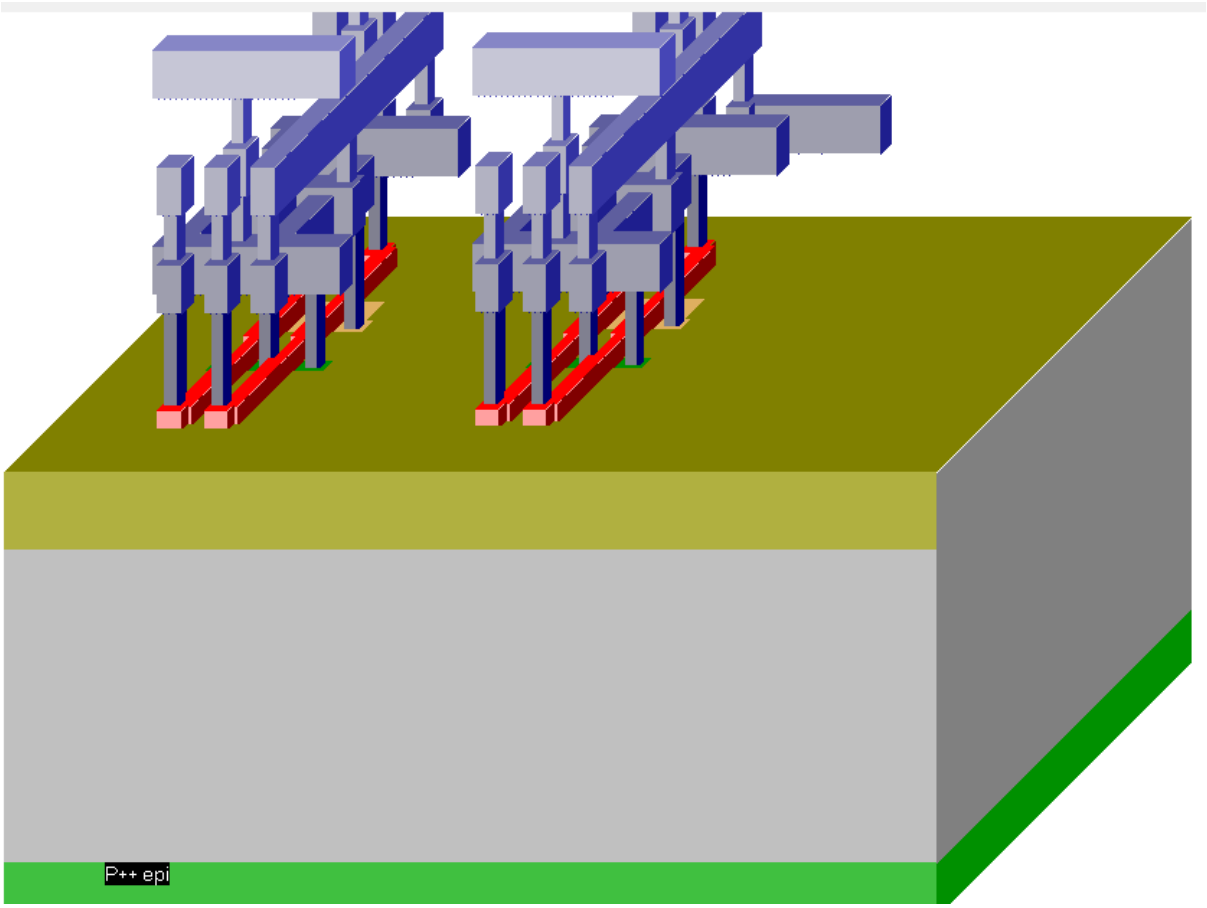




Layout NOR circuit:




a
b
nor2
nor2_10fF
1.20
0.0
47ps
48ps
50ps
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
Time(ns)
Display
delay
Gain
between...
and...
Evaluate
min/max
Frequency
FFT
Time scale
2 ns
Step(ps)
0.300
More
Reset
Stop
Print
Close
P=17.382μW


P++ epi

Pre Lab Questions:

1 Why inputs of a NAND gate are connected together?
2 Why inputs of a NOR gate are connected together?

Post Lab Questions:

1. How the NAND occupy less area than the NOR?

Conclusion: Thus, we have designed the layout of NAND and NOR gates as per the design rules of micro wind and observed the output.

CO's Covered:

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Expt. No. 6

Title: To design layout of SR flip-flop using Micro-wind.

Rubric	*Score (1 to 5)*
Timely Submission	
Result Interpretation and Output	
Program Logic and Concept Understanding	
Total	

Performed On: _____

Sign:_____

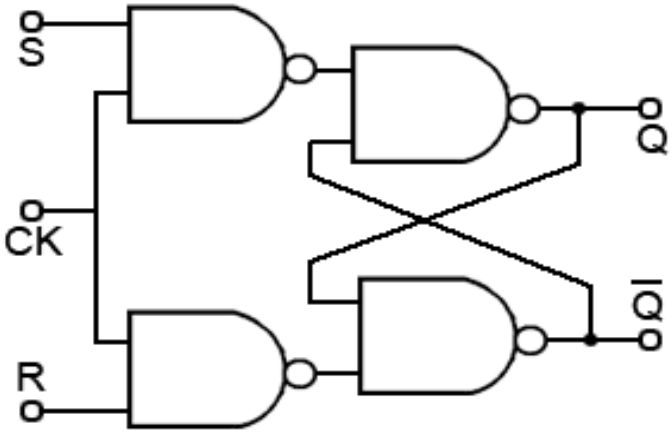
Experiment No. 6

Aim: To design layout of SR flip-flop using DSCH and Micro-wind.

Software Required: DSCH, Micro-wind software

Hardware: computer

Circuit Diagram:



Theory:

SR flip-flop can be implemented using cross NAND structure. SR flip- flop shown in which does not match with synchronous. A hacked version of the latch is shown. It consist of CMOS inverter of transistor to device flip-flop from one state to another private synchronous control is steady state where o the one is slow.

1. N-well/p-well spacing = 6A
2. N-well p-well width = 6A
3. Active spacing = 3A
4. Metal width = 2A
5. Metal to metal = 3A
6. Poly width = 3A
7. Mask = 3A

Procedure:

1] Start micro-wind software for layout**.**

2] Draw the SR flip-flop following the design.

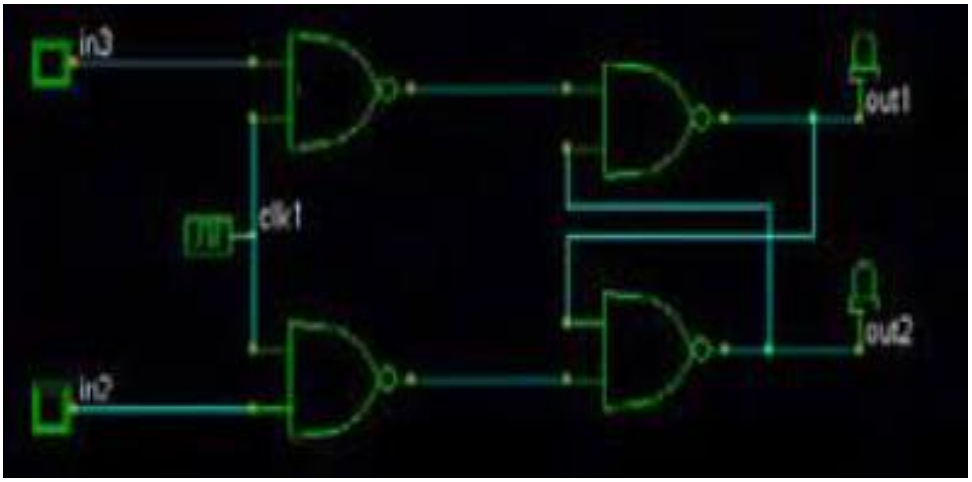
3] Click on DRC at every stage of design to avoid error.

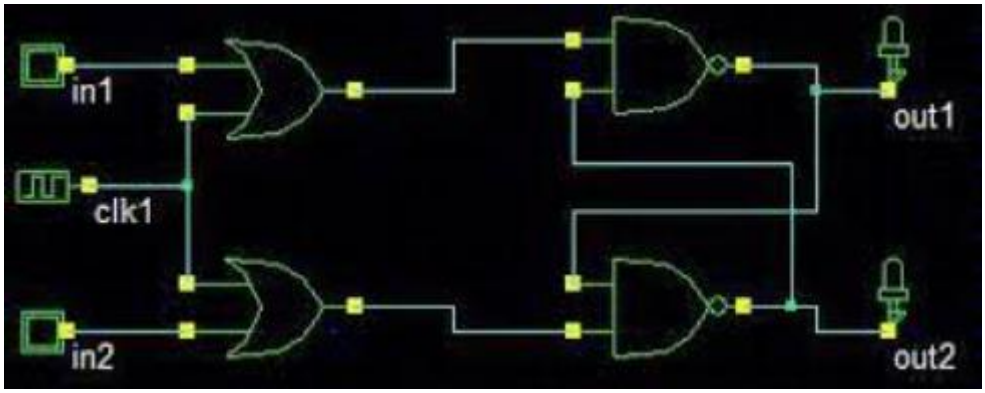
4] Observe various output while using simulation output.

5] Software design and finally run a DRC.

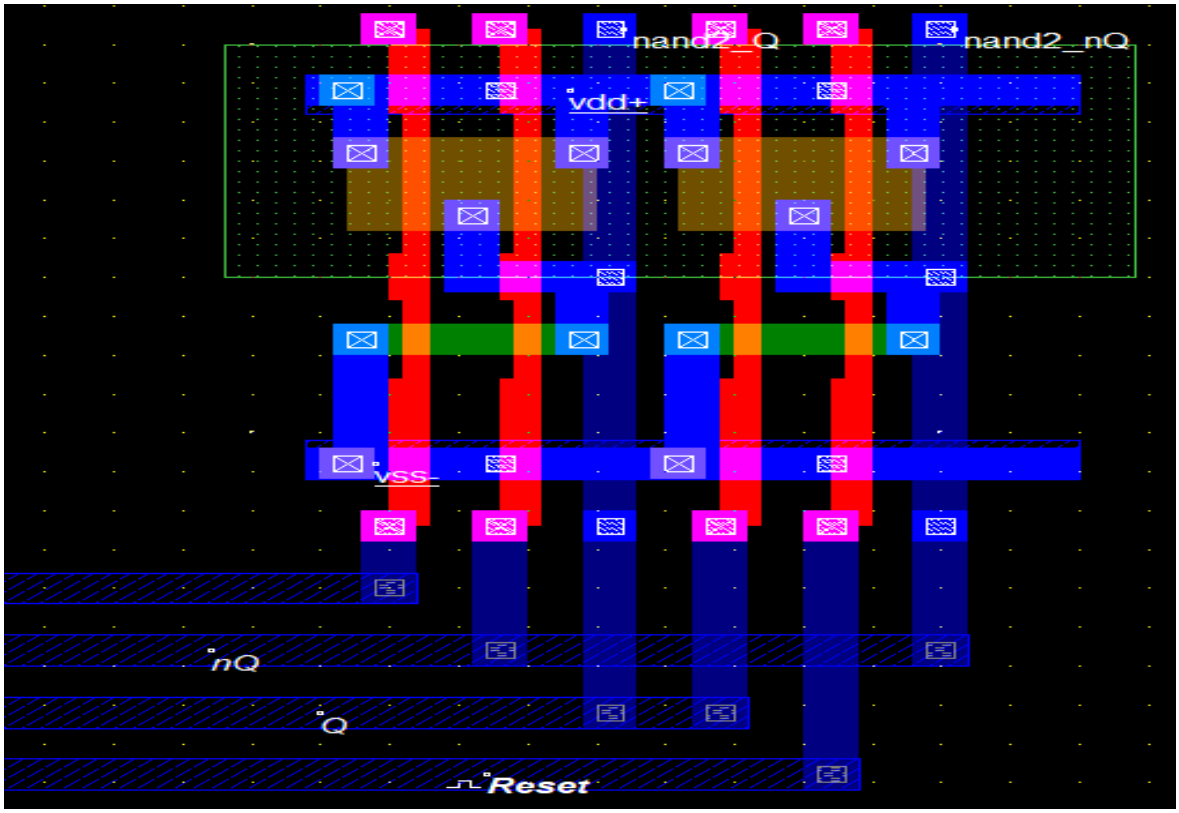
6] Record the output waveform observed.

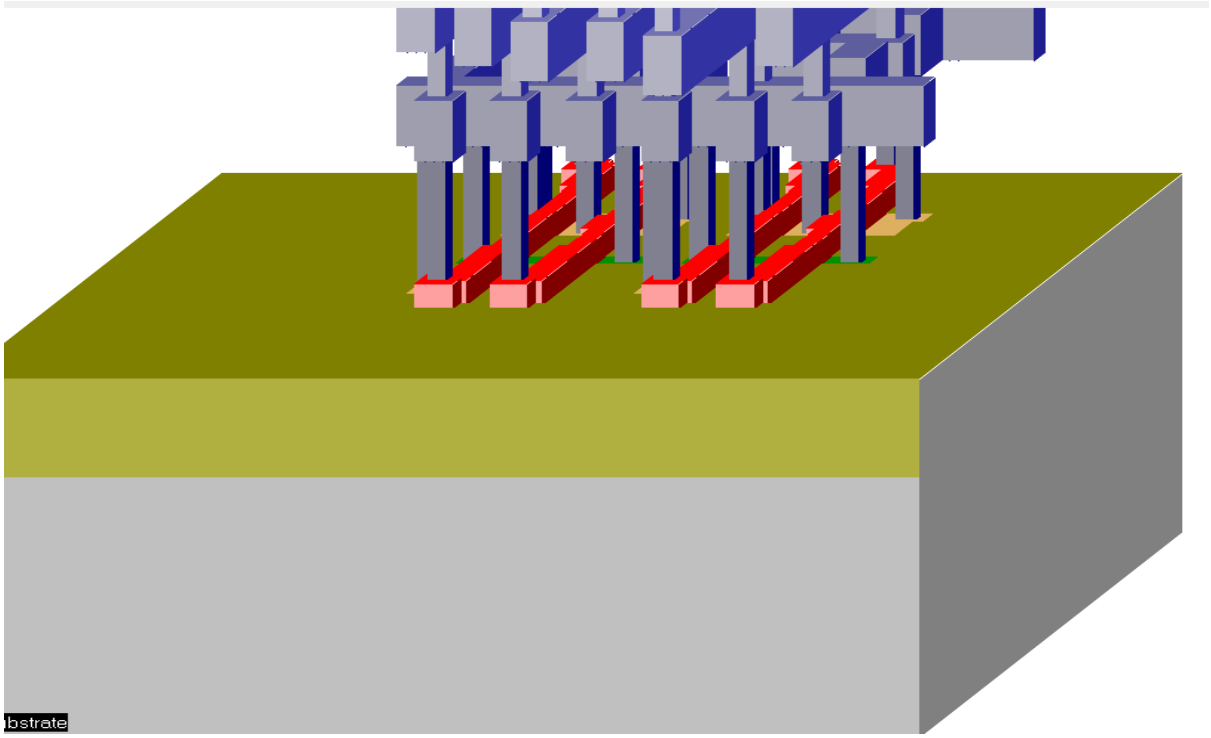
DSCH:


in3
clk1
in2
out1
out2


in1
clk1
in2
out1
out2

Layout:


nand2_Q
nand2_nQ
vdd+
vss-
nQ
Q
Reset



Pre Lab-Questions:

1. Why S-R flip-flop can be constructed by cross-coupling?
2. Why S-R latch composed of NAND gates?

Post Lab Questions:

1 Why Is the implant in PMOS?
2 Why metal 1 and metal 2 layer Indicate?

Conclusion: Thus, we have designed layout of SR flip flop and verified layout design

CO's covered:

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Expt. No. 7

Title: To design the layout of the D latch using micro-wind.

Rubric	*Score (1 to 5)*
Timely Submission	
Result Interpretation and Output	
Program Logic and Concept Understanding	
Total	

Performed On: _____

Sign:_____

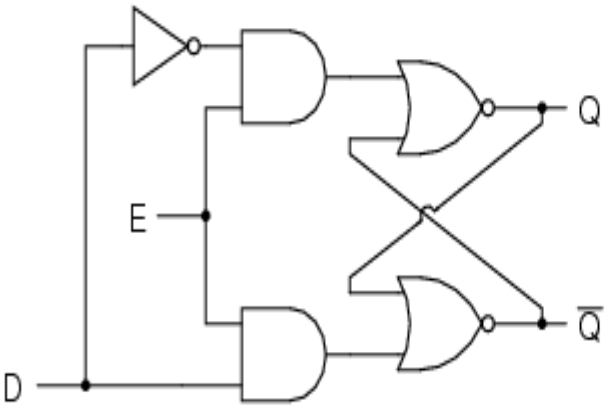
Experiment No. 7

Aim: To design the layout of the D latch using micro-wind.

Software Required: DSCH, Micro-wind software

Hardware: computer

Circuit Diagram:



E	D	Q	$\overline{Q}$
0	0	latch	latch
0	1	latch	latch
1	0	0	1
1	1	1	0

Theory:

Latch is an electronic device that can be used to store one bit of information. The D latch is used to capture, or 'latch' the logic level which is present on the Data line when the clock input is high. If the data on the D line changes state while the clock pulse is high, then the output, Q, follows the input, D. When the CLK input falls to logic 0, the last state of the D input is trapped and held in the latch.

CMOS circuit can be implemented using either logic diagram or
Structured description and direct transition shown A physical layer this can be created by 2 NOR cell adding a custom layout could be probably assume all area enable control can be added to basic

1. N-well/p-well spacing = 6A
2. N-well p-well width = 6A
3. Active spacing = 3A
4. Metal width = 2A
5. Metal to metal = 3A
6. Poly width = 3A
7. Mask = 3A

Procedure:

1] Start micro-wind software for layout.

2] Draw the D Latch following the design.

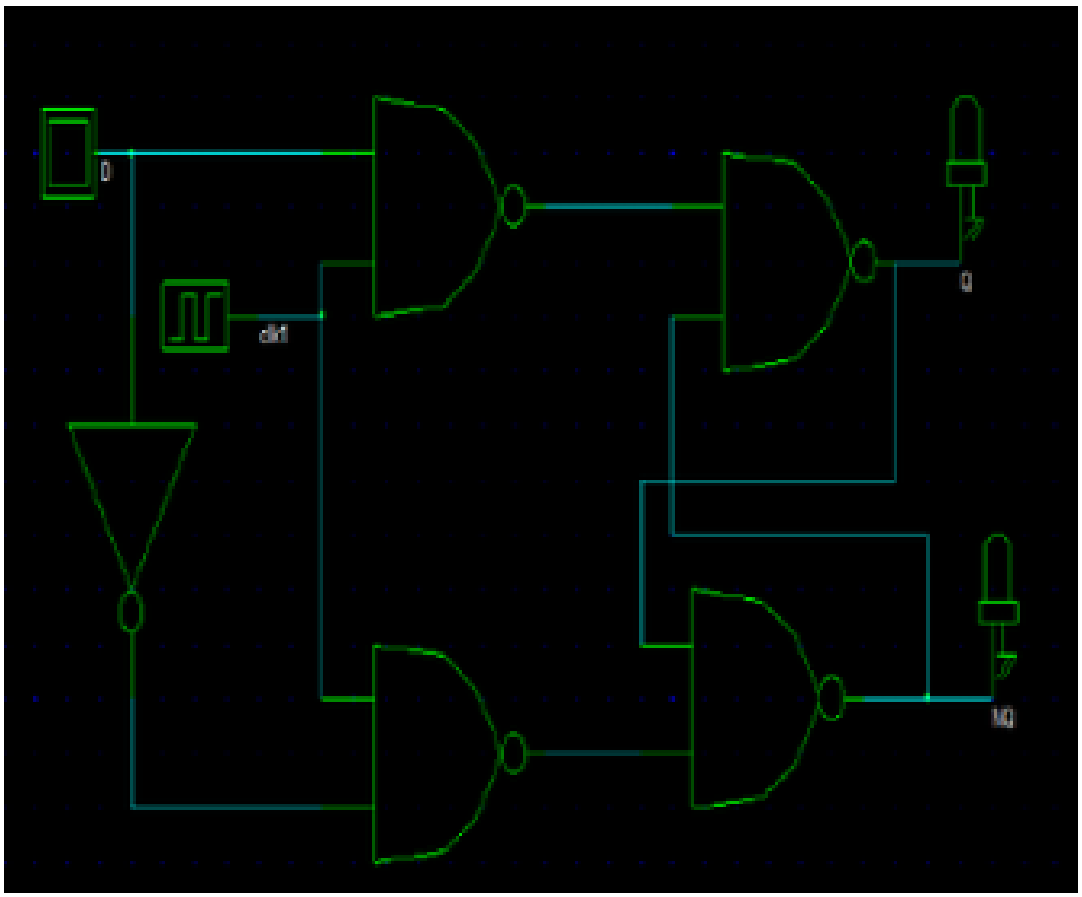
3] Click on DRC at every stage of design to avoid error.

4] Observe various output while using simulation output.

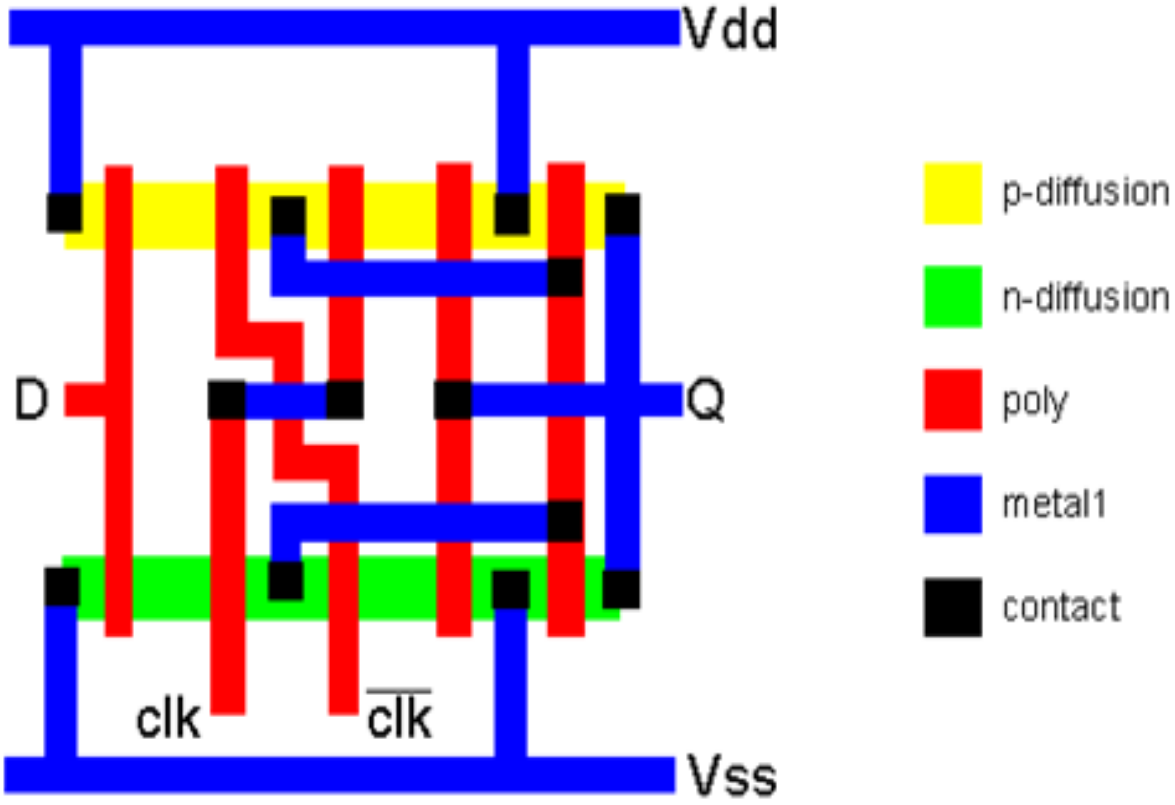
5] Software design and finally run a DRC.

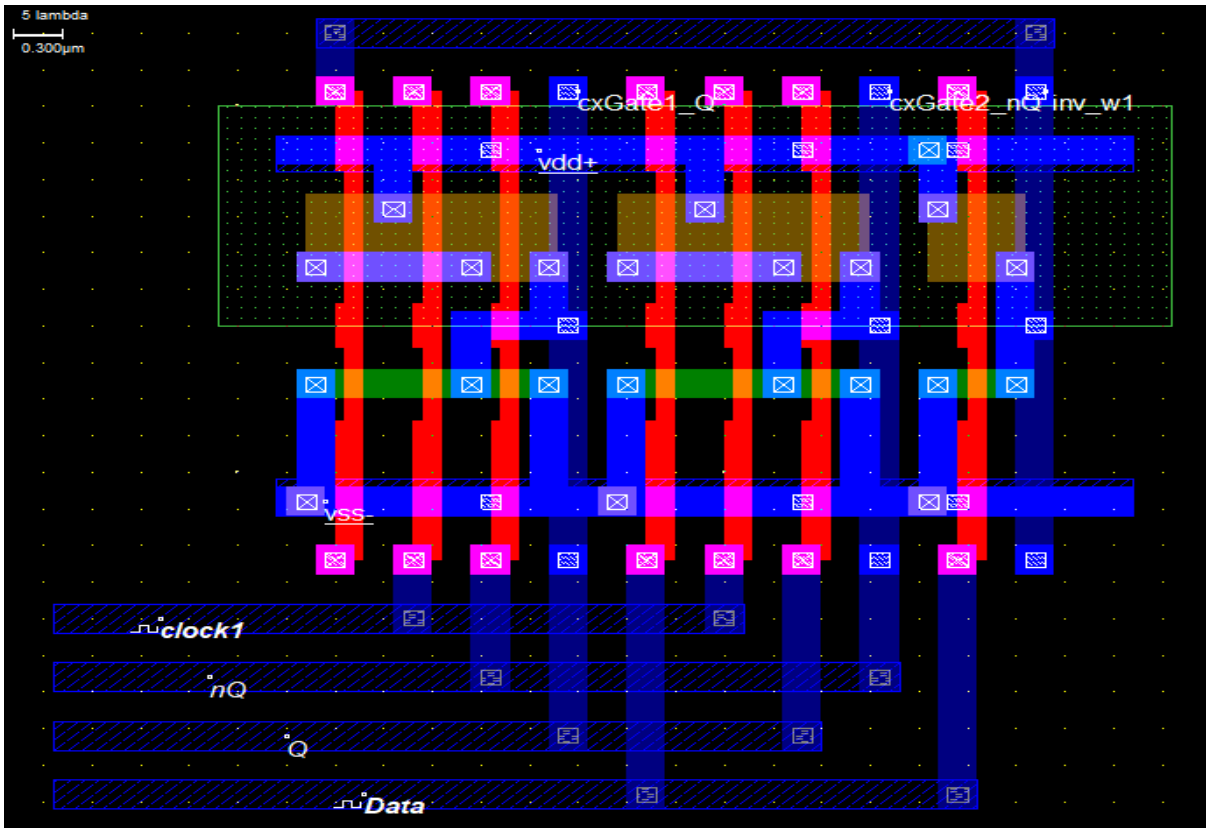
6] Record the output waveform observed.

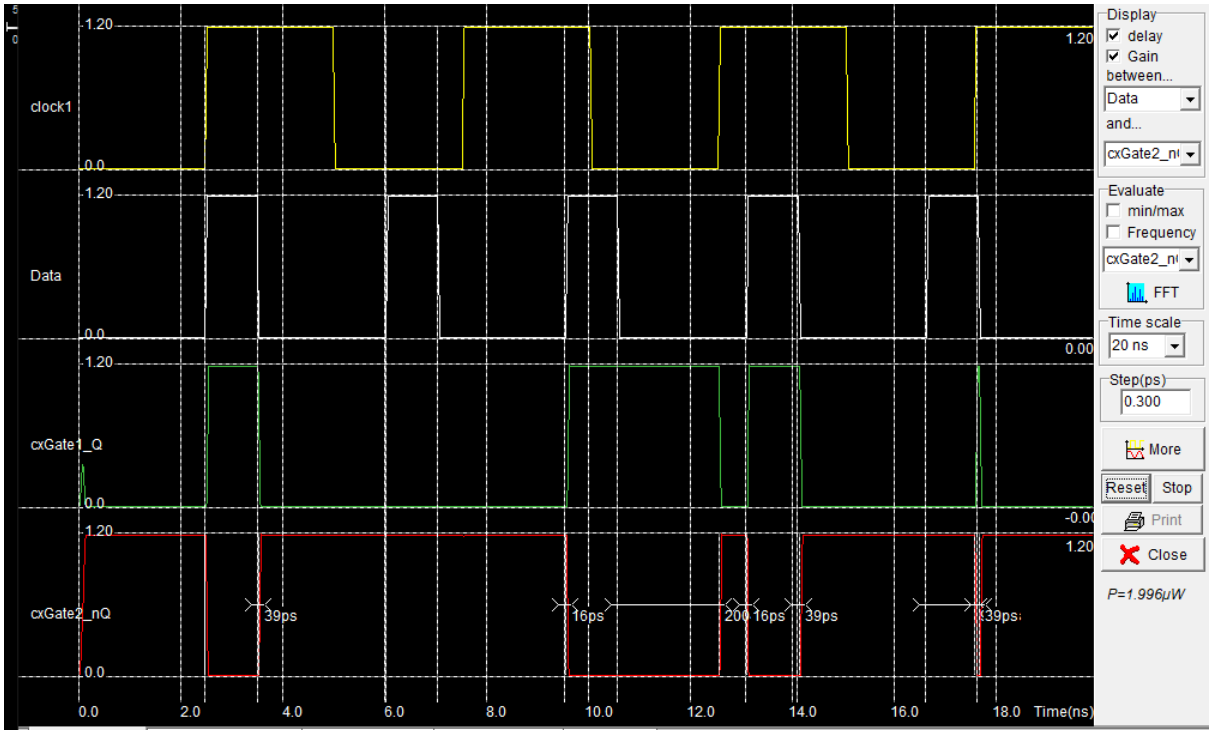
DSCH:

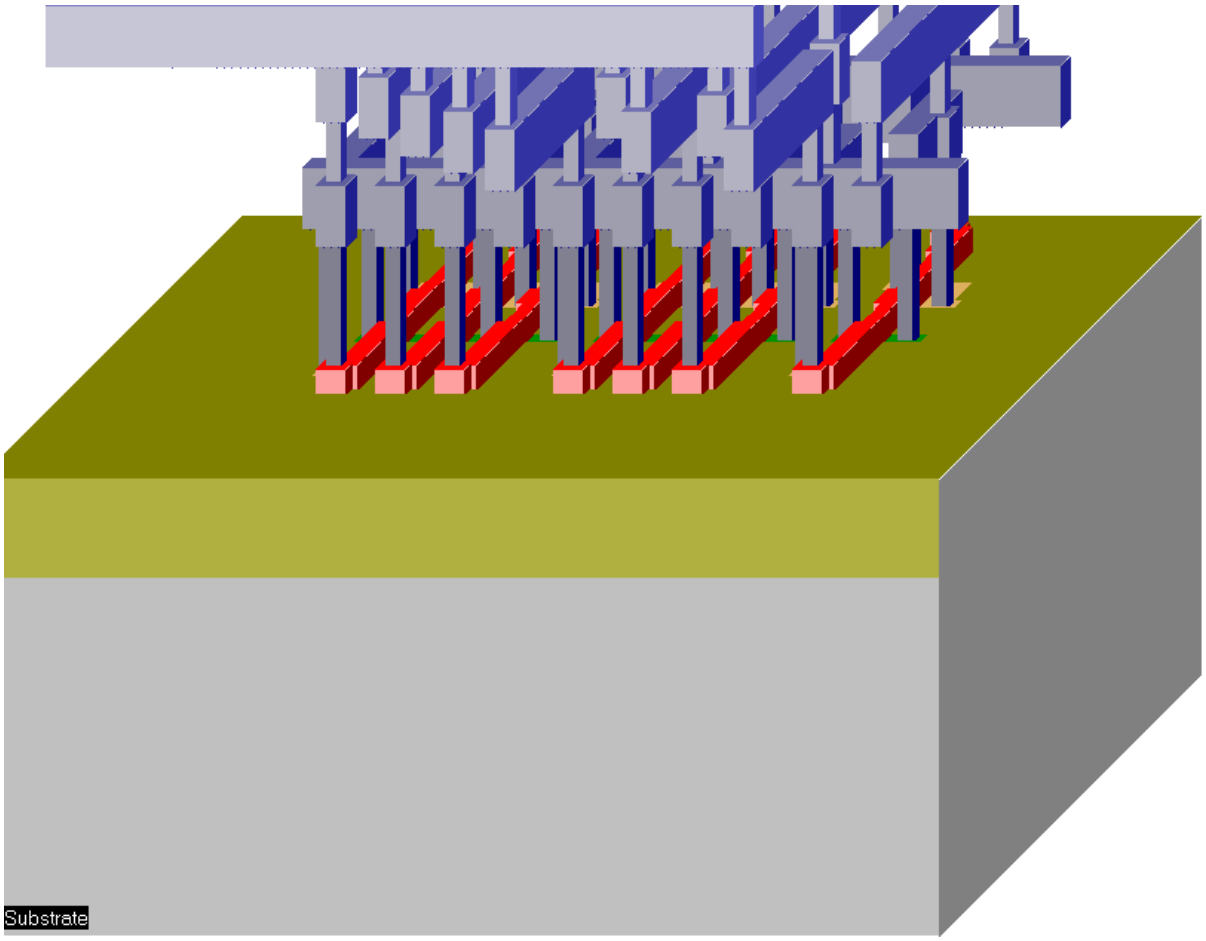


Layout:


Vdd
D
Q
clk
clk
Vss
p-diffusion
n-diffusion
poly
metal1
contact


5 lambda
0.300µm
cxGate1_Q
cxGate2_nQ
inv_w1
vdd
vss
clock1
nQ
Q
Data


clock1
Data
cxGate1_Q
cxGate2_nQ
1.20
0.0
39ps
16ps
200
16ps
39ps
39ps
0.0
2.0
4.0
6.0
8.0
10.0
12.0
14.0
16.0
18.0
Time(ns)
Display
delay
Gain
between...
Data
and...
cxGate2_nQ
Evaluate
min/max
Frequency
FFT
Time scale
20 ns
Step(ps)
0.300
More
Reset
Stop
Print
Close
P=1.996µW


Substrate

Pre Lab Questions:

1. Which flip-flop plays a vital role by functioning as the basic building block of a ripple counter?
2. Is D Flip flop is level Triggered?

Post Lab Questions:

3. How to construct D Flip flop?

Conclusion: Thus, we have designed layout of D Latch using Micro wind and verified layout using design rules.

CO's covered:

……
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Expt. No. 8

Title: To design the layout of 6T SRAM using micro-wind.

Rubric	*Score (1 to 5)*
Timely Submission	
Result Interpretation and Output	
Program Logic and Concept Understanding	
Total	

Performed On: _____

Sign:_____

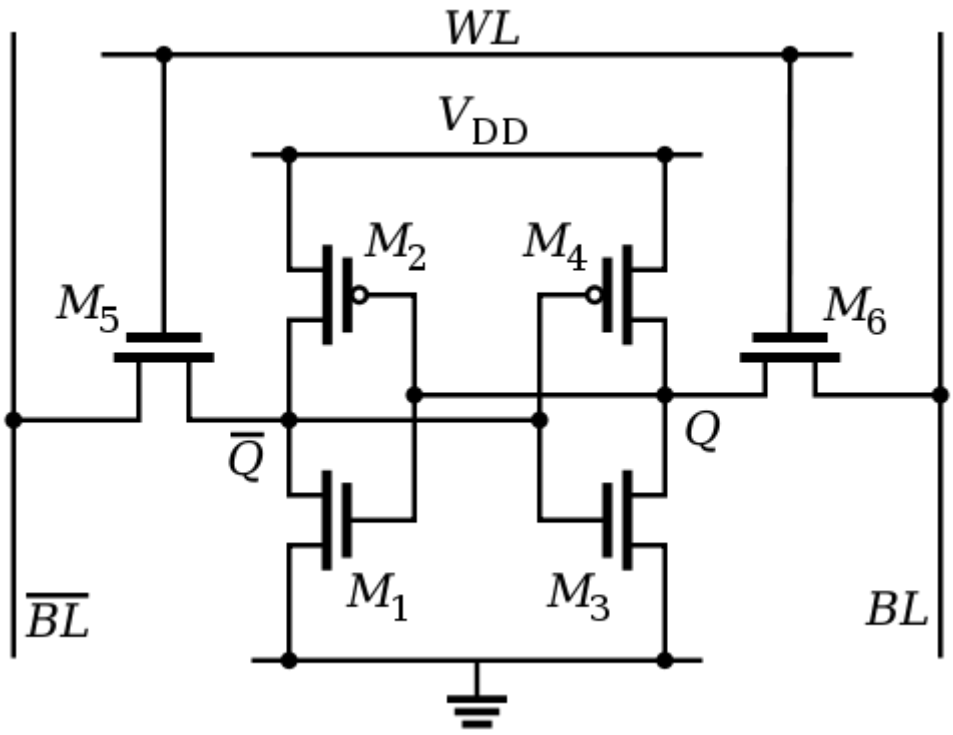
Experiment No. 8

Aim: To design the layout of 6T SRAM using micro-wind.

Software Required: Micro-wind software

Hardware: computer

Circuit Diagram:



Theory:

The memory cell shown here forms the basis for most static random-access memories in CMOS technology. It uses six transistors to store and access one bit. The four transistors in the center form two cross-coupled inverters. In actual devices, these transistors are made as small as possible to save chip-area, and are very weak. Due to the feedback structure, a low input value on the first inverter will generate a high value on the second inverter, which amplifies (and stores) the low value on the second inverter. Similarly, a high input value on the first inverter will generate a low input value on the second inverter, which feeds back the low input value onto the first inverter. Therefore, the two inverters will store their current logical value, whatever value that is.

The two lines between the inverters are connected to two separate bit-lines via two n-channel pass-transistors (left and right of the cell). The gates of those transistors are driven

by a word-line. In a larger SRAM, the word-line is used to address and enable all bits of one memory word (e.g. all 32 bits at address 0xcafe from a 64Kx32 SRAM chip). As long as the word-line is kept low, the SRAM cell is disconnected from the bit-lines. The inverters keep feeding themselves, and the SRAM stores its current value.

1) NWELL/PWELL width =6lambda
2) NWELL/PWELL spacing=6lambda
3) Active spacing =2lambda
4) Poly-width =2lambda
5) Poly-spacing=2lambda
6) Metal width =3lambda
7) Metal to metal width =3lambda
8) Mask=2lambda*2lambda

Procedure:

1] Start micro-wind software for layout**.**

2] Draw the 6T SRAM following the design.

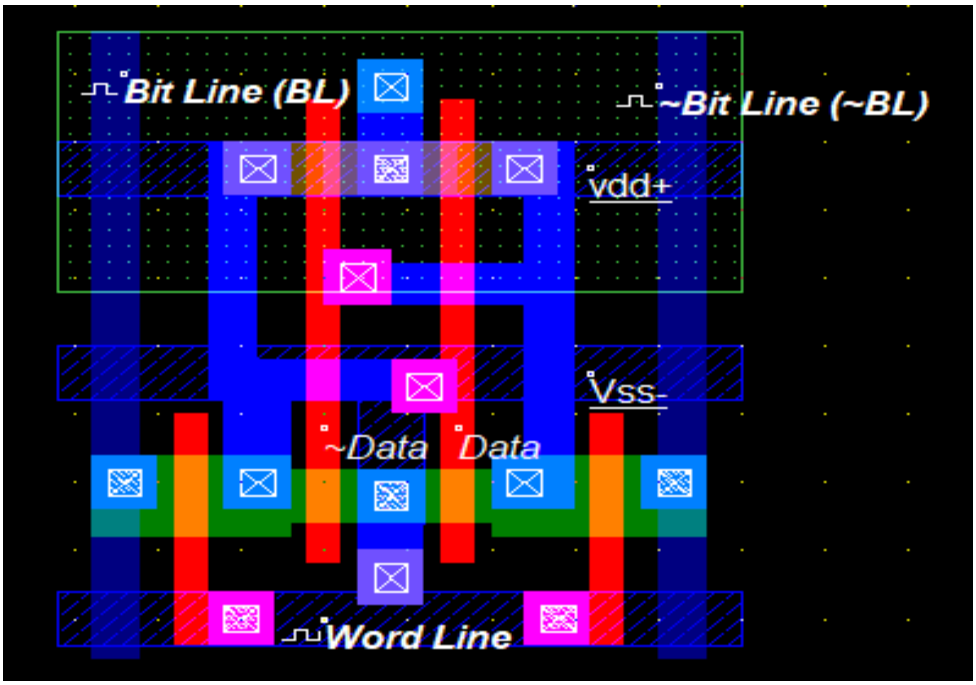
3] Click on DRC at every stage of design to avoid error.

4] Observe various output while using simulation output.

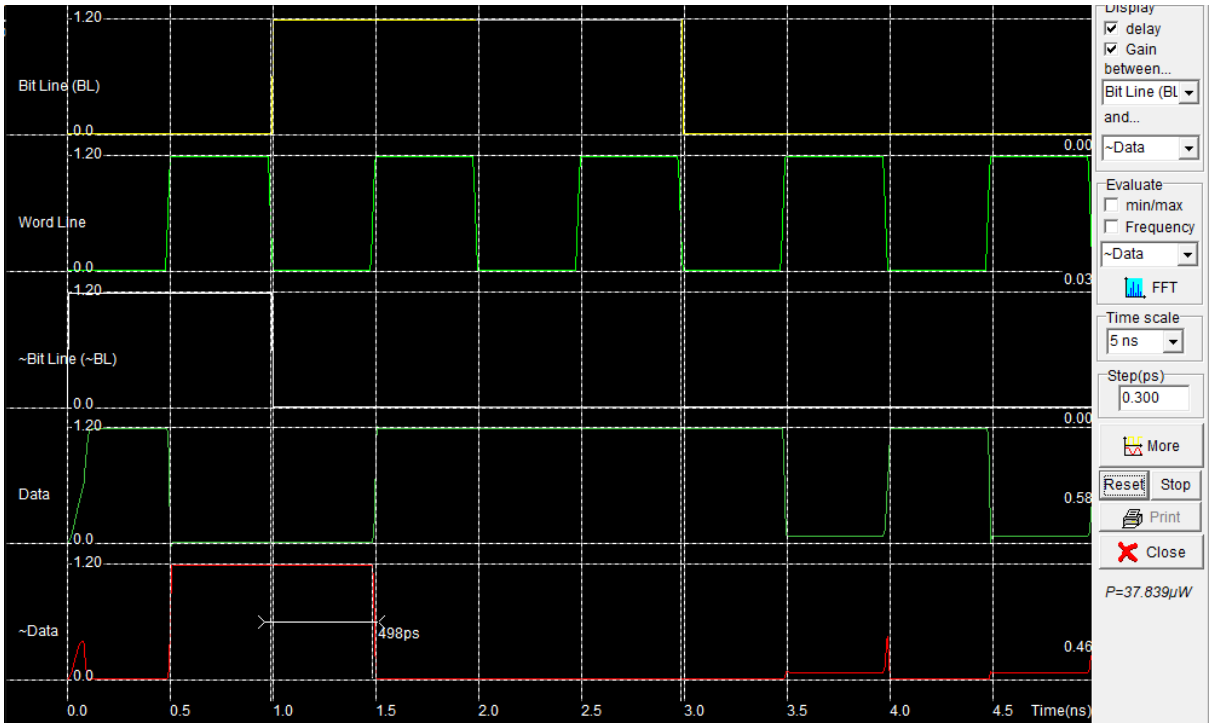
5] Software design and finally run a DRC.

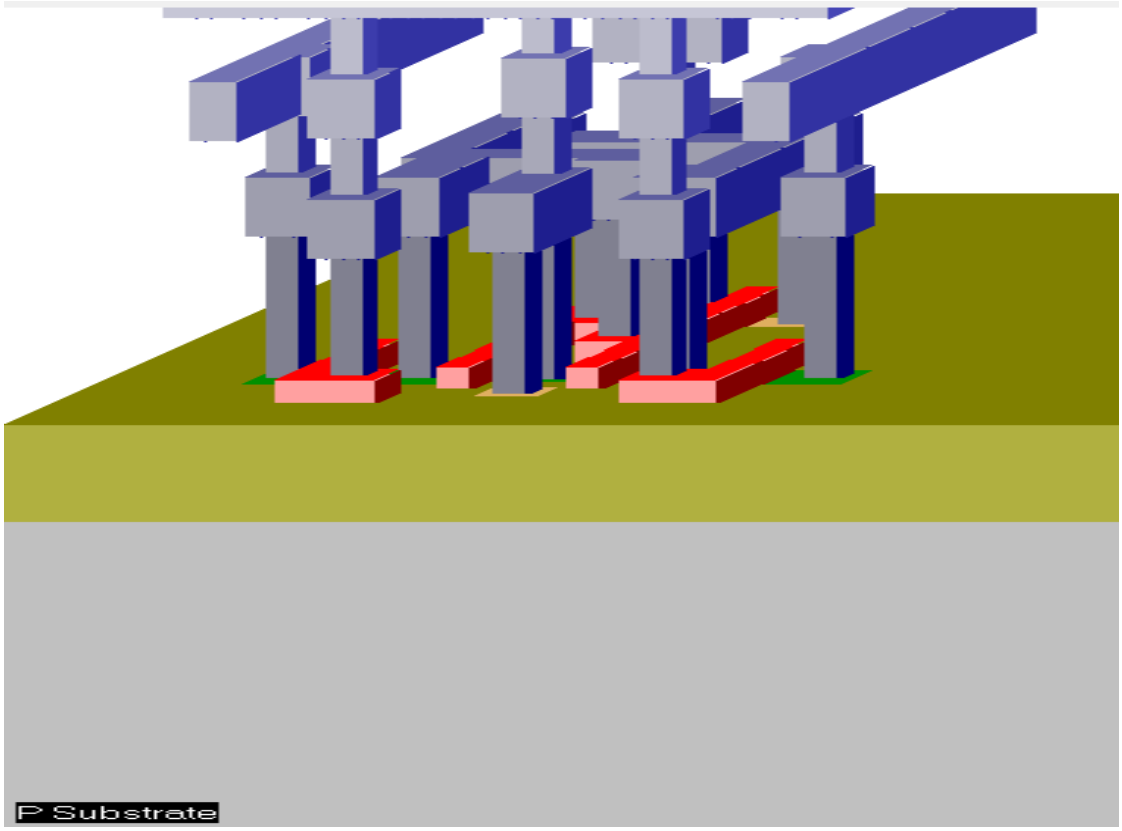
6] Record the output waveform observed.

Layout:



Output:


Bit Line (BL)
Word Line
~Bit Line (~BL)
Data
~Data
1.20
0.0
498ps
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
Time(ns)
Display
delay
Gain
between...
Bit Line (BL
and...
~Data
Evaluate
min/max
Frequency
~Data
FFT
Time scale
5 ns
Step(ps)
0.300
More
Reset
Stop
Print
Close
P=37.839µW


P Substrate

Pre Lab-Questions:

1 Describe the various effects of scaling in 6T SRAM**?**
2 What are the limitations in increasing the power supply to reduce delay?

Post Lab Questions:

3 What's the critical path in a SRAM?
4 How does the size of PMOS Pull Up transistors (for bit & bit- lines) affect SRAM's performance?

Conclusion: Thus, we have designed the layout of the 6T SRAM using micro-wind

CO's covered:

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Expt. No. 9

Title: To design the layout of Ripple Carry Adder using DSCH and Micro-wind.

Rubric	*Score (1 to 5)*
Timely Submission	
Result Interpretation and Output	
Program Logic and Concept Understanding	
Total	

Performed On: _____

Sign:_____

Experiment No. 9

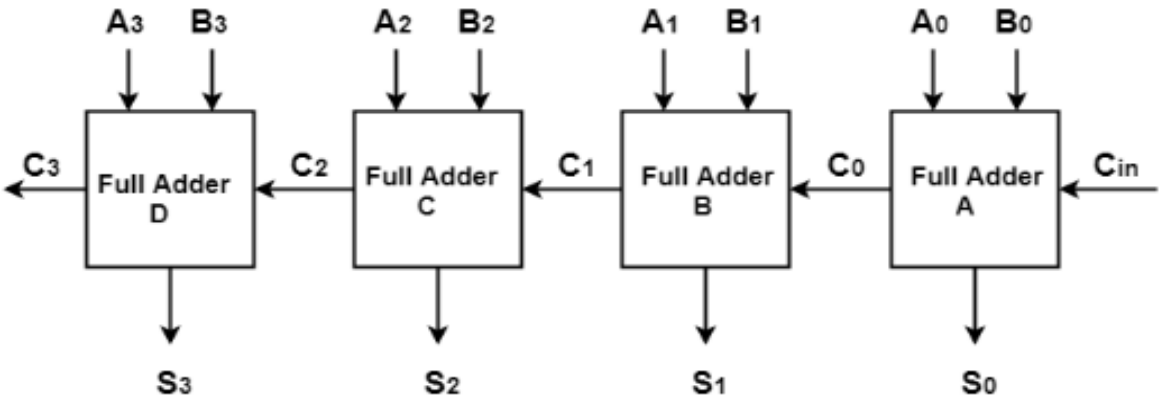
Aim: To design the layout of Ripple Carry Adder using DSCH and Micro-wind

.

Software Required: DSCH and Micro-wind software

Hardware: computer

Circuit Diagram:



Logic Diagram for 4-bit Ripple Carry Adder

Theory:

A ripple carry adder also known as "n-bit parallel adder" is a combinational logic circuit used for the purpose of adding two n-bit binary numbers and requires 'n' full adders in the circuit.4-bit ripple carry adder is used for the purpose of adding two 4-bit binary numbers.

In mathematics, any two 4-bit binary numbers $A_3A_2A_1A_0$ and $B_3B_2B_1B_0$ will be added as- As shown, Ripple Carry Adder works in different stages where the carry out produced by each full adder as output serves as the carry in input for its adjacent most significant full adder. When the carry in becomes available to the full adder, it activates that full adder and it comes into operation.

In Ripple Carry Adder, the carry out produced by each full adder as output serves as the carry in input for its next most significant full adder. Since in ripple carry adder, each carry bit ripples or waves into the next stage, that's why it is called by the name

Calculations:

Working of a 4-bit Ripple Carry Adder-

Suppose we want to add two 4 bit binary numbers 0101 ($A_3A_2A_1A_0$) and 1010 ($B_3B_2B_1B_0$). Using ripple carry adder, this addition will be carried out as explained below-

Stage-01:

When C_{in} will be fed as input to full Adder A, it will activate full adder A.

Then, At Full Adder A,

- $A_0 = 1$
- $B_0 = 0$
- $C_{in} = 0$

The sum bit and carry bit produced as output by full adder A will be calculated by full adder A as-

Calculating S_0–

$S_0 = A_0 \oplus B_0 \oplus C_{in}$

$= 1 \oplus 0 \oplus 0$

$= 1$

$$\therefore S_0 = 1$$

Calculating C_0–

$C_0 = A_0B_0 \oplus B_0C_{in} \oplus C_{in}A_0$

$= 1.0 \oplus 0.0 \oplus 0.1$

$= 0 \oplus 0 \oplus 0$

$= 0$

$$\therefore C_0 = 0$$

Stage-02:

Now, when C_o will be fed as input to full adder B by full adder A, it will activate full adder B.

Then, At Full Adder B,

- $A_1 = 0$
- $B_1 = 1$
- $C_0 = 0$

The sum bit and carry bit produced as output by full adder B will be calculated by full adder B as-

Calculating S_1–

$S_1 = A_1 \oplus B_1 \oplus C_0$

$= 0 \oplus 1 \oplus 0$

$= 1$

$$\therefore S_1 = 1$$

Calculating C_1–

$C_1 = A_1B_1 \oplus B_1C_0 \oplus C_0A_1$

$= 0.1 \oplus 1.0 \oplus 0.0$

$= 0 \oplus 0 \oplus 0$

$= 0$

$$\therefore C_1 = 0$$

Stage-03:

Now, when C_1 will be fed as input to full adder C by full adder B, it will activate full adder C.

Then, At Full Adder C,

- $A_2 = 1$
- $B_2 = 0$
- $C_1 = 0$

The sum bit and carry bit produced as output by full adder C will be calculated by full adder C as-

Calculating S_2–

$S_2 = A_2 \oplus B_2 \oplus C_1$

$= 1 \oplus 0 \oplus 0$

$= 1$

$$\therefore S_2 = 1$$

Calculating C_2–

$C_2 = A_2B_2 \oplus B_2C_1 \oplus C_1A_2$

$= 1.0 \oplus 0.0 \oplus 0.1$

$= 0 \oplus 0 \oplus 0$

$= 0$

$$\therefore C_2 = 0$$

Stage-04:

Now, when C_2 will be fed as input to full adder D by full adder C, it will activate full adder D.

Then, At Full Adder D,

- $A_3 = 0$
- $B_3 = 1$
- $C_2 = 0$

The sum bit and carry bit produced as output by full adder D will be calculated by full adder D as-

Calculating S_3–

$S_3 = A_3 \oplus B_3 \oplus C_2$

$= 0 \oplus 1 \oplus 0$

$= 1$

$$\therefore S_3 = 1$$

Calculating C_3–

$C_3 = A_3B_3 \oplus B_3C_2 \oplus C_2A_3$

$= 0.1 \oplus 1.0 \oplus 0.0$

$= 0 \oplus 0 \oplus 0$

$= 0$

$$\therefore C_3 = 0$$

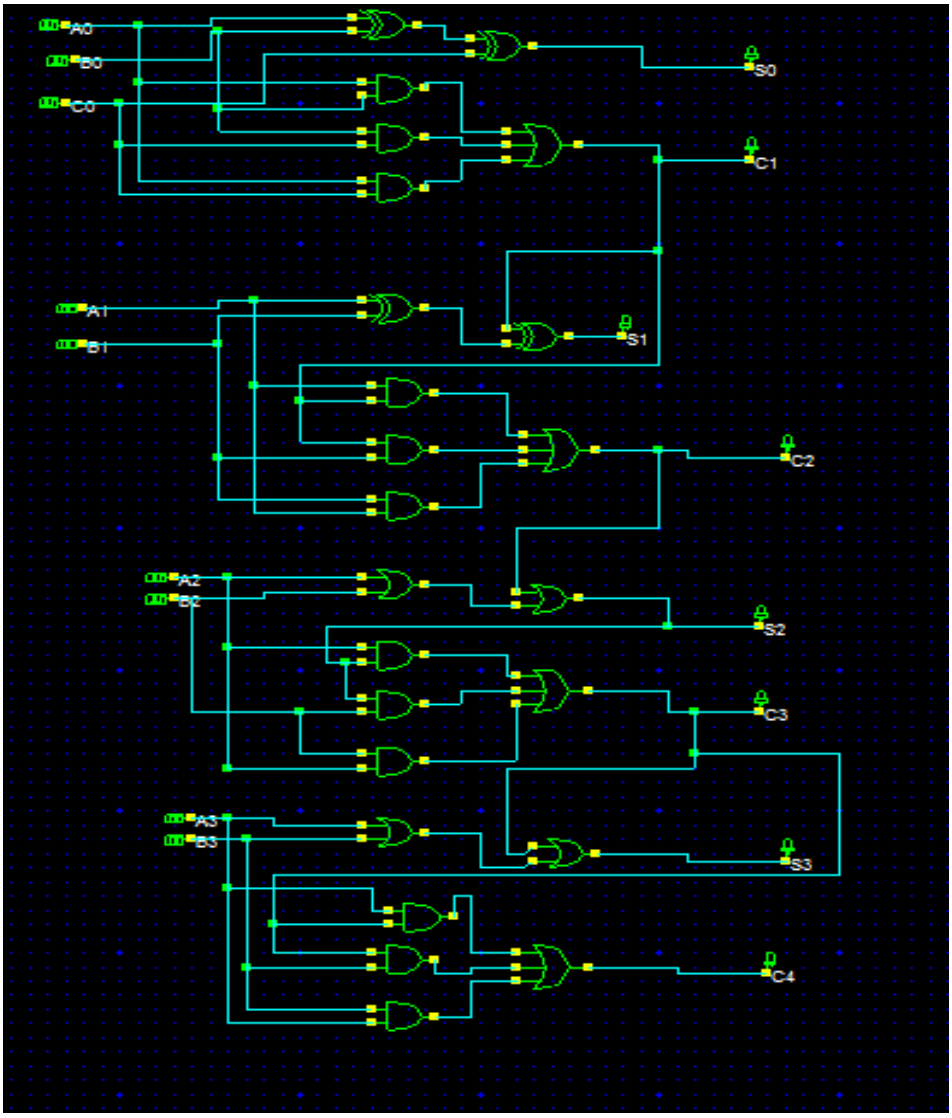
Thus finally,

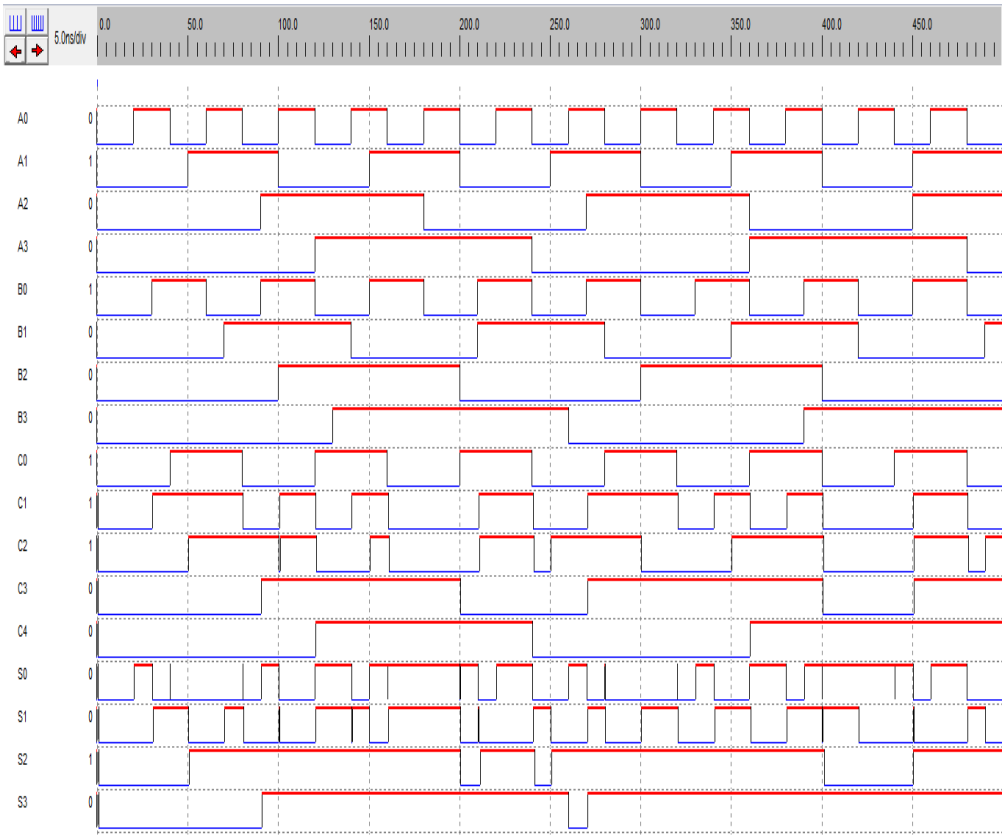
- Output Sum = $S_3S_2S_1S_0 = 1111$
  Output Carry = $C_3 = 0$ "Ripple Carry Adder".

Procedure:

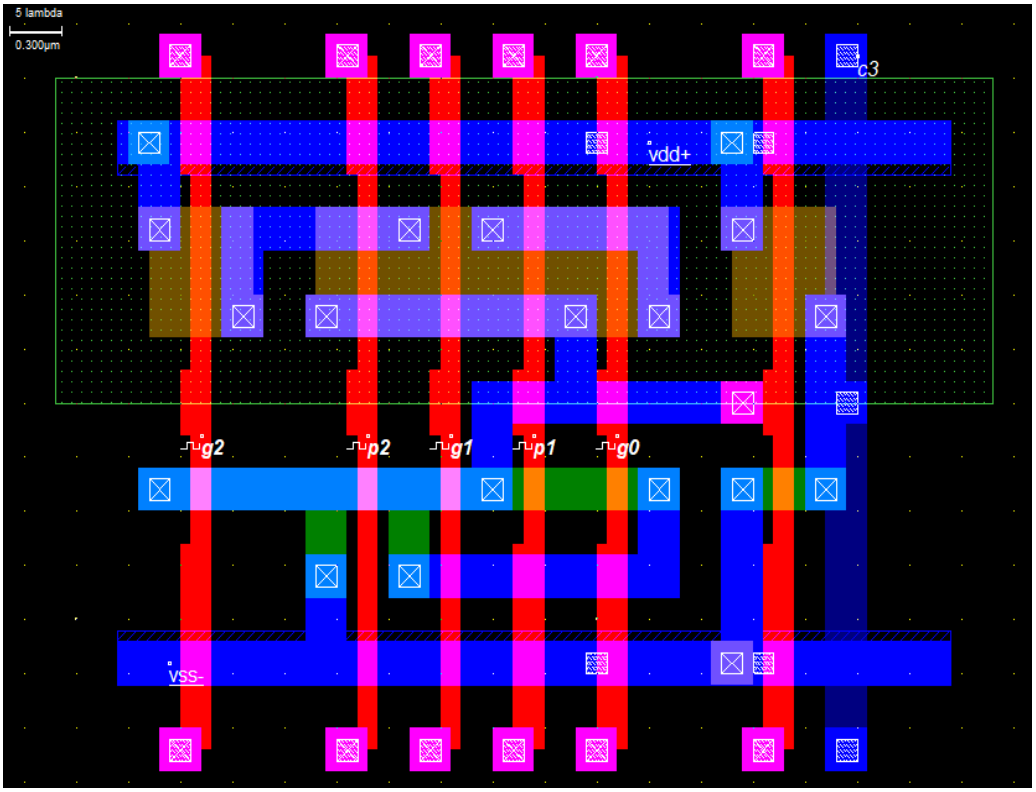
1. Open Micro-wind DSCH Software.
2. Draw the Ripple Carry Adder following the design
3. Save the circuit and make Verilog file.
4. Open micro-wind and compile the Verilog file.
5. Observe layout and run simulation.

Output: Ripple Carry Adder

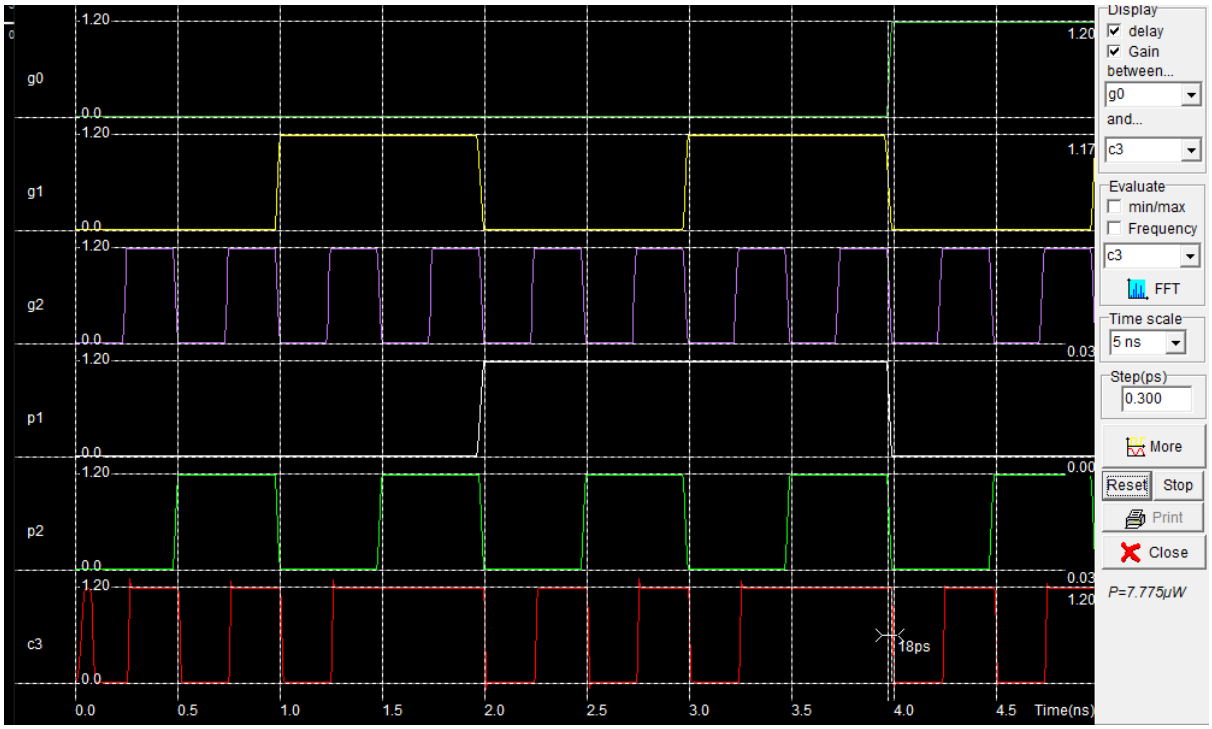


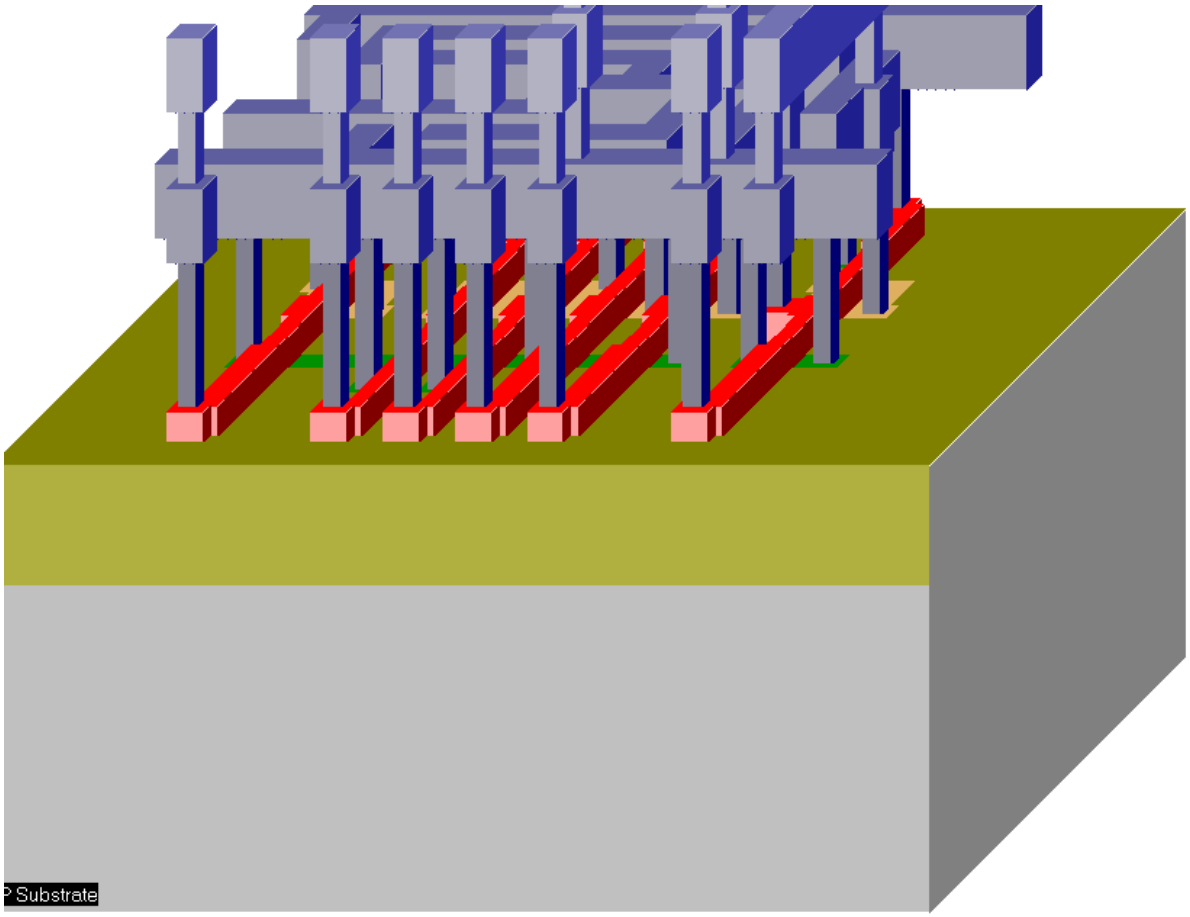

5.0ns/div
0.0
50.0
100.0
150.0
200.0
250.0
300.0
350.0
400.0
450.0
A0
A1
A2
A3
B0
B1
B2
B3
C0
C1
C2
C3
C4
S0
S1
S2
S3

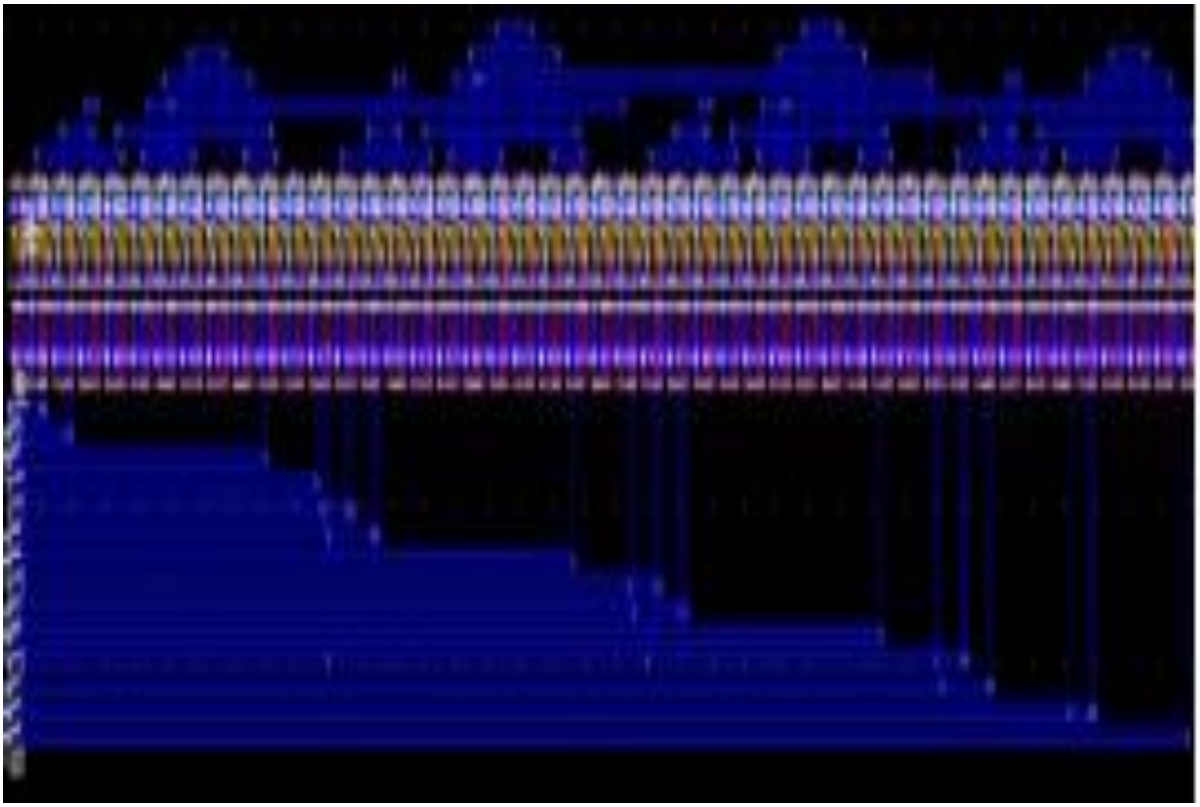
Layout:


5 lambda
0.300µm
c3
vdd+
g2
p2
g1
p1
g0
vss-

Output:


g0
g1
g2
p1
p2
c3
1.20
0.0
1.17
0.03
0.00
18ps
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
Time(ns)
Display
delay
Gain
between...
g0
and...
c3
Evaluate
min/max
Frequency
c3
FFT
Time scale
5 ns
Step(ps)
0.300
More
Reset
Stop
Print
Close
P=7.775μW


P Substrate



Pre Lab-Questions:

1 What is ripple carry adder?
2 Which Adder is used In Ripple carry adder?

Post Lab Questions:

1. What is one disadvantage of the ripple-carry adder?

Conclusion: Four-bit Ripple Carry Adder has been successfully implemented on DSCH and Code has been tested in micro-wind and layout has been recorded.

CO's covered:

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Expt. No. 10

Title: To design the layout of Carry look ahead adder using DSCH and Micro-wind.

Rubric	*Score (1 to 5)*
Timely Submission	
Result Interpretation and Output	
Program Logic and Concept Understanding	
Total	

Performed On: _____

Sign:_____

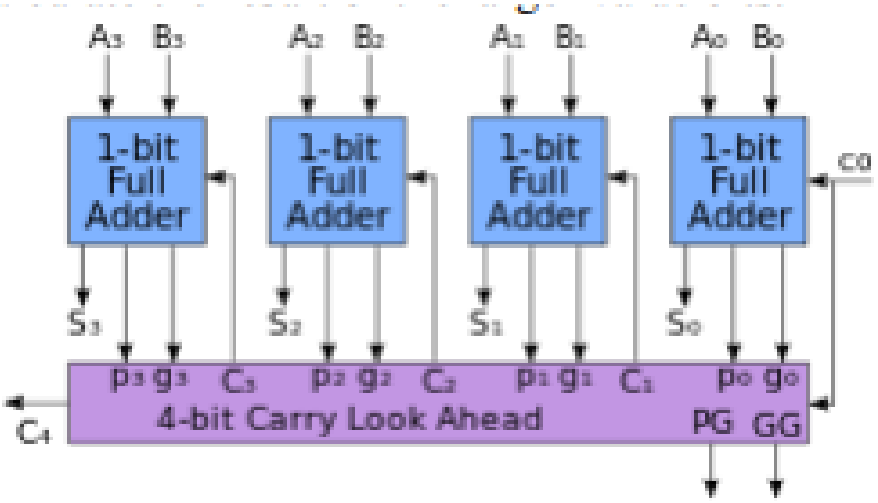
Experiment No. 10

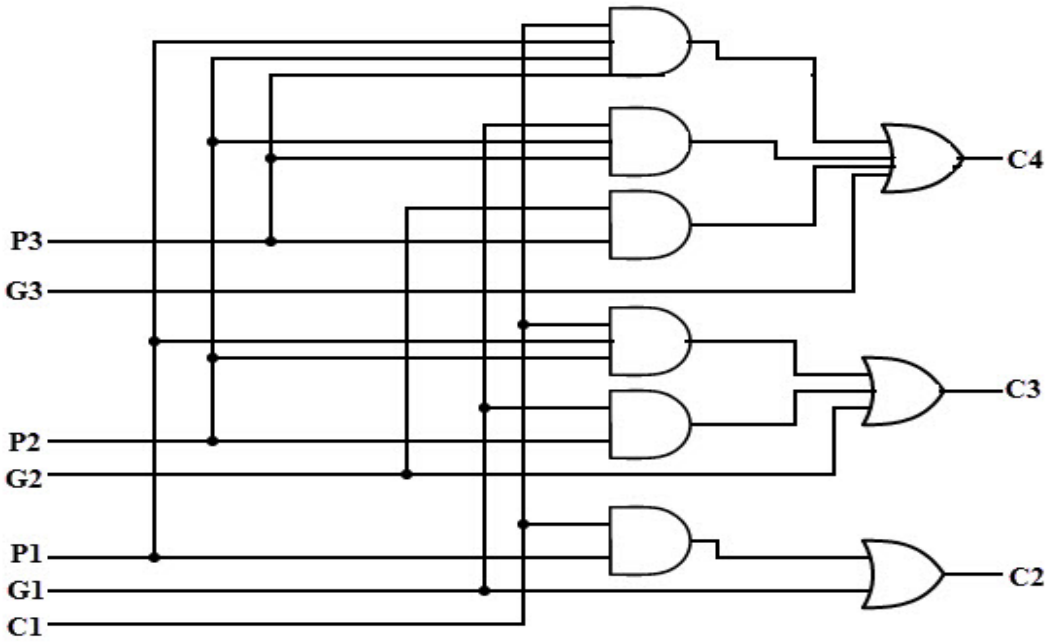
Aim: To design the layout of Carry look ahead adder using DSCH and Micro-wind

Software Required: DSCH and Micro-wind software

Hardware: computer

Circuit Diagram:



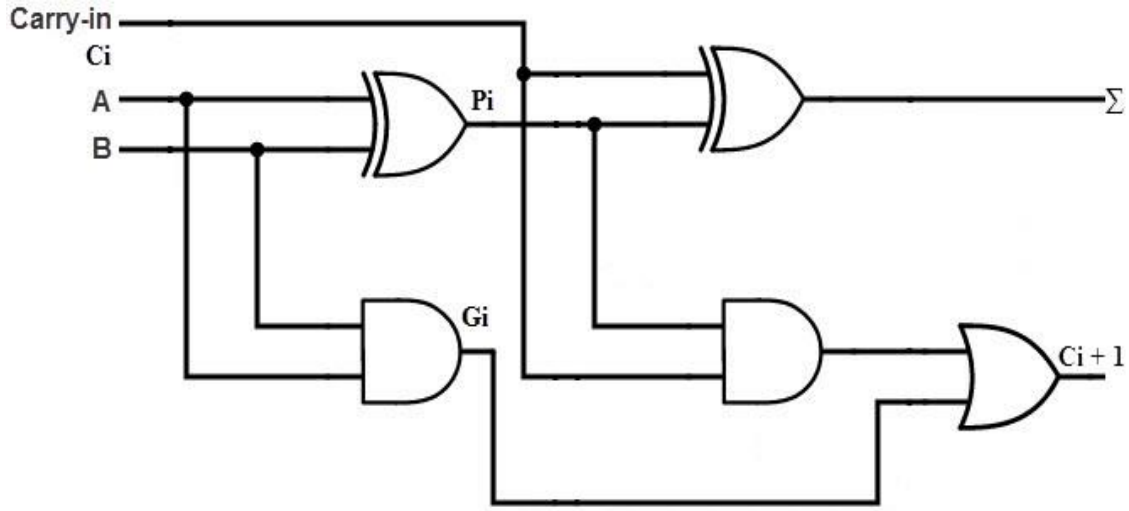


Theory:

Carry-Look ahead Adder

A carry-Lookahead adder is a fast parallel adder as it reduces the propagation delay by more complex hardware, hence it is costlier. In this design, the carry logic over fixed groups of bits of the adder is reduced to two-level logic, which is nothing but a transformation of the ripple carry design.

This method makes use of logic gates so as to look at the lower order bits of the augend and addend to see whether a higher order carry is to be generated or not. Let us discuss in detail.



A	B	Ci	C i +1	Condition
0	0	0	0	No carry generate
0	0	1	0	
0	1	0	0	
0	1	1	1	No carry propagate
1	0	0	0	
1	0	1	1	
1	1	0	1	Carry generate
1	1	1	1	

Consider the full adder circuit shown above with corresponding truth table. If we define two variables as carry generate Gi and carry propagate Pi then,

$P_i = A_i \oplus B_i$

Gi = Ai Bi

The sum output and carry output can be expressed as

$S_i = P_i \oplus C_i$

$C_{i+1} = G_i + P_i C_i$

Where Gi is a carry generate which produces the carry when both Ai, Bi are one regardless of the input carry. Pi is a carry propagate and it is associate with the propagation of carry from Ci to Ci +1.

The carry output Boolean function of each stage in a 4 stage carry-Lookahead adder can be expressed as

$C1 = G0 + P0\ Cin$

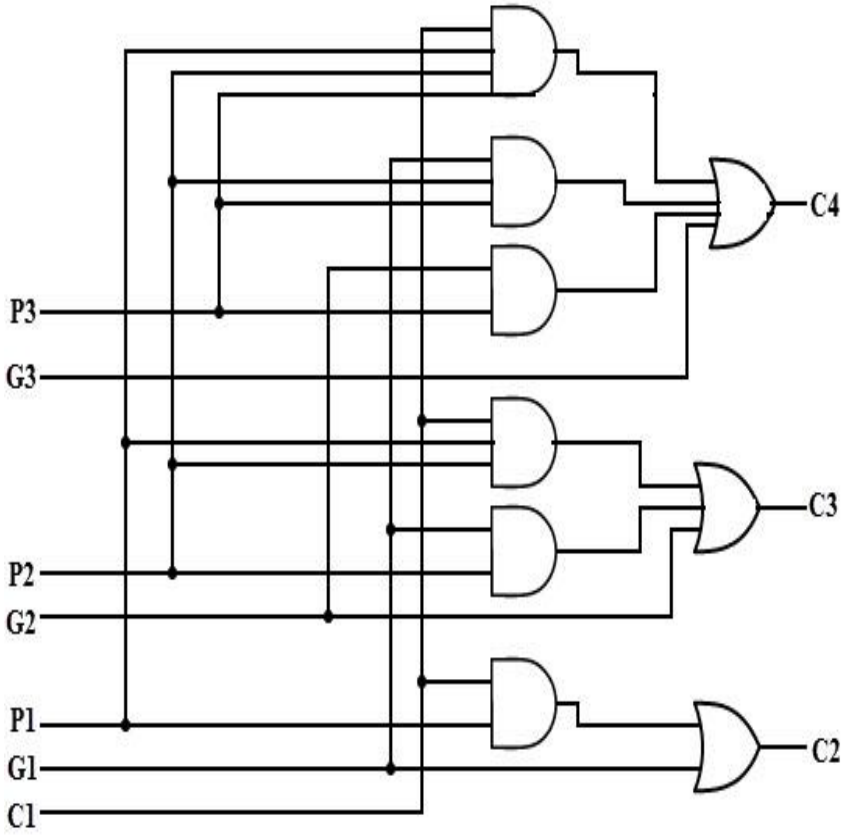
$C2 = G1 + P1\ C1 = G1 + P1\ G0 + P1\ P0\ Cin$

$C3 = G2 + P2\ C2 = G2 + P2\ G1 + P2\ P1\ G0 + P2\ P1\ P0\ Cin$

$C4 = G3 + P3\ C3 = G3 + P3\ G2 + P3\ P2\ G1 + P3\ P2\ P1\ G0 + P3\ P2\ P1\ P0\ Cin$

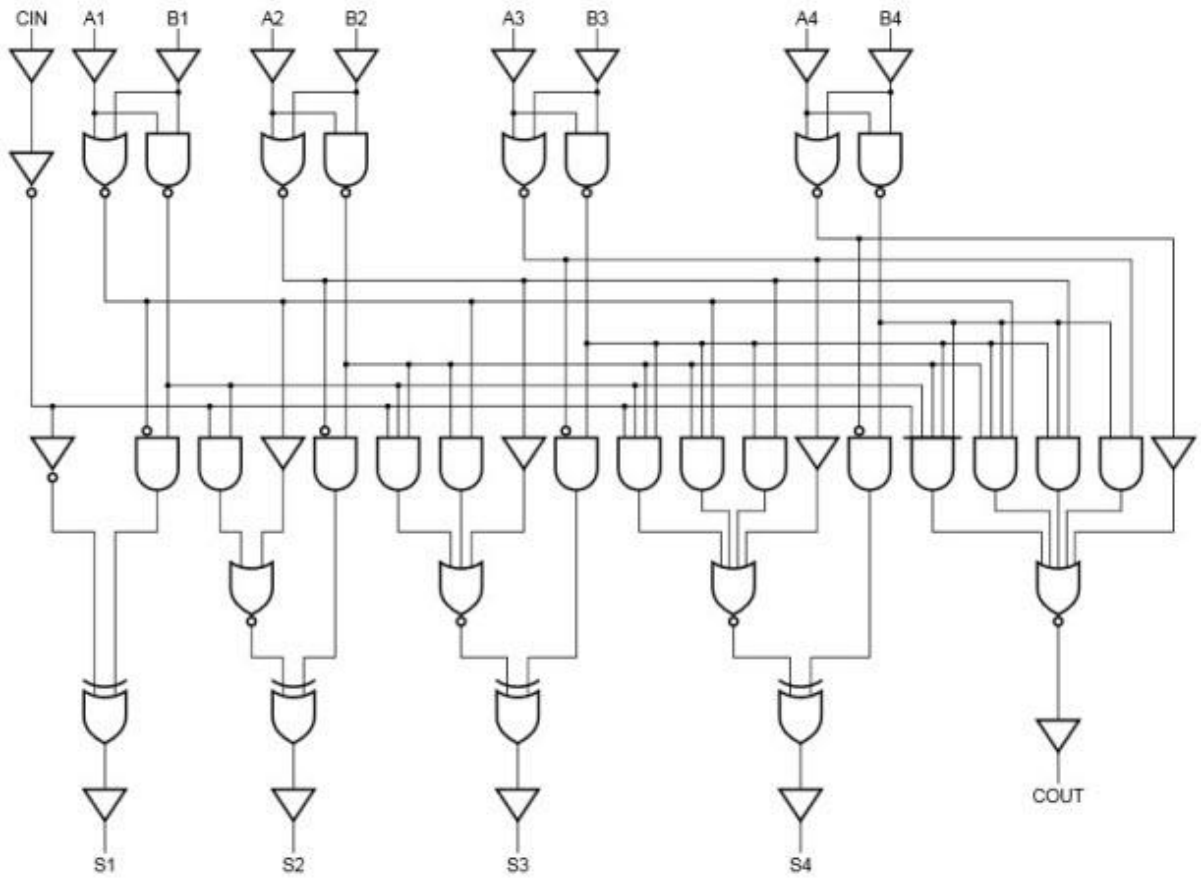
From the above Boolean equations, we can observe that C4 does not have to wait for C3 and C2 to propagate but actually C4 is propagated at the same time as C3 and C2. Since the Boolean expression for each carry output is the sum of products so these can be implemented with one level of AND gates followed by an OR gate.

The implementation of three Boolean functions for each carry output (C2, C3 and C4) for a carry-Lookahead carry generator shown in below figure.



Therefore, a 4 bit parallel adder can be implemented with the carry-Lookahead scheme to increase the speed of binary addition as shown in below figure. In this, two Ex-OR gates are required by each sum output. The first Ex-OR gate generates Pi variable output and the AND gate generates Gi variable.

Hence, in two gates levels all these P's and G's are generated. The carry-Lookahead generators allows all these P and G signals to propagate after they settle into their steady state values and produces the output carriers at a delay of two levels of gates. Therefore, the sum outputs S2 to S4 have equal propagation delay times.



It is also possible to construct 16 bit and 32 bit parallel adders by cascading the number of 4 bit adders with carry logic. A 16 bit carry-Lookahead adder is constructed by cascading the four 4 bit adders with two more gate delays, whereas the 32 bit carry-Lookahead adder is formed by cascading of two 16 bit adders.

In a 16 bit carry-Lookahead adder, 5 and 8 gate delays are required to get C16 and S15 respectively, which are less as compared to the 9 and 10 gate delay for C16 and S15 respectively in cascaded four bit carry-Lookahead adder blocks. Similarly, in 32 bit adder, 7 and 10 gate delays are required by C32 and S31 which are less compared to 18 and 17 gate delays for the same outputs if the 32 bit adder is implemented by eight 4 bit adders.

Carry-Look ahead Adder ICs

The high-speed carry-Lookahead adders are integrated on integrated circuits in different bit configurations by several manufacturers. There are several individual carry generator ICs are available so that we have to make connection with logic gates to perform the addition operation.

A typical carry-Lookahead generator IC is 74182 which accept four pairs of active low carry propagate (as P0, P1, P2 and P3) and carry generate (Go, G1, G2 and G3) signals and an active high input (Cn).

It provides active high carriers (Cn+x, Cn+y, Cn+z) across the four groups of binary adders. This IC also facilitates the other levels of look ahead by active low propagate and carry generate outputs.

The logic expressions provided by the IC 74182 are

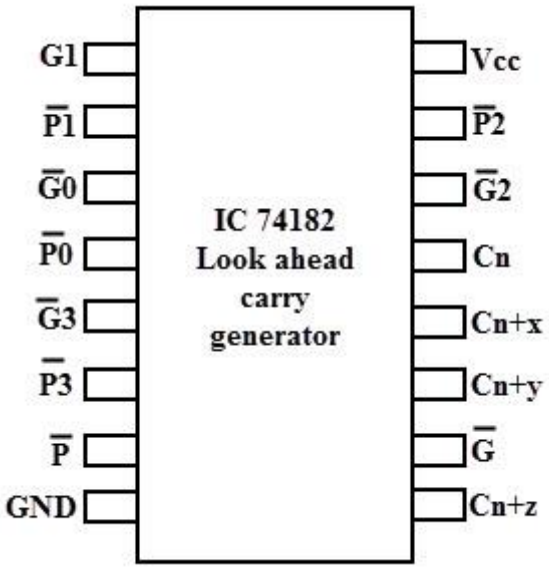
$$C_{n+x} = G_0 + P_0 C_n$$

$$C_{n+y} = G_1 + P_1 G_0 + P_1 P_0 C_n$$

$$C_{n+z} = G_2 + P_2 G_1 + P_2 P_1 G_0$$

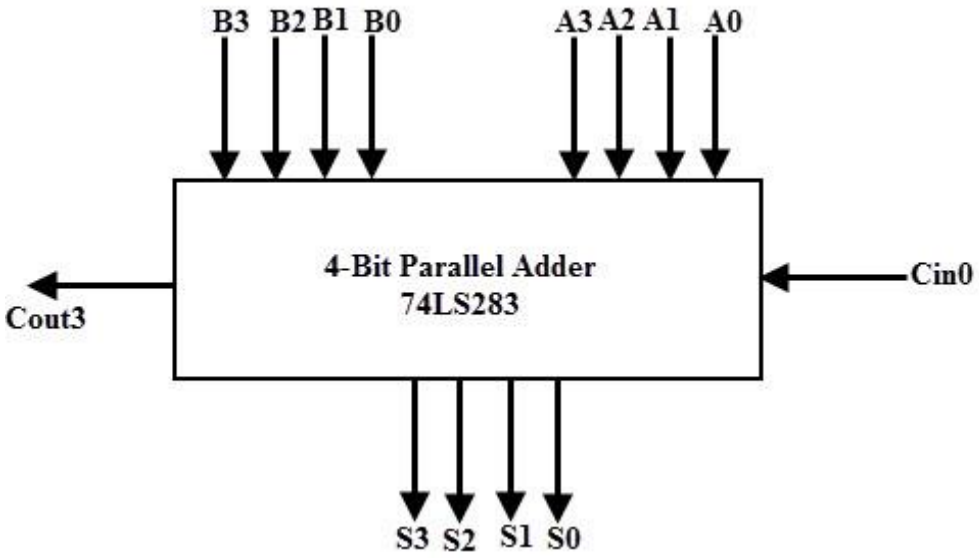
$$\overline{G} = \overline{G3 + P3\,G2 + P3\,P2\,G1 + P3\,P2\,P1\,G0}$$

$$\overline{P} = \overline{P3\,P2\,P1\,P0}$$



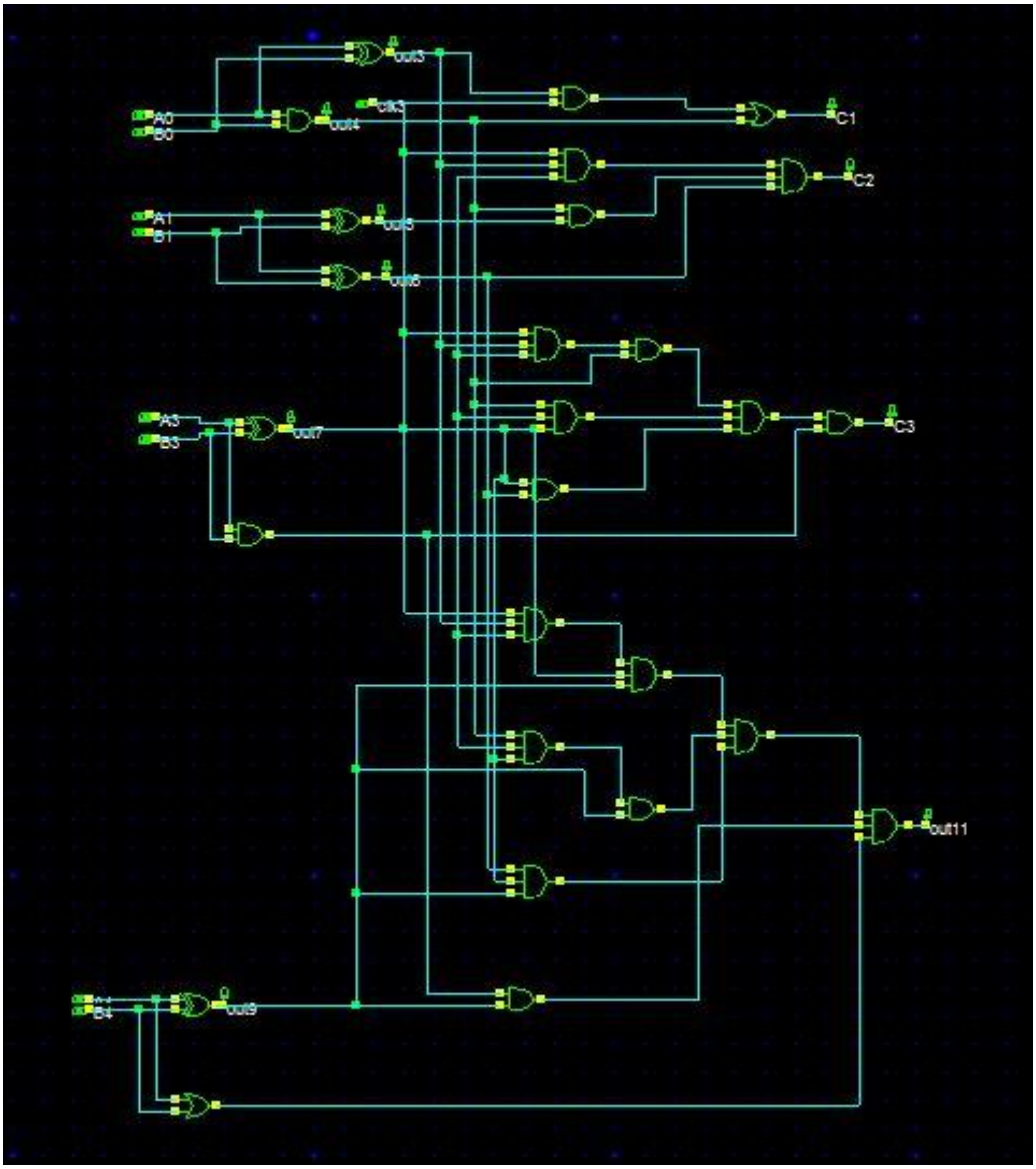
On the other hand, there are many high speed adder ICs which combine a set of full adders with carry-Lookahead circuitry. The most popular form of such IC is 74LS83/74S283 which is a 4 bit parallel adder high speed IC that contains four interconnected full adders with a carry-Lookahead circuitry.

The functional symbol for this type of IC is shown in below figure. It accepts the two 4 bit numbers as A3A2A1A0 and B3B2B1B0 and input carry Cin0 into the LSB position. This IC produce output sum bits as S3S2S1S0 and the carry output Cout3 into the MSB position.



y cascading two or more parallel adder ICs we can perform the addition of larger binary numbers such as 8 bit, 24-bit and 32 bit addition.

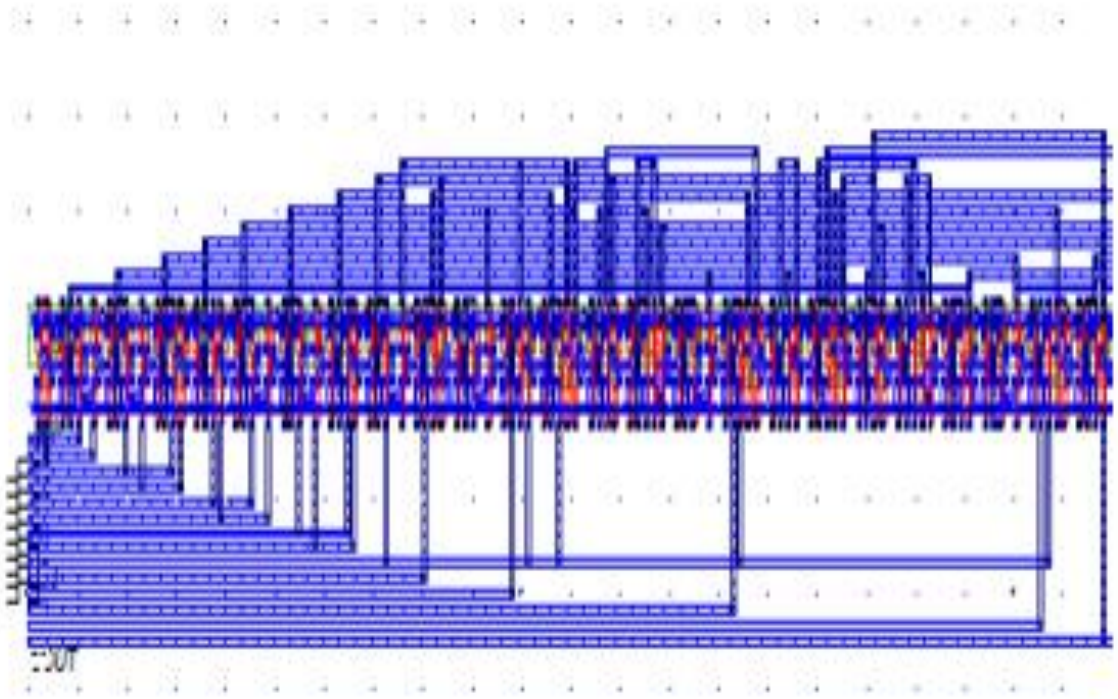
Diagram:



Procedure:

1. Open Micro-wind DSCH Software.
2. Draw the Ripple Carry Adder following the design
3. Save the circuit and make verilog file.
4. Open micro-wind and compile the verilog file.
5. Observe layout and run simulation.

Layout:



Pre-Lab Questions:

1 What distinguishes the look-ahead-carry adder?

Post Lab Questions:

1. Carry look-ahead logic uses the concepts of _____

Conclusion: Four bit Carry look ahead adder has been successfully implemented on DSCH and Code has been tested in micro-wind and layout recorded.

CO's covered:

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ASSIGNMENTS